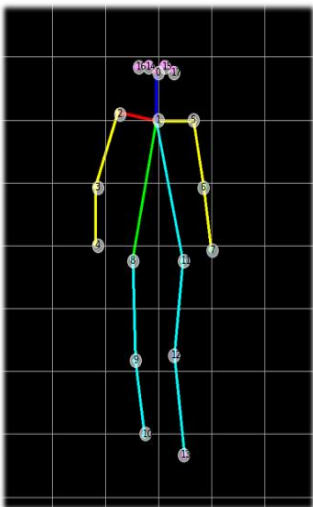
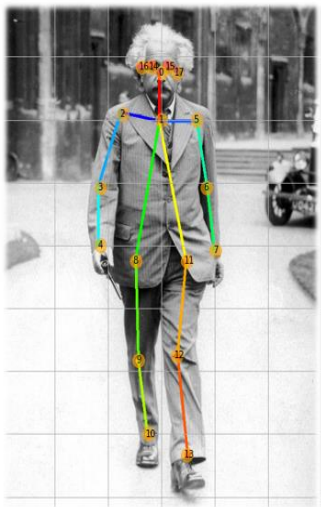




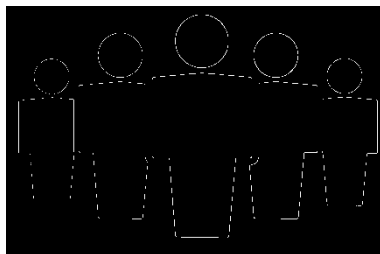
DIGITAL IMAGE PROCESSING

การใช้การประมวลผลทางคอมพิวเตอร์ (Computer Algorithm) ในการจัดการรูปภาพ มีจุดประสงค์การทำงานที่หลากหลาย เช่น การปรับปรุงรูปภาพ (Enhancement) การตรวจจับวัตถุ (Detection) การแบ่งแยกบริเวณในภาพ (Segmentation) เป็นต้น





IMAGE

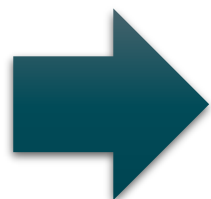


IMAGE

OUTPUT



IMAGE



5 OBJECTS

HEIGHT 1.5 M

DESCRIPTOR



IMAGE



A cat that is lying
down and looking
at the camera

RECOGNITION
/ CLASSIFICATION

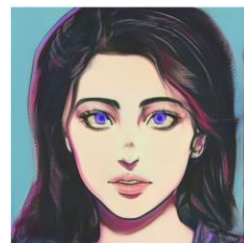
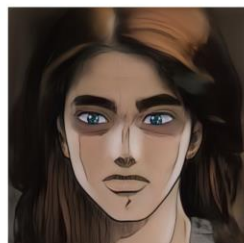
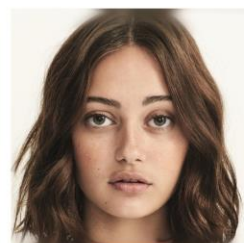
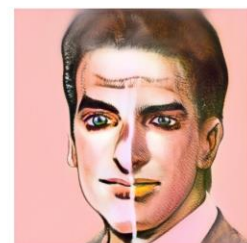
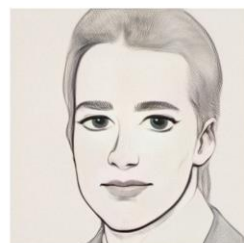
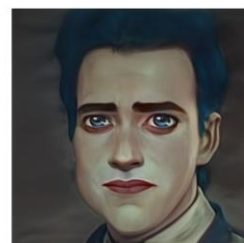
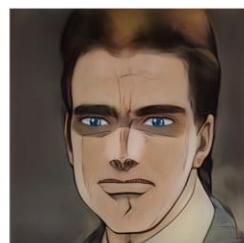
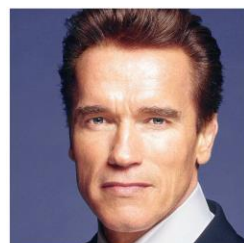
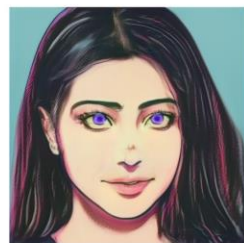
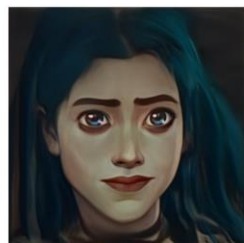
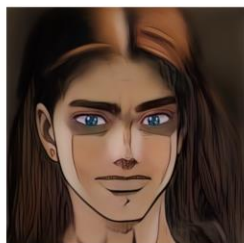
References



OOD stylizations



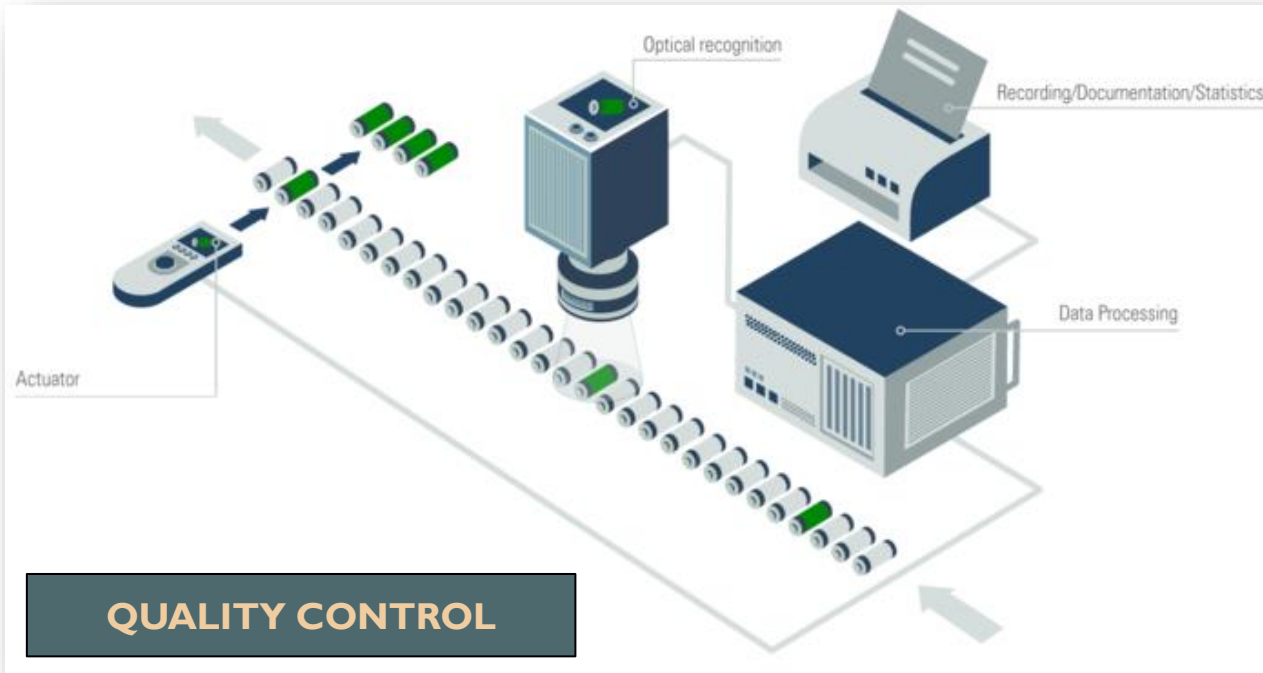
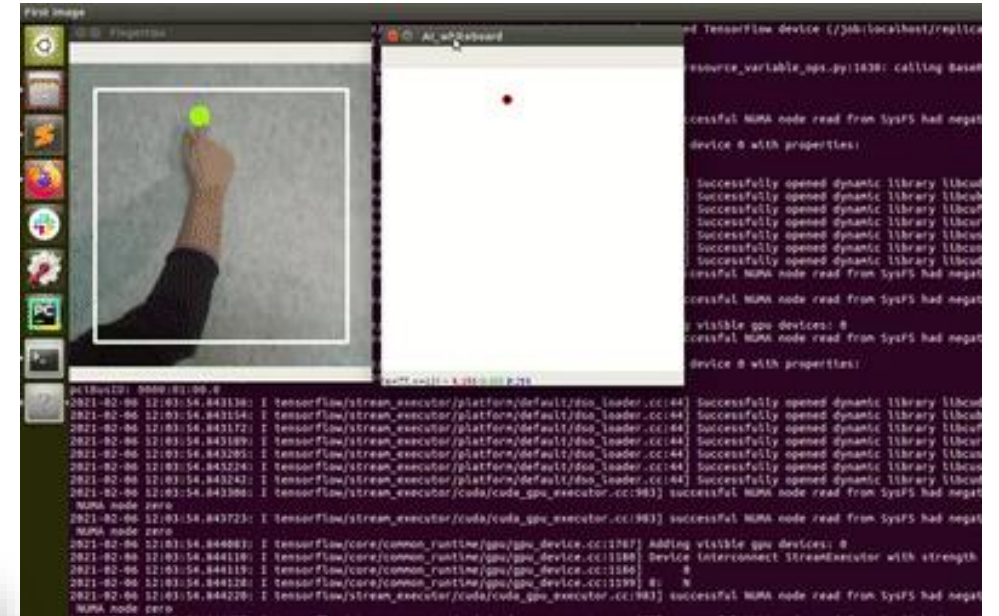
Inputs



Continuous stylizations



LICENSE PLATE RECOGNITION

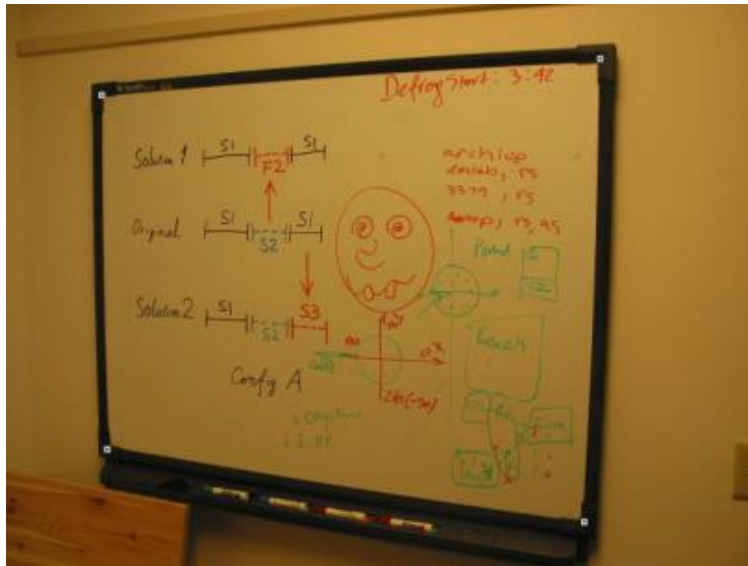
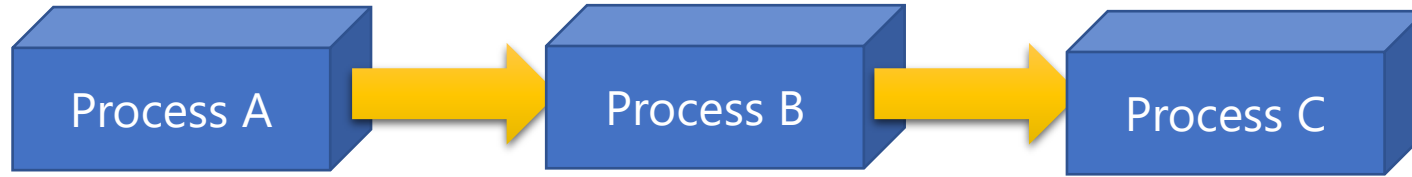


QUALITY CONTROL

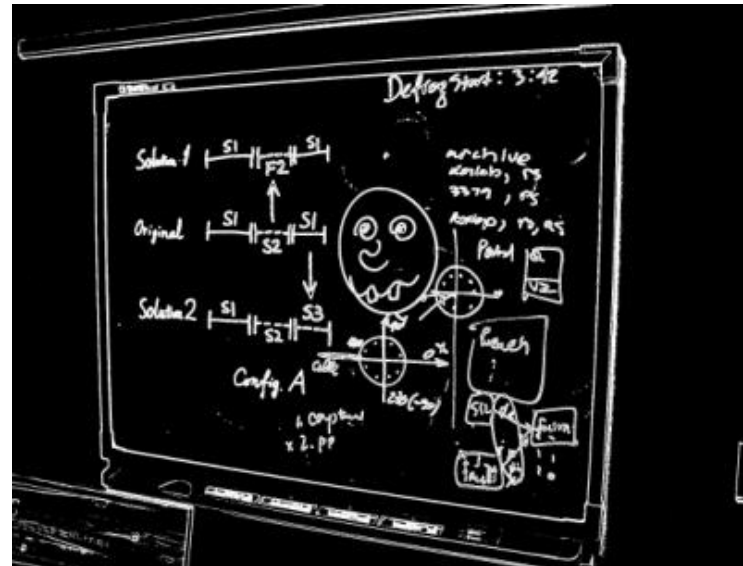


IMAGE RESTORATION

*** ถ้าโจทย์ปัญหายาก ให้ใช้วิธีกำหนดเงื่อนไข, ขอบเขตของงาน, แบ่งเป็นงานย่อย



TAKE IMAGE FROM CAMERA



EDGE & CORNER DETECTION

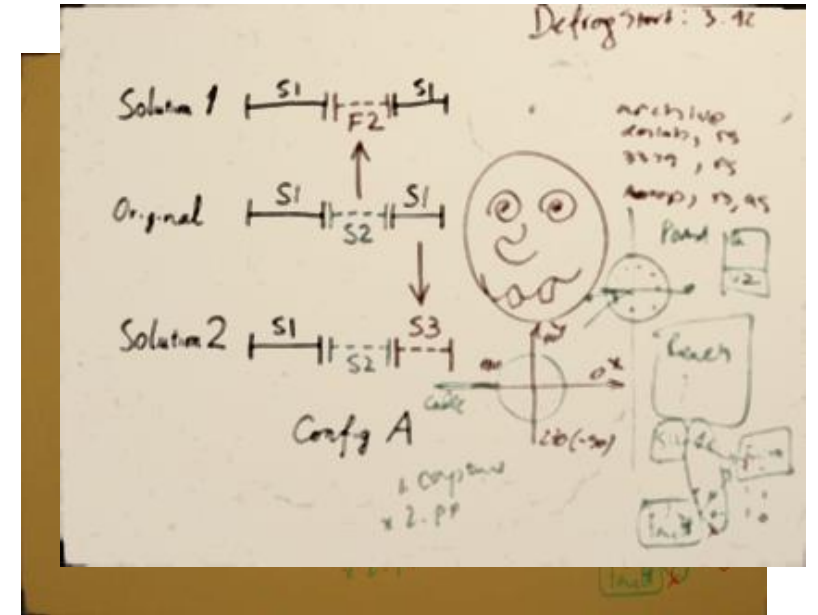
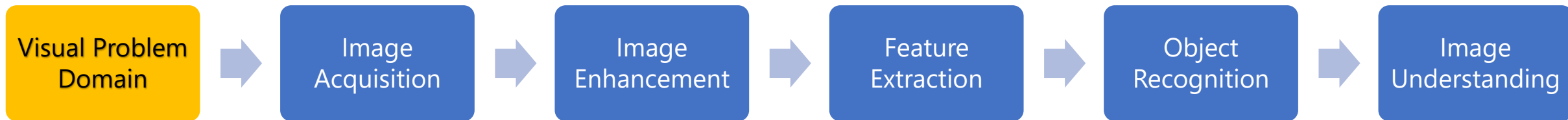


IMAGE TRANSFORMATION

ตัวอย่างการใช้เครื่องมือ IMAGE PROCESSING

Class Projects



แนวทางการหาโจทย์ปัญหา งานทางด้านการประมวลผลภาพ (อย่างง่าย)

- 1.) งานที่ใช้ตา (มอง) และทำการตัดสินใจ
- 2.) งานที่มีการทำงานอย่างต่อเนื่อง => ต้องการระบบอัตโนมัติ
- 3.) งานที่จำเป็นต้องมีการแก้ไข ปรับปรุงภาพ

การคัดแยกวัตถุ

การตรวจสอบความสมบูรณ์

การหาวัตถุที่กำหนดในภาพ

การนับชิ้นวัตถุ

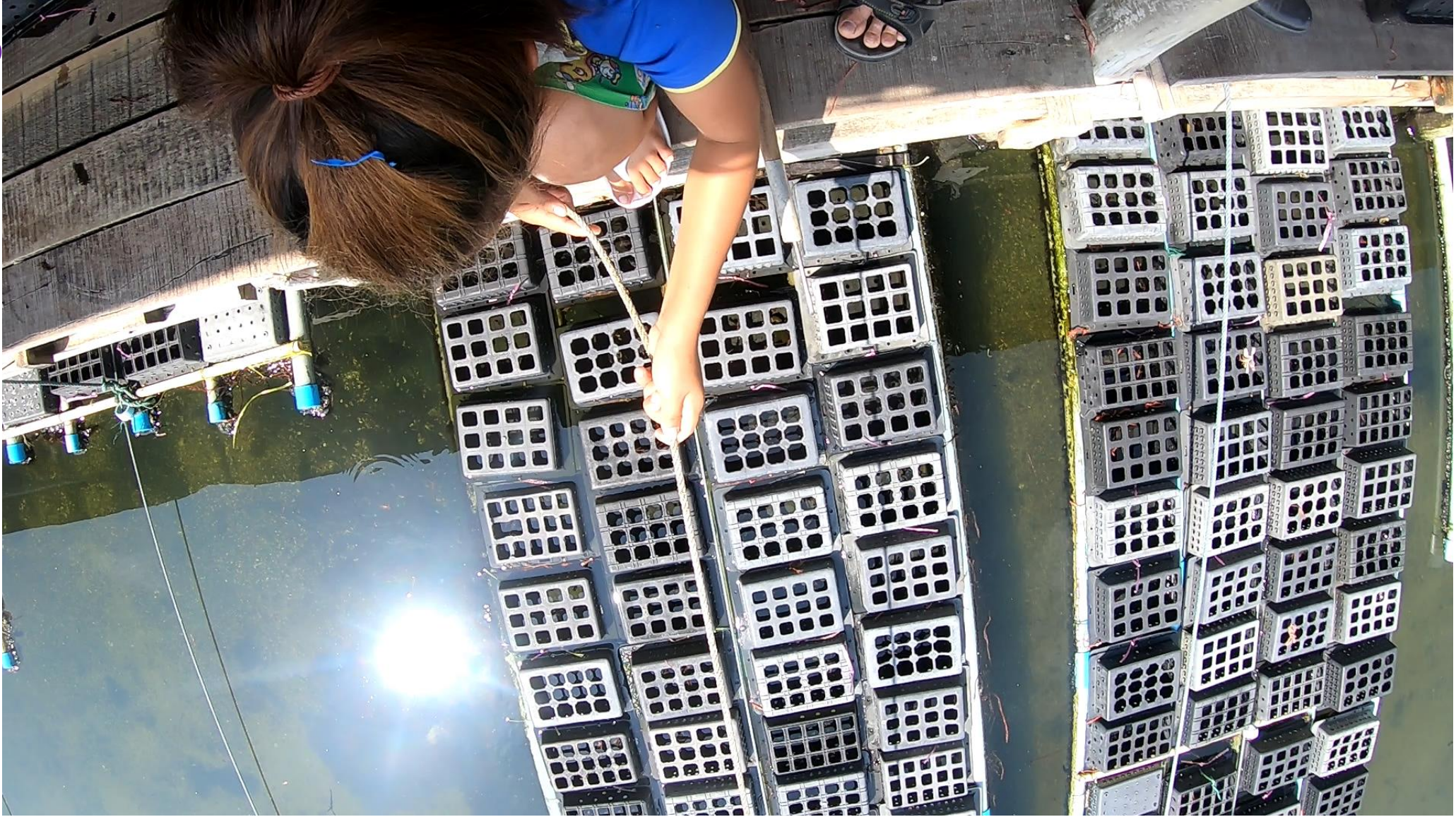
การวัดปริมาณจากสิ่งที่เห็น

Problem

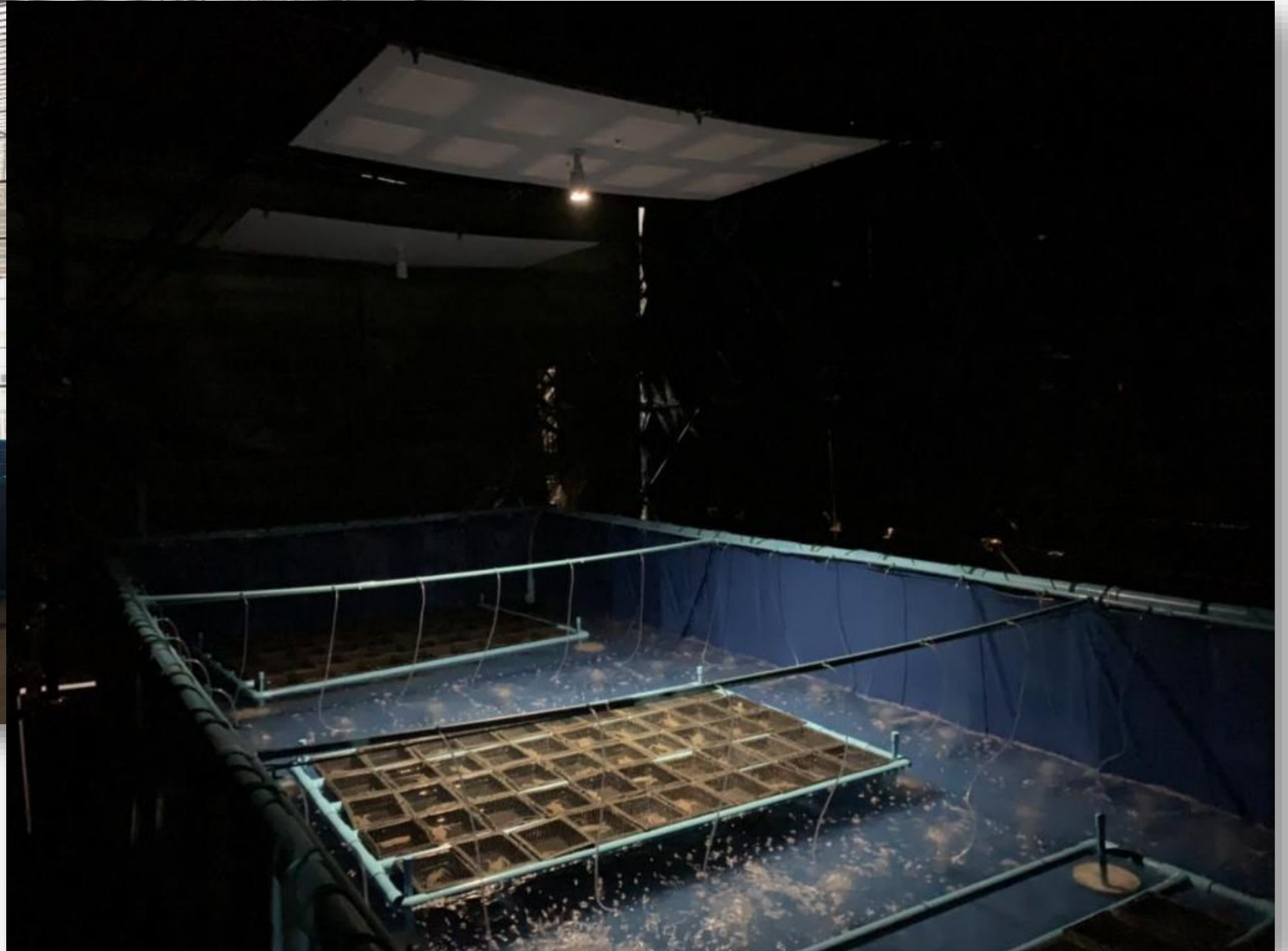


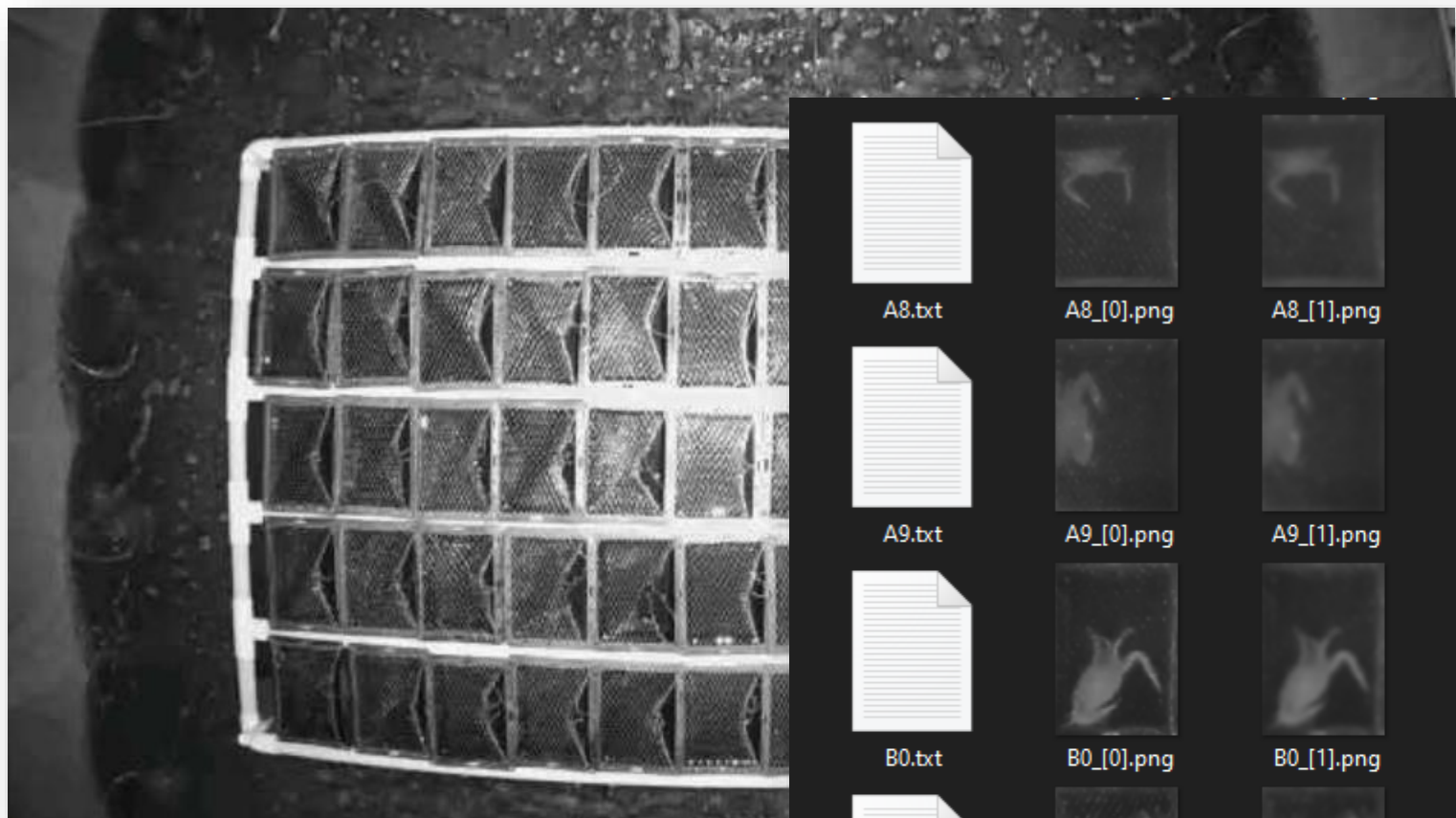
24 x 7 Workload

Prob

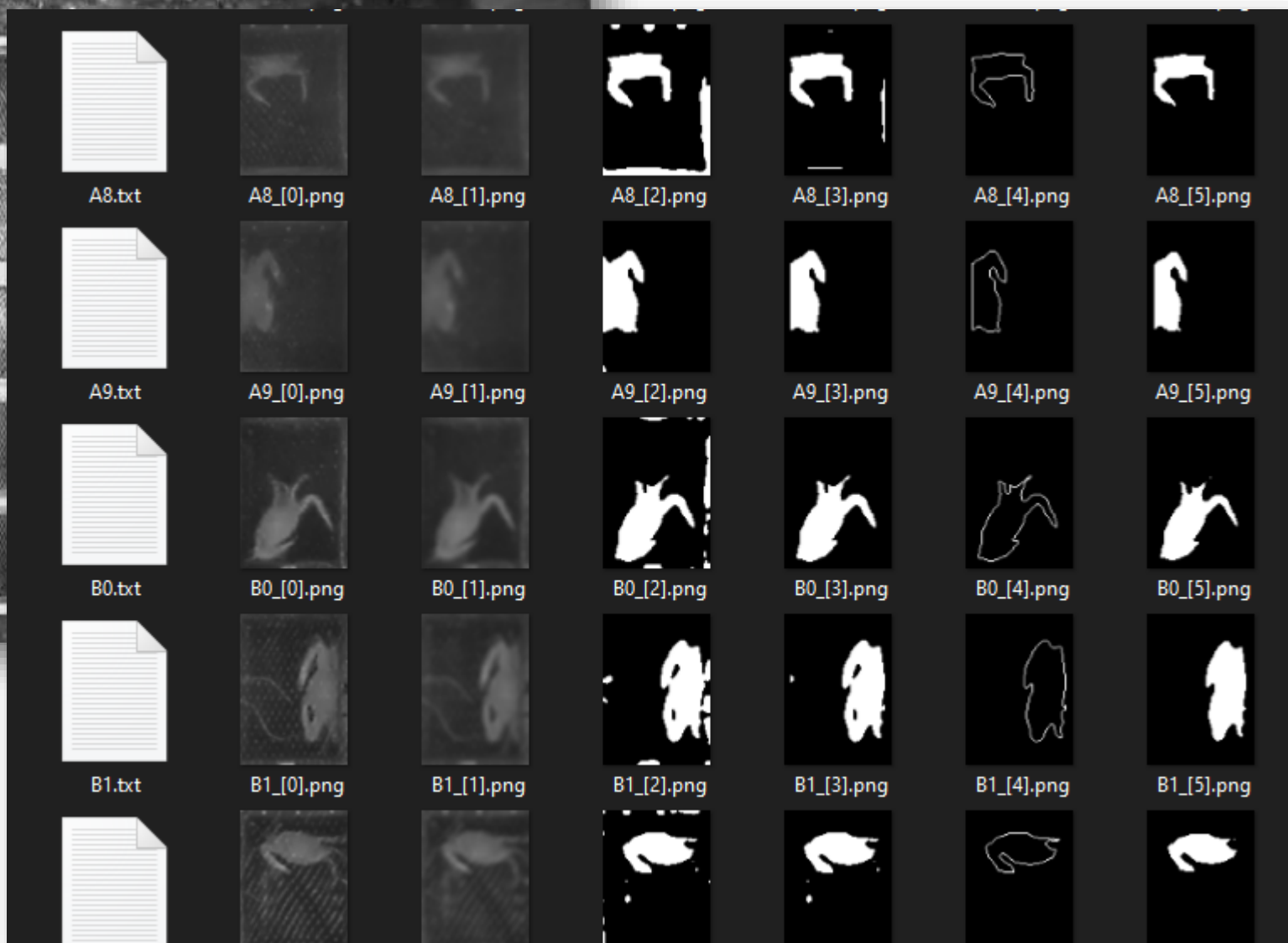


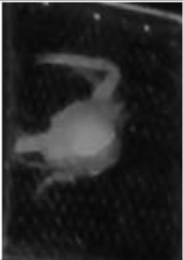
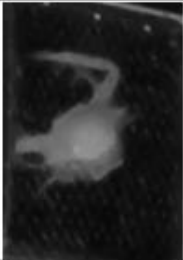
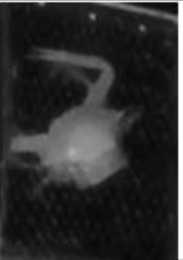
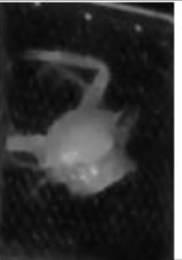
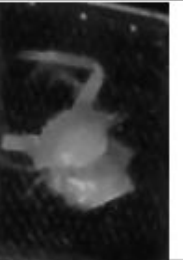
skilled worker (need to train)



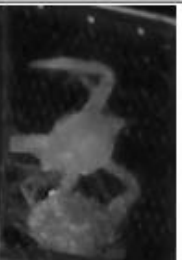
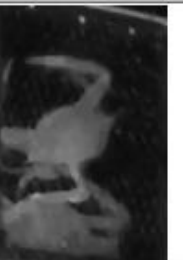
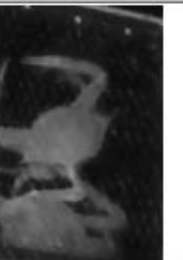


Algorithm

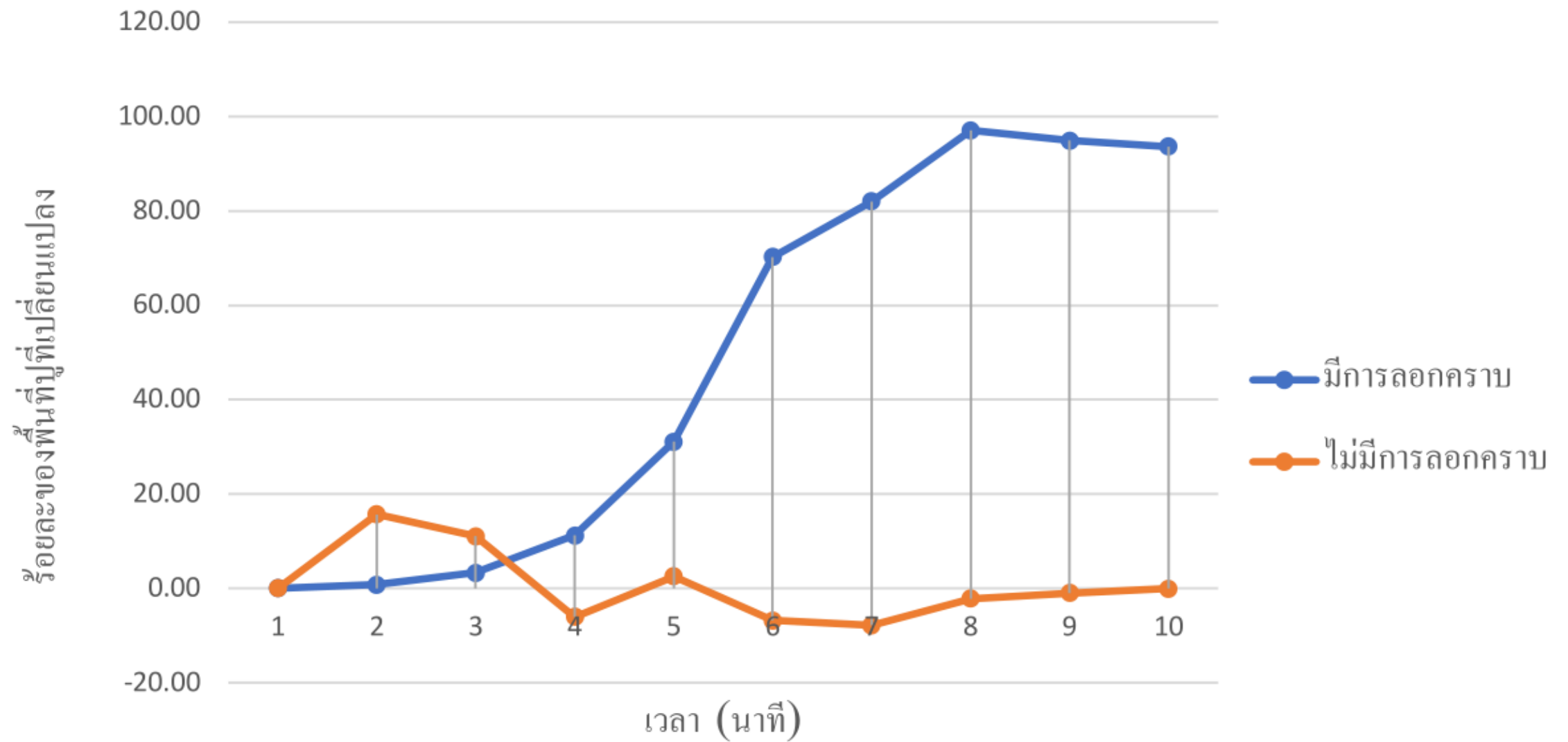


ภาพกล้อง					
ภาพตรวจจับ					
เวลา (นาทื)	1	2	3		
พื้นที่ (ร้อยละ)	15.02	15.14	15.51		

Experiment

ภาพกล้อง					
ภาพตรวจจับ					
เวลา (นาทื)	6	7	8	9	10
พื้นที่ (ร้อยละ)	25.57	27.34	29.60	29.28	29.09

Conclusion



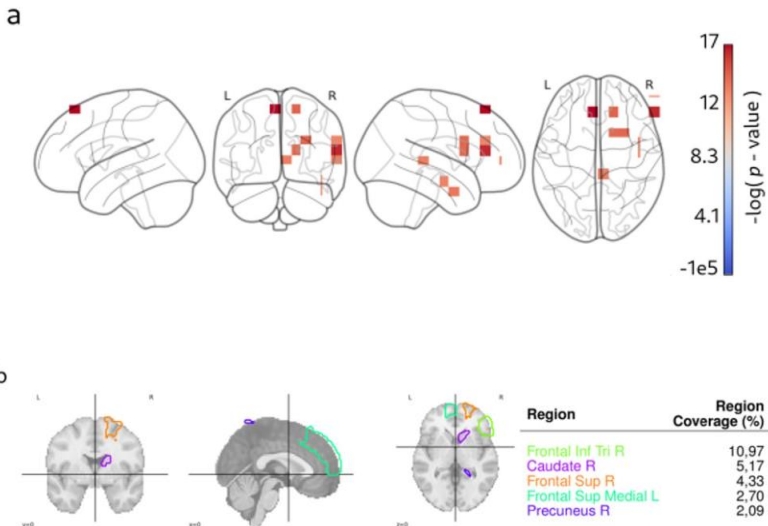
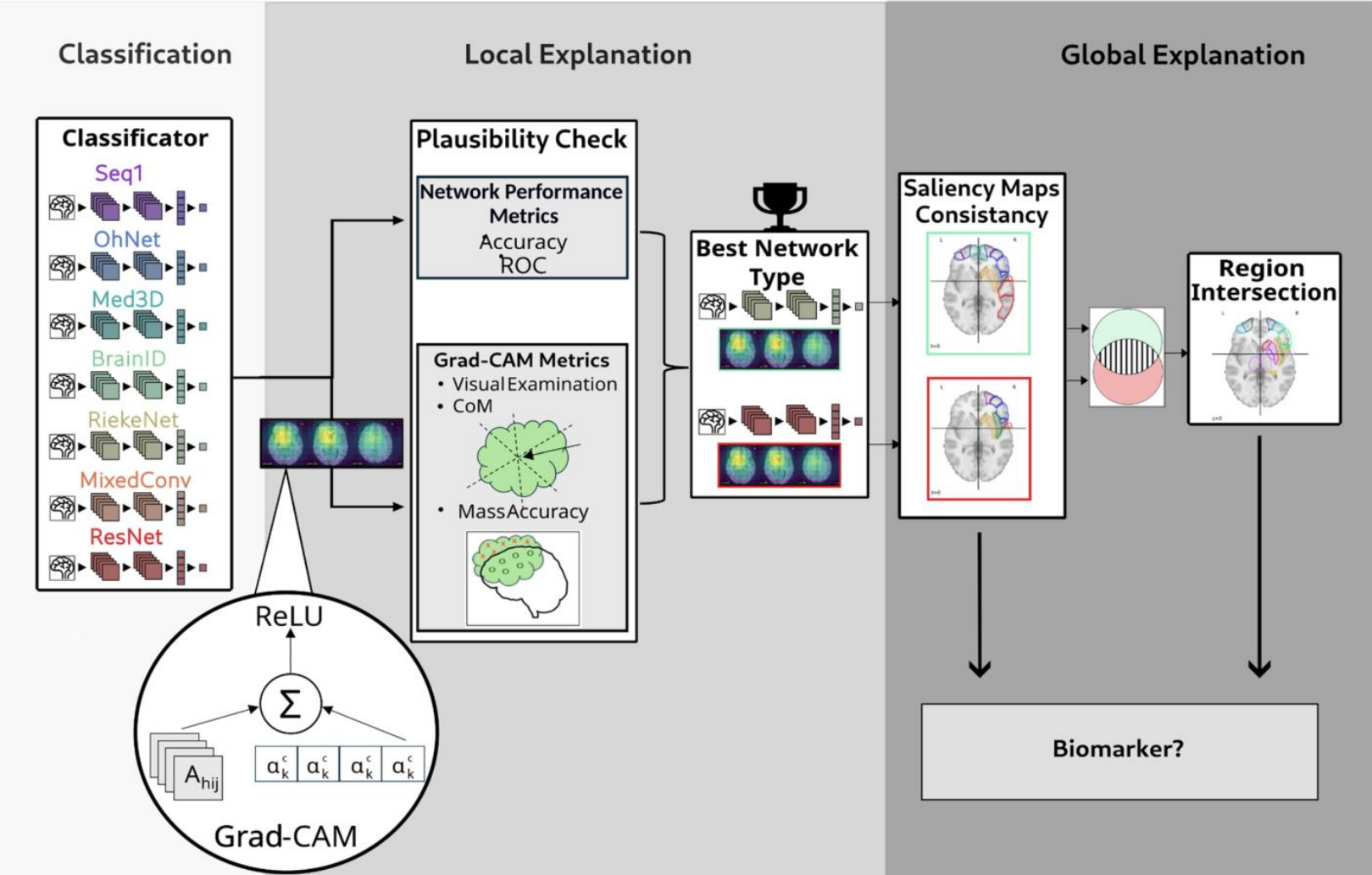


Research in Image Processing

Image Processing (310-3311)

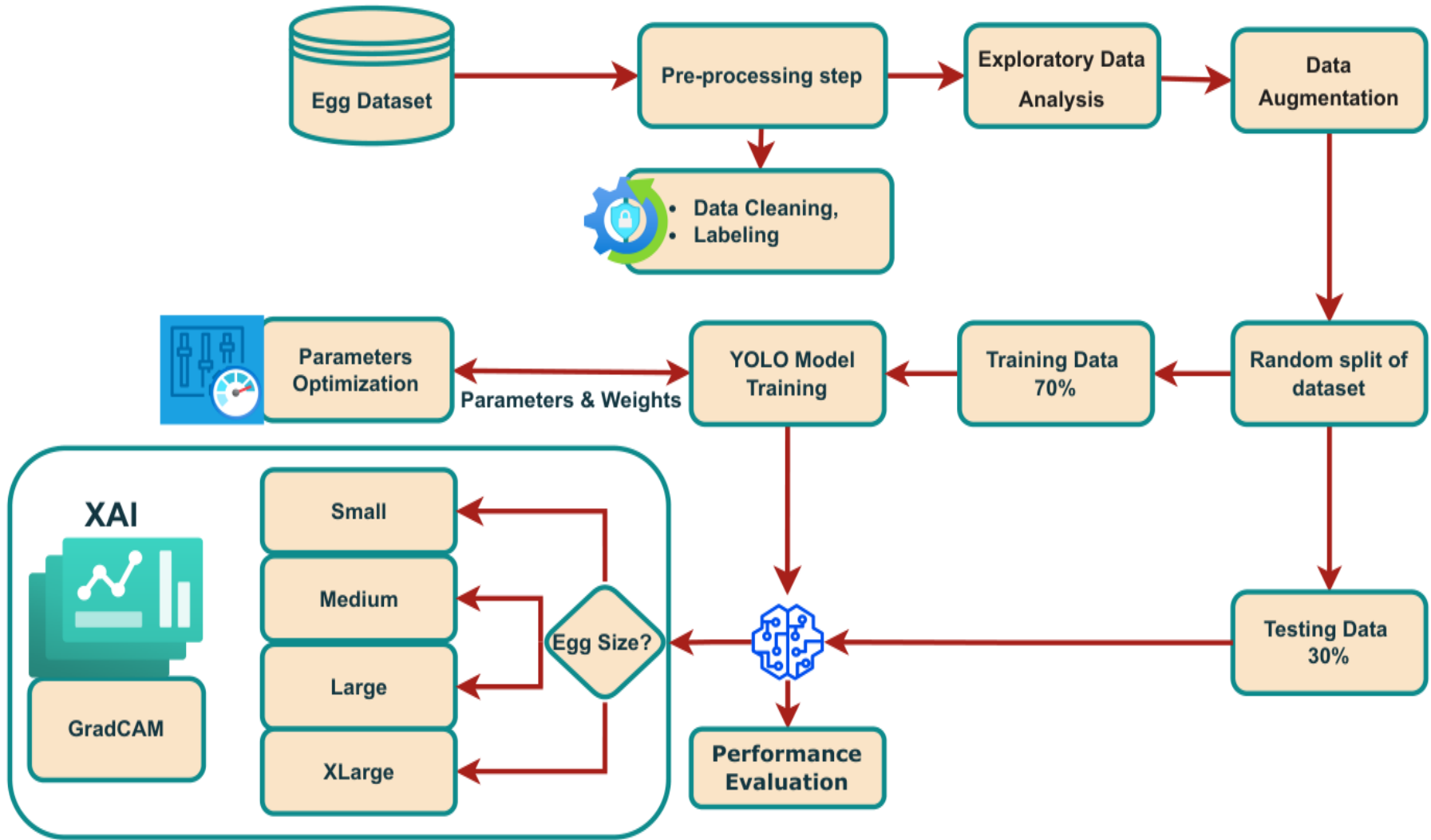


Decision processes in 3D structural MRI schizophrenia classification evaluated with saliency maps [2026]

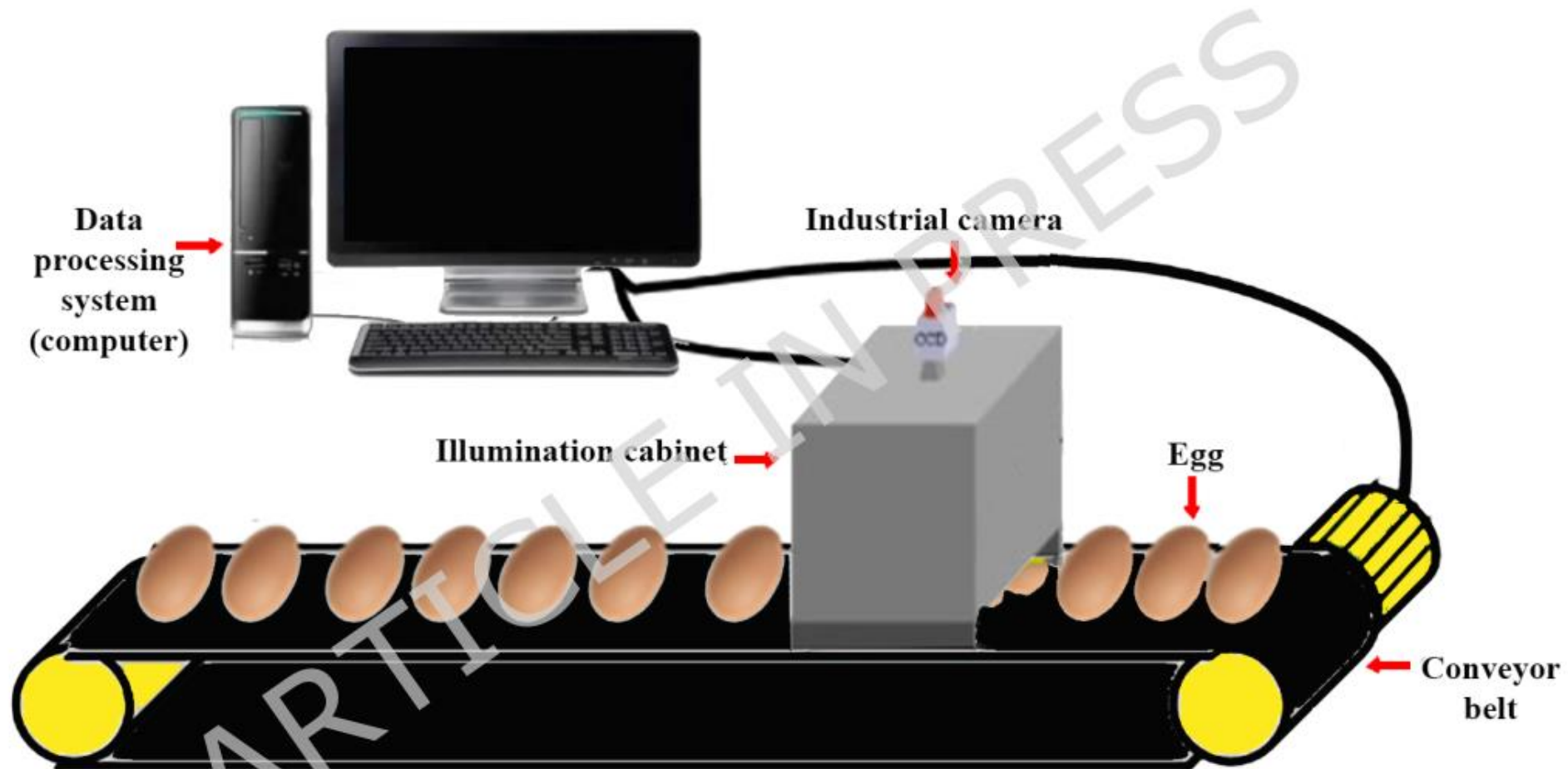


	Schizophrenia	Control
Number of subjects	101	91
Age (years) mean $\pm$ STD	34.2 $\pm$ 11.2	32.3 $\pm$ 11.8
Gender (male/female)	78/23	61/30

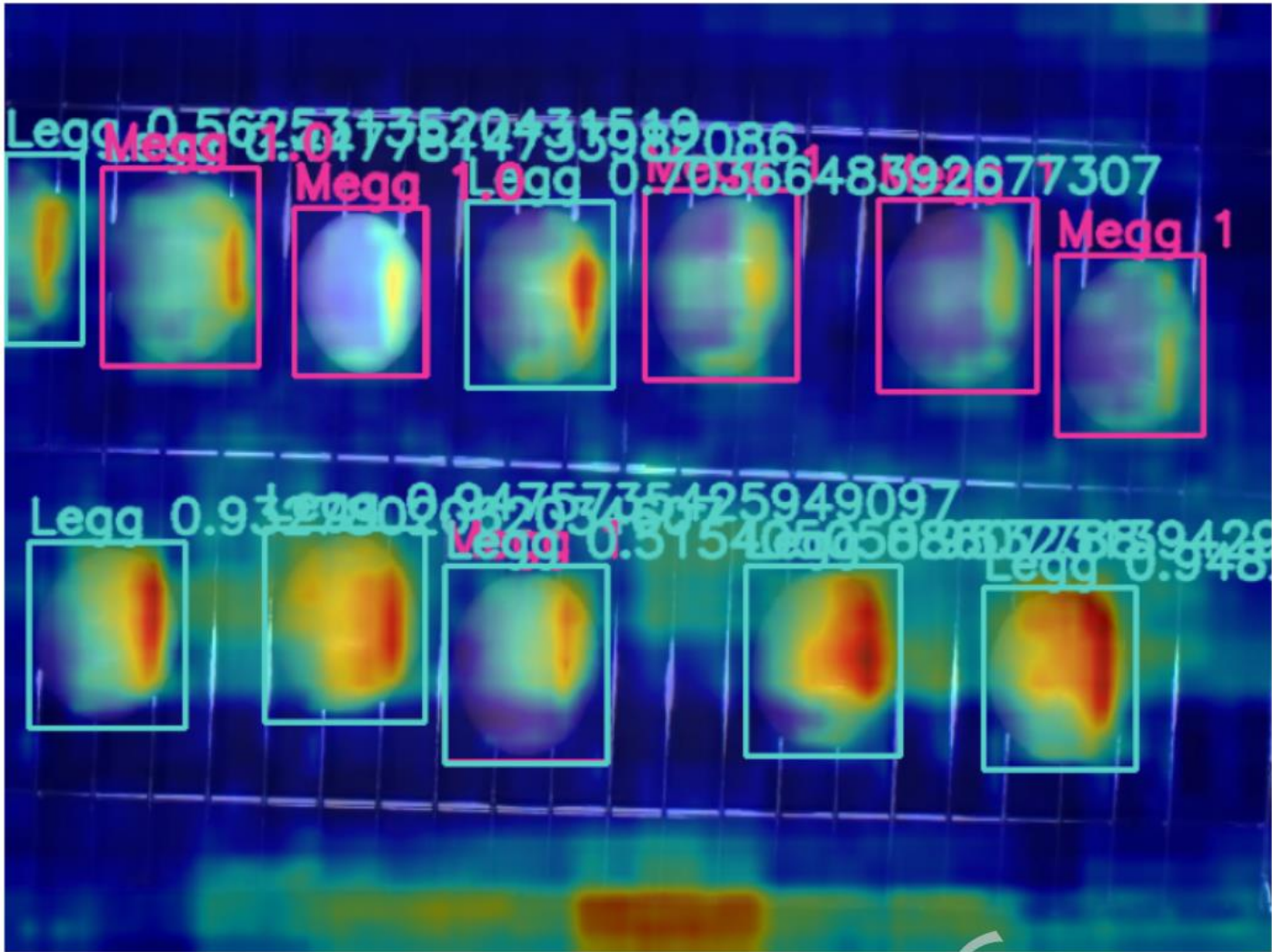
Explainable YOLO architectures for egg size measurement and classification [2026]



Explainable YOLO architectures for egg size measurement and classification [2026]



Explainable YOLO architectures for egg size measurement and classification [2026]



Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization

Visual Examples

What makes the network think the image label is 'pug, pug-dog'	What makes the network think the image label is 'tabby, tabby cat'

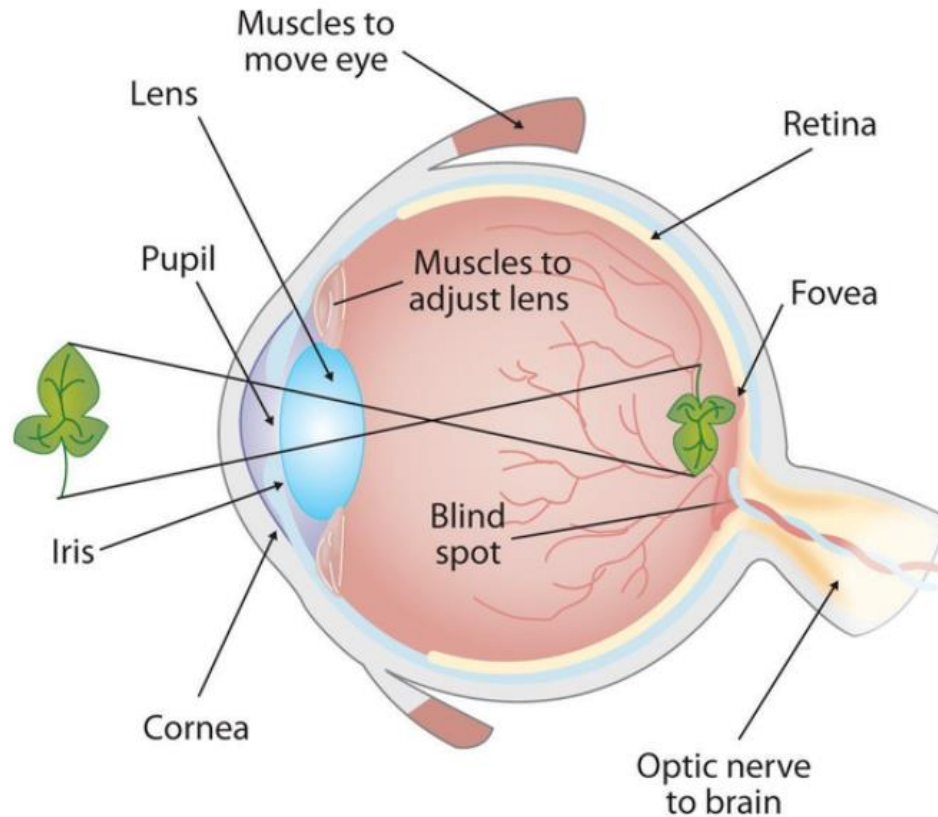


Human Visual Perception

Image Processing



Why do we have to study about our eyes in image processing class ?

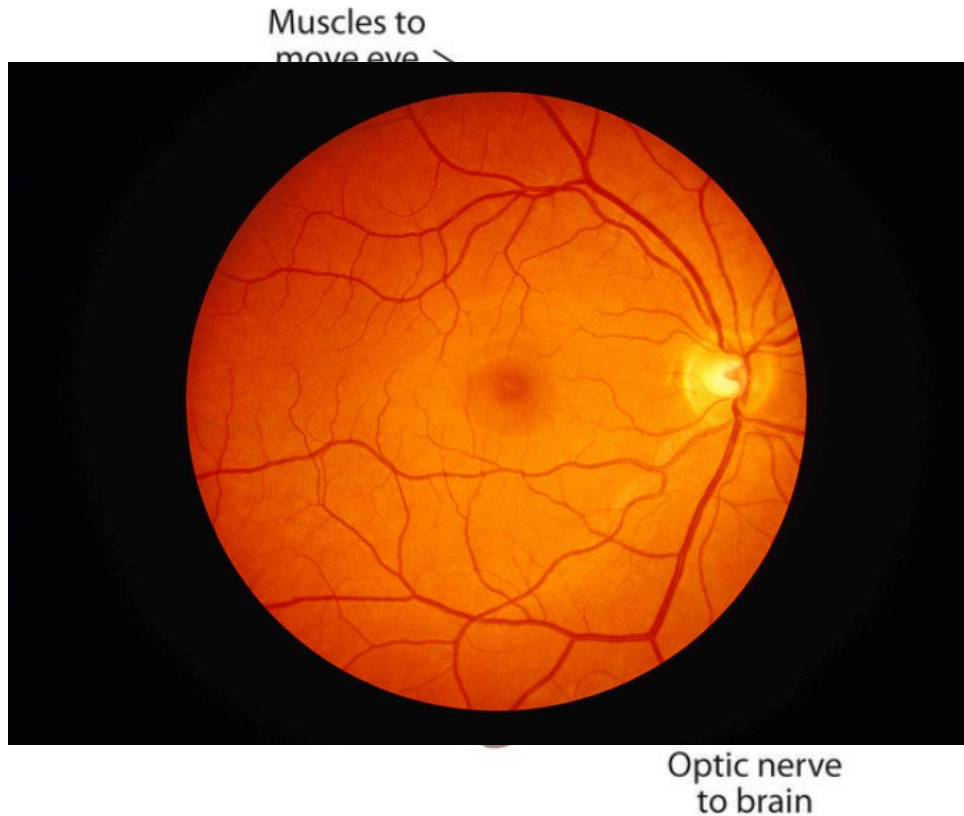


Cornea: A tough, transparent tissue that covers the anterior surface of the eye.

Lens: Makes up of concentric layers of fibrous cells and contains 60-70% of water. To focus on objects near or far the eye, the controlling muscles cause the lens to be thicker or thinner.

Retina: innermost membrane lines the inside of the wall's entire posterior portion. Retina contains **cones** and **rods**.

Why do we have to study about our eyes in image processing class ?

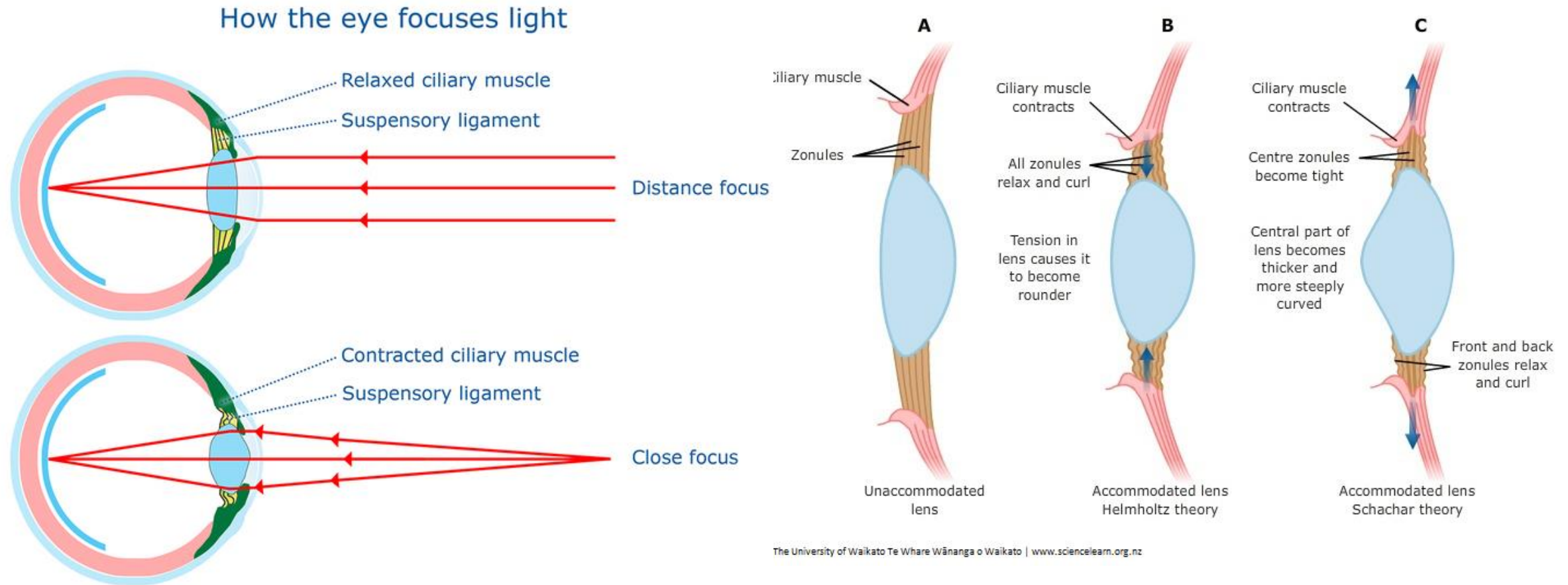


Cornea: A tough, transparent tissue that covers the anterior surface of the eye.

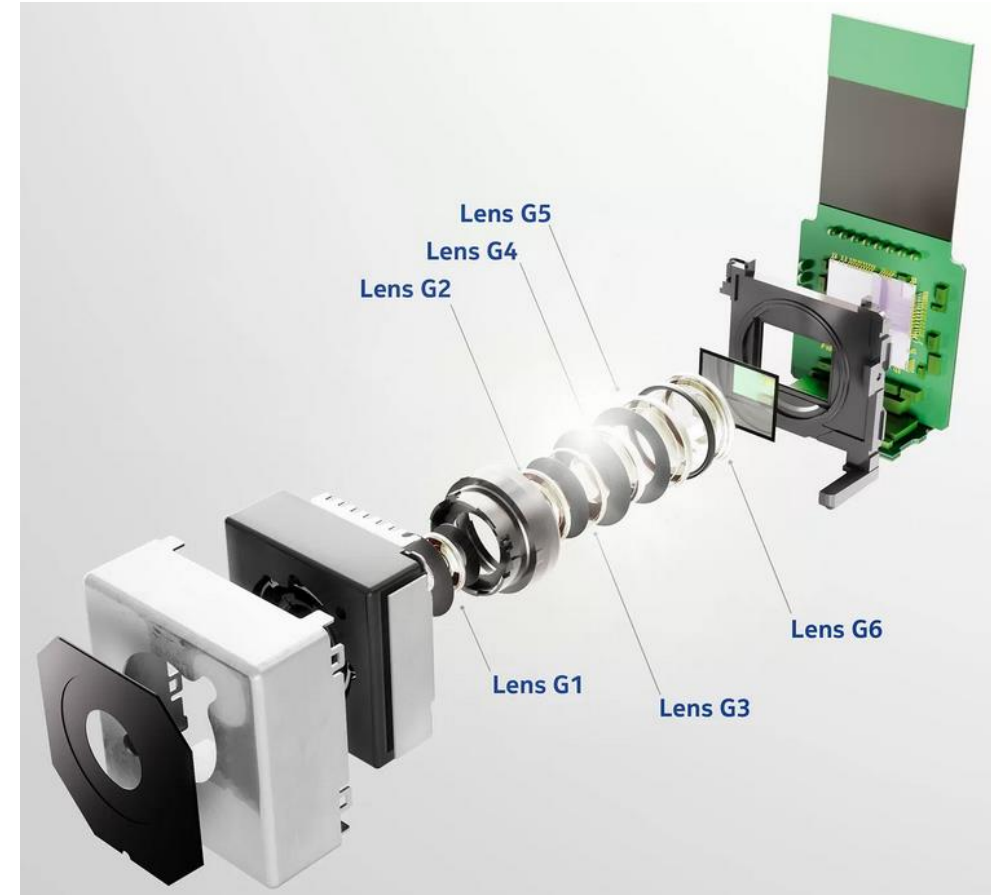
Lens: Makes up of concentric layers of fibrous cells and contains 60-70% of water. To focus on objects near or far the eye, the controlling muscles cause the lens to be thicker or thinner.

Retina: innermost membrane lines the inside of the wall's entire posterior portion. Retina contains **cones** and **rods**.

Comparing between lens of your eye and camera, what is big the difference ?

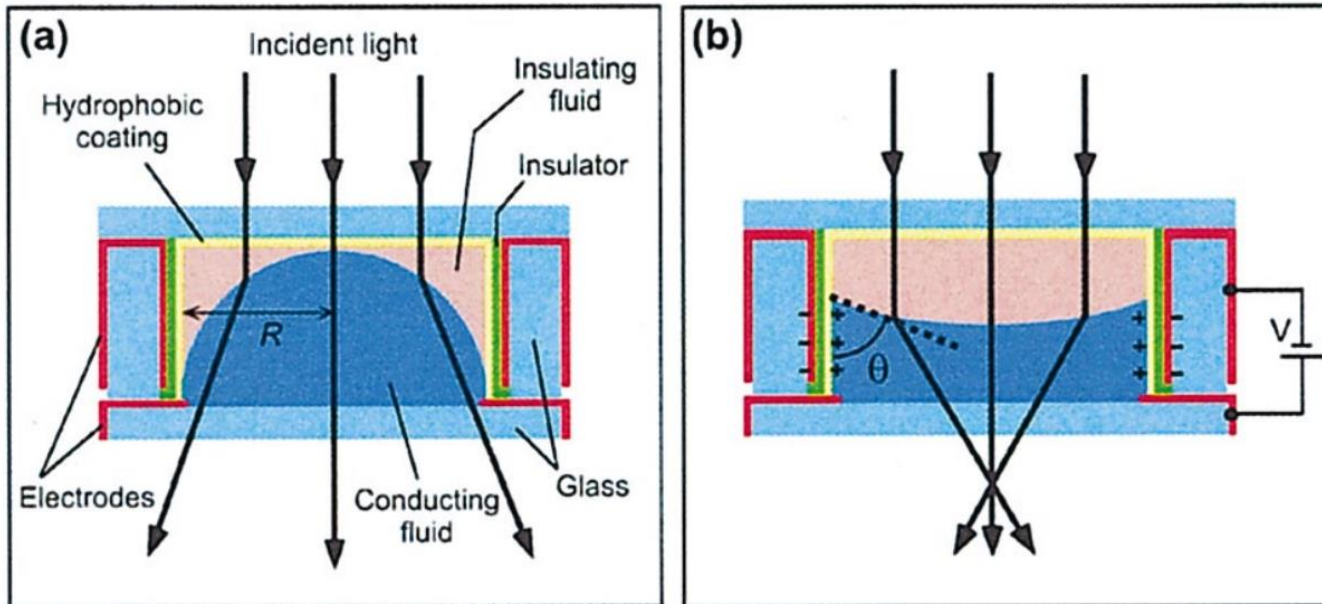


Comparing between lens of your eye and camera, what is big the difference ?



Smartphone camera

Why could we design our camera lens like our eyes?



Water & Oil



On Smartphone ?

Xiaomi Mi Mix Fold (available in China only)

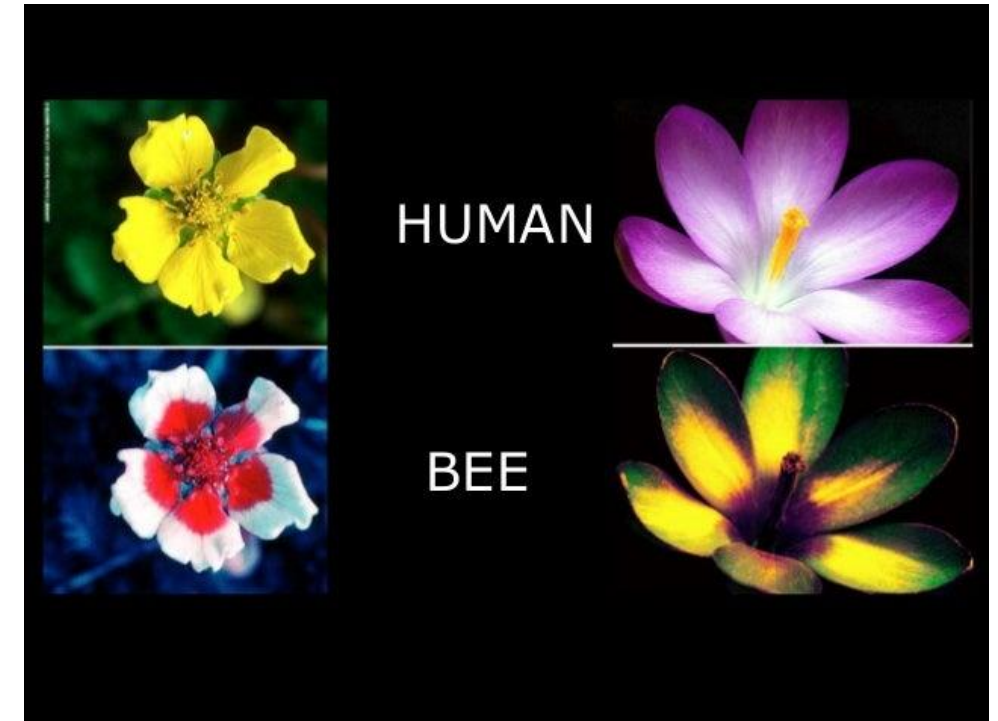
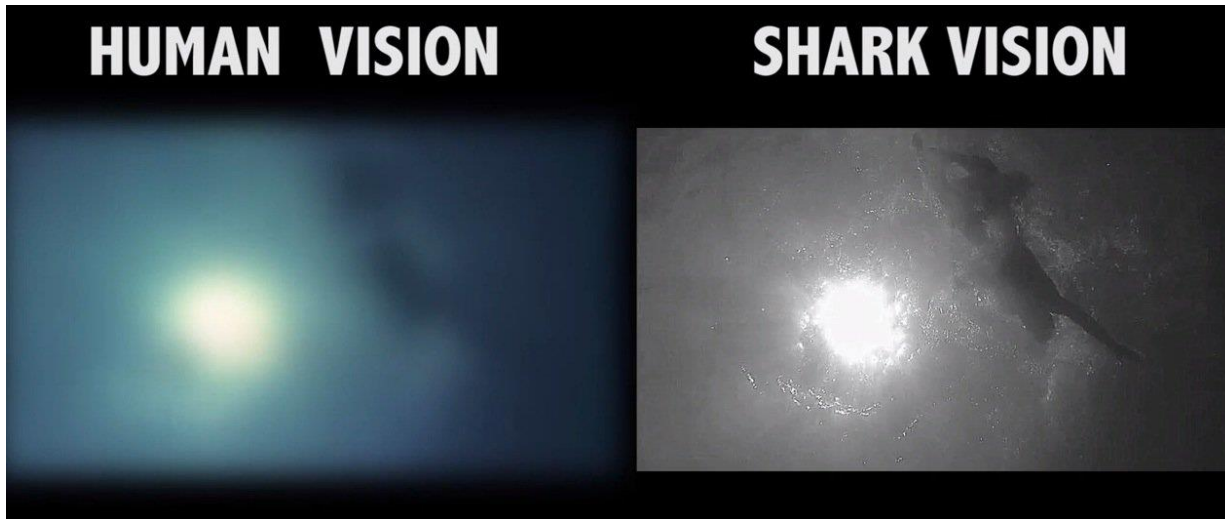
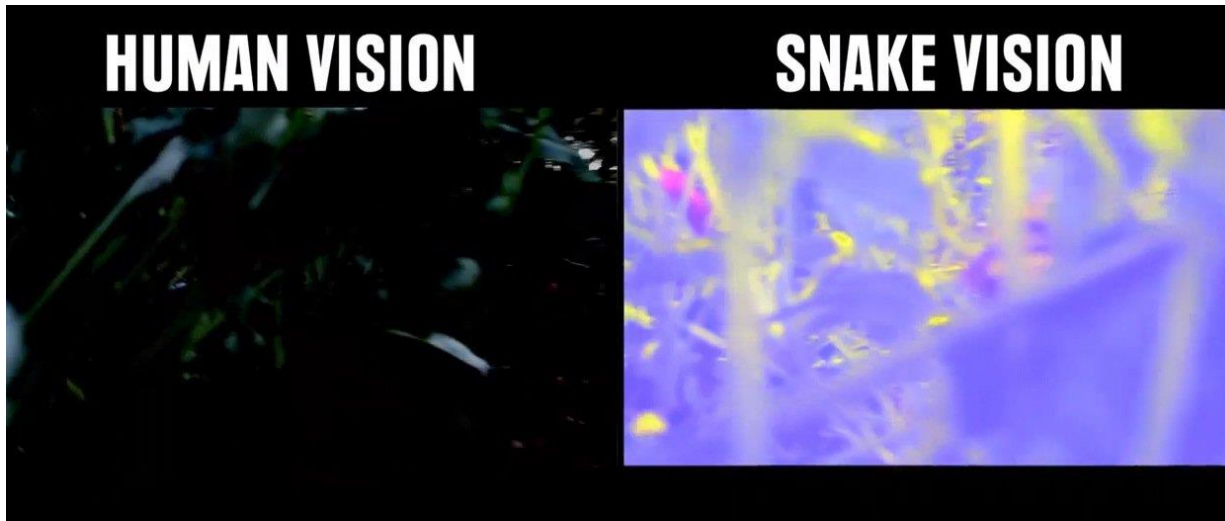


telephoto and macro by the same liquid lens

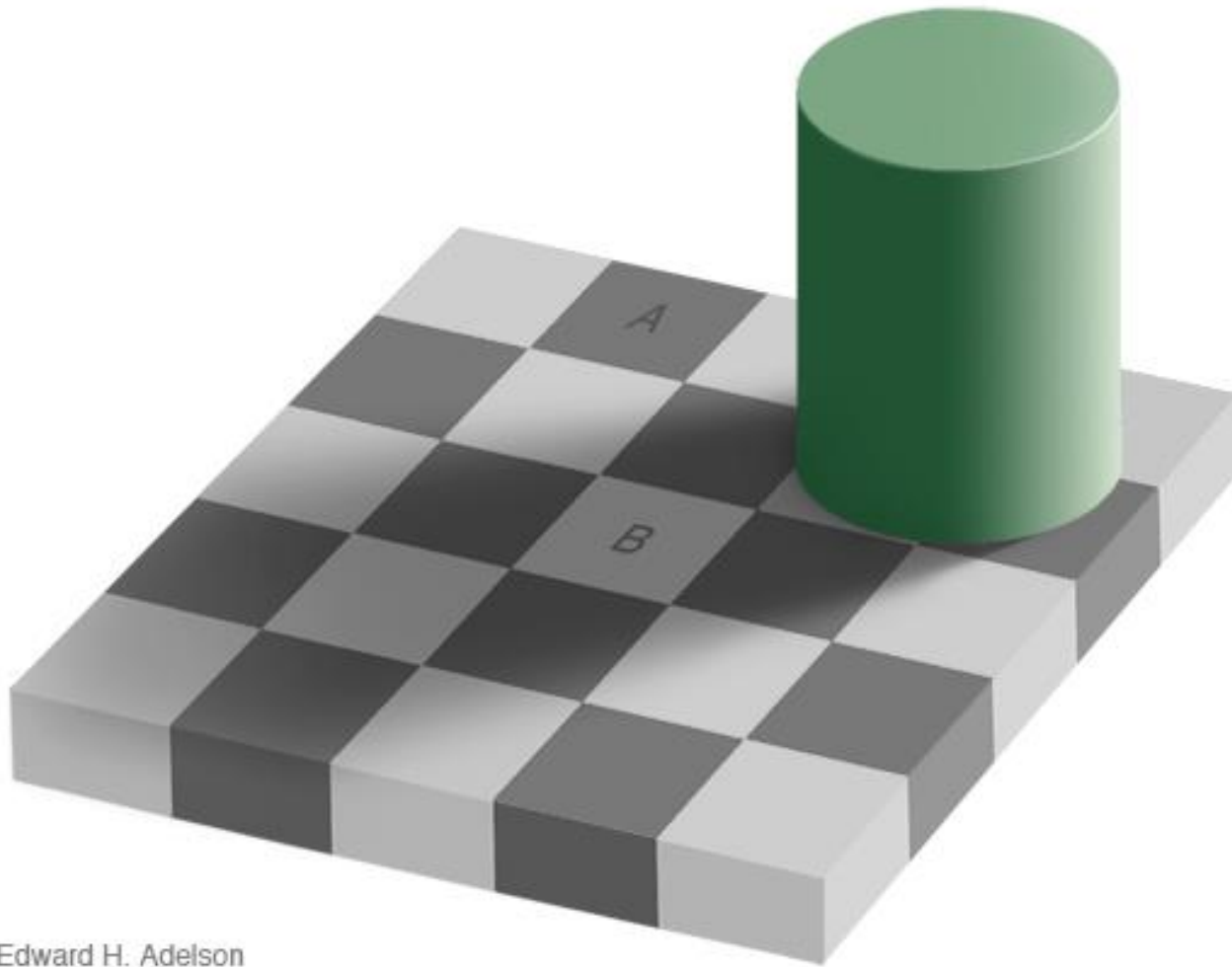


Using C1 professional imaging chip.
Mass production of mobile phone with
liquid lenses.

Animals Eyes



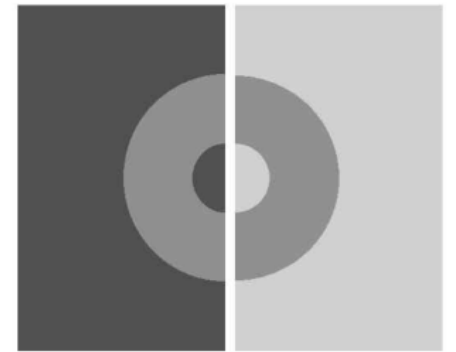
Can we trust our eyes?



Edward H. Adelson



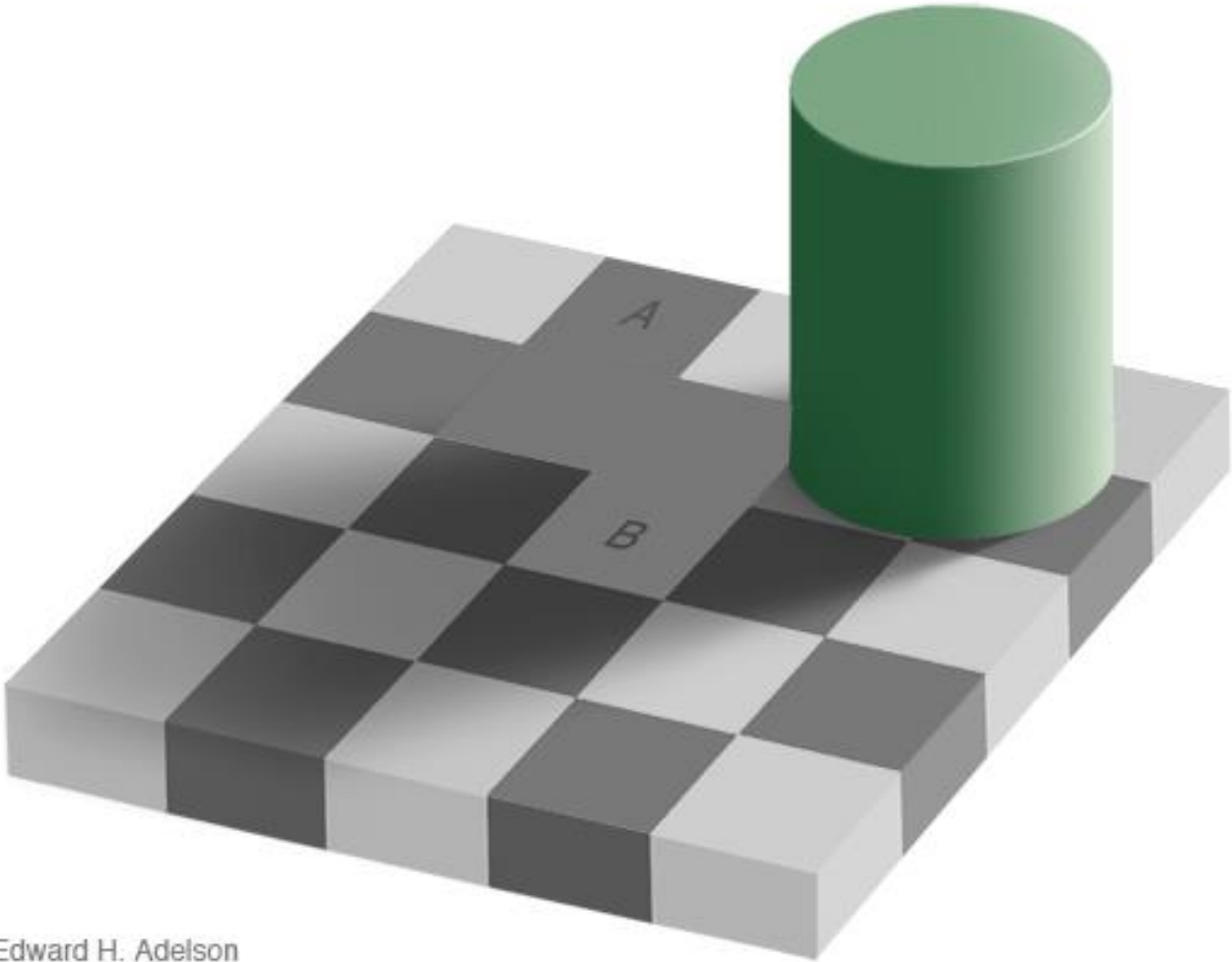
a



b

Koffka ring

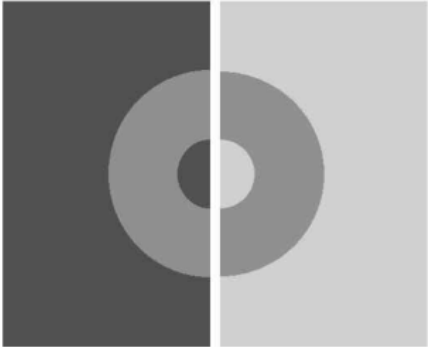
Can we trust our eyes?



Edward H. Adelson



a

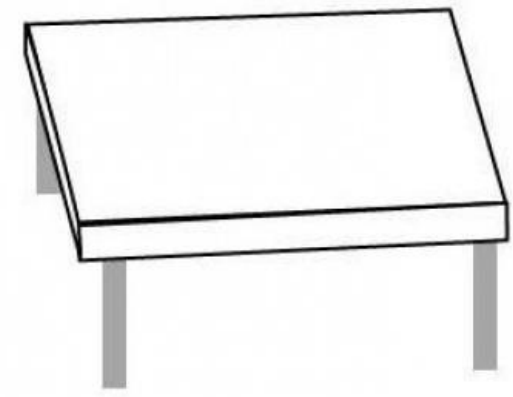
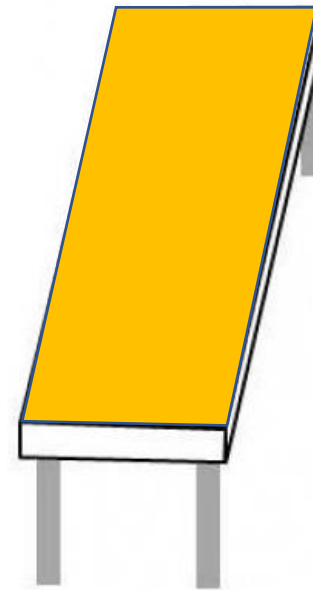
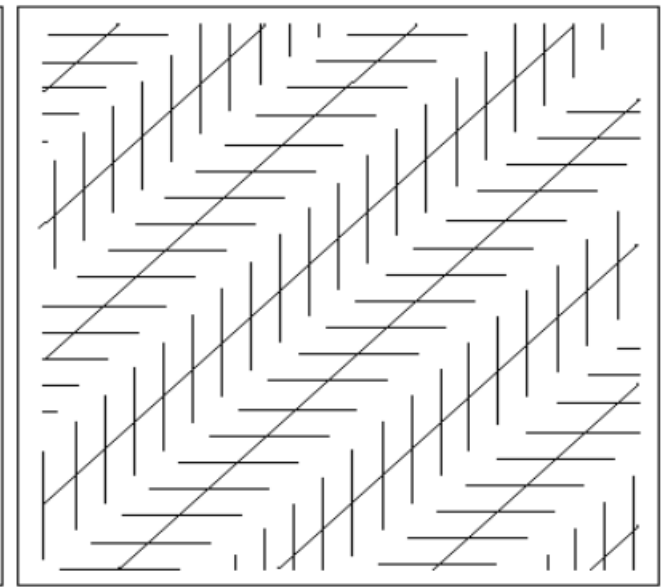
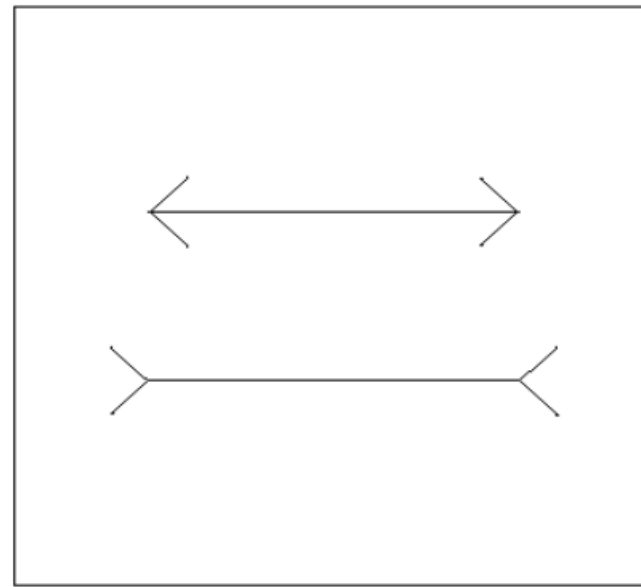
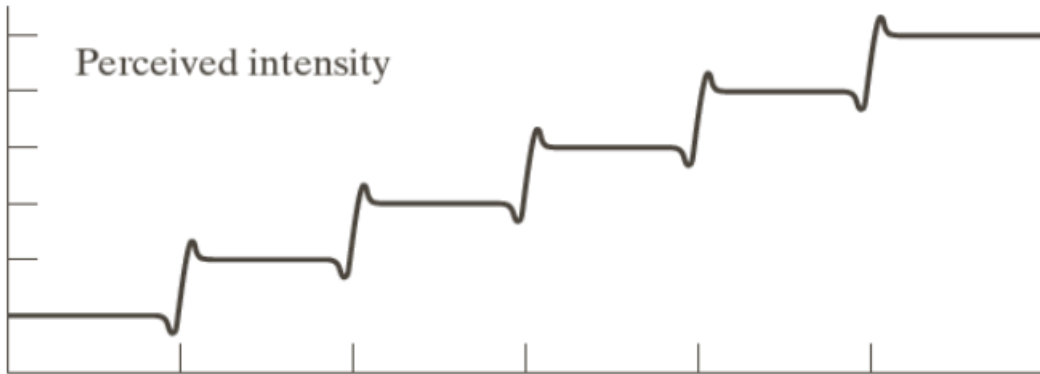
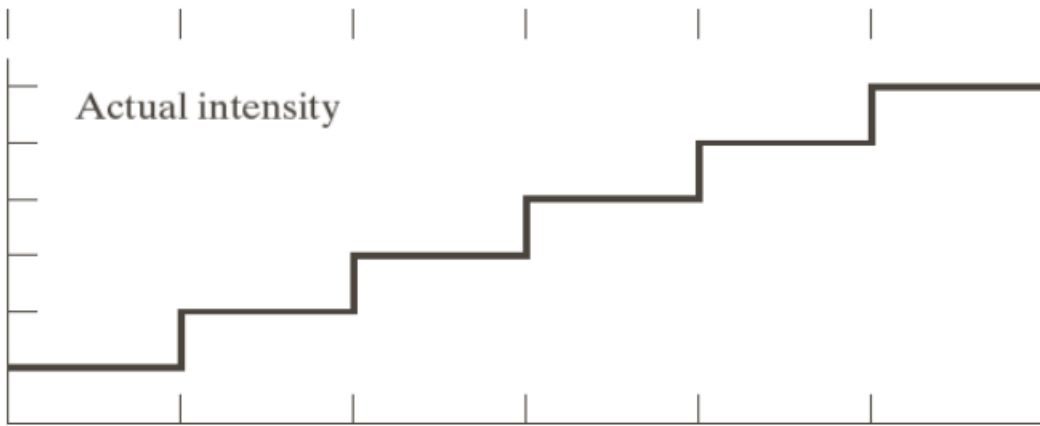


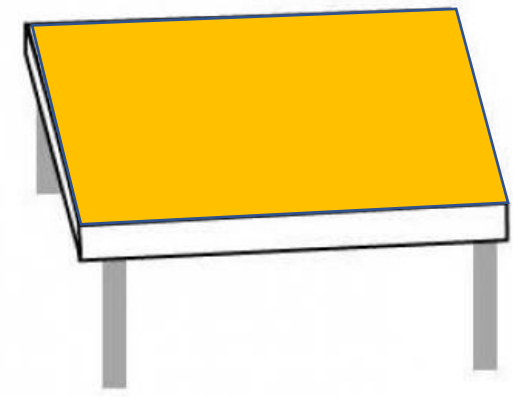
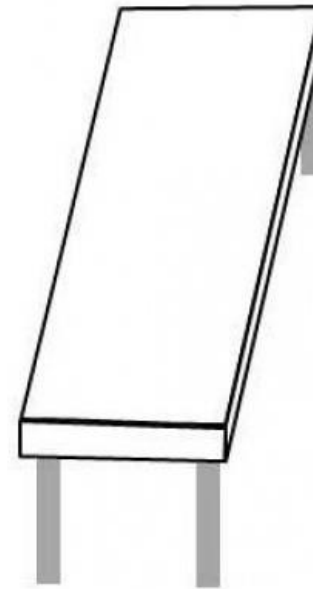
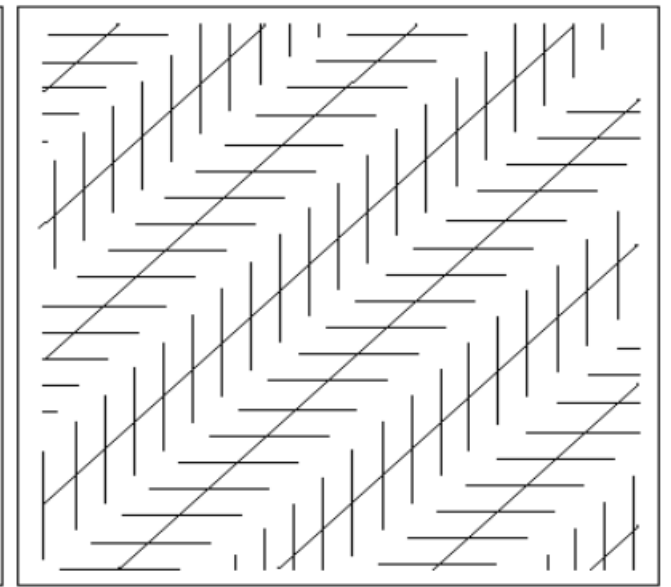
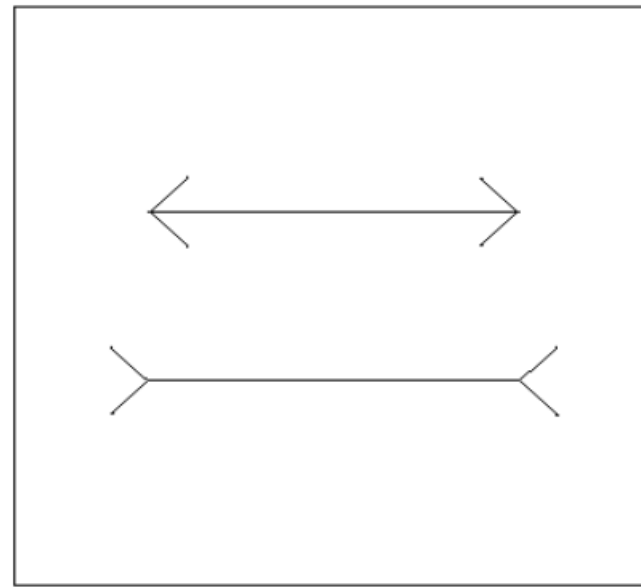
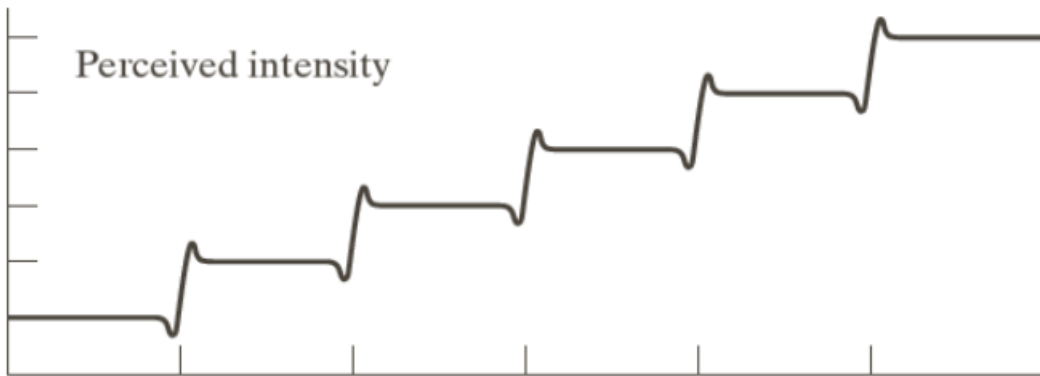
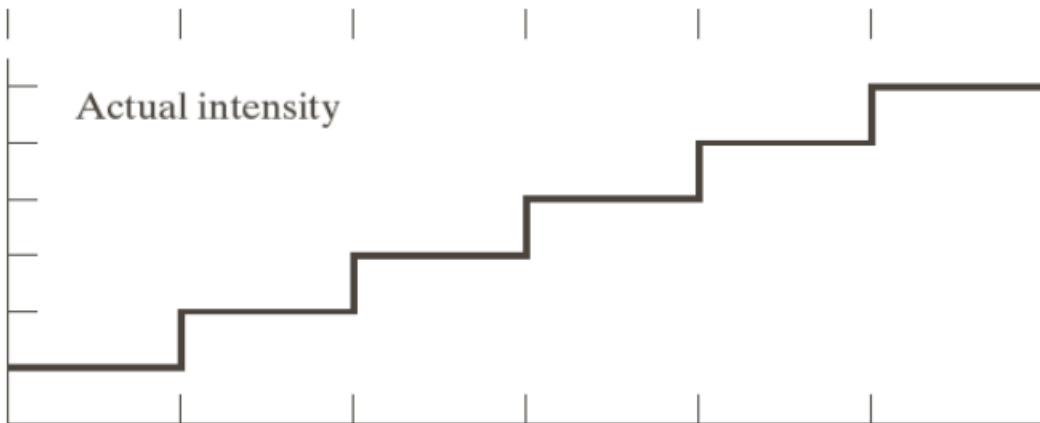
b



c

Koffka ring







Fundamental Operation

Image Processing



Digital Image (Grayscale)



We see “image” like this.

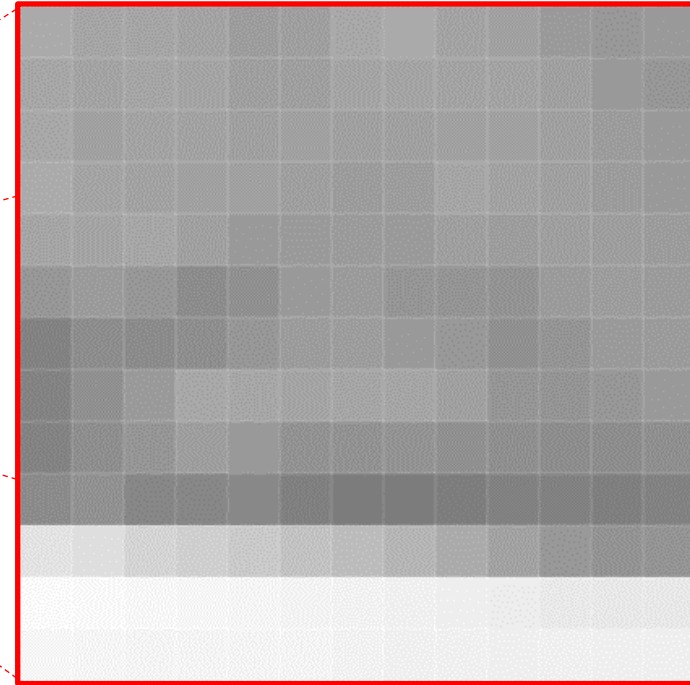
148	123	52	107	123	162	172	123	64	89	...
147	130	92	95	98	130	171	155	169	163	...
141	118	121	148	117	107	144	137	136	134	...
82	106	93	172	149	131	138	114	113	129	...
57	101	72	54	109	111	104	135	106	125	...
138	135	114	82	121	110	34	76	101	111	...
138	102	128	159	168	147	116	129	124	117	...
113	89	89	109	106	126	114	150	164	145	...
120	121	123	87	85	70	119	64	79	127	...
145	141	143	134	111	124	117	113	64	112	...
:	:	:	:	:	:	:	:	:	:	:

But, this is what **computer** see.

Digital Image (Grayscale)



We see “image” like this.



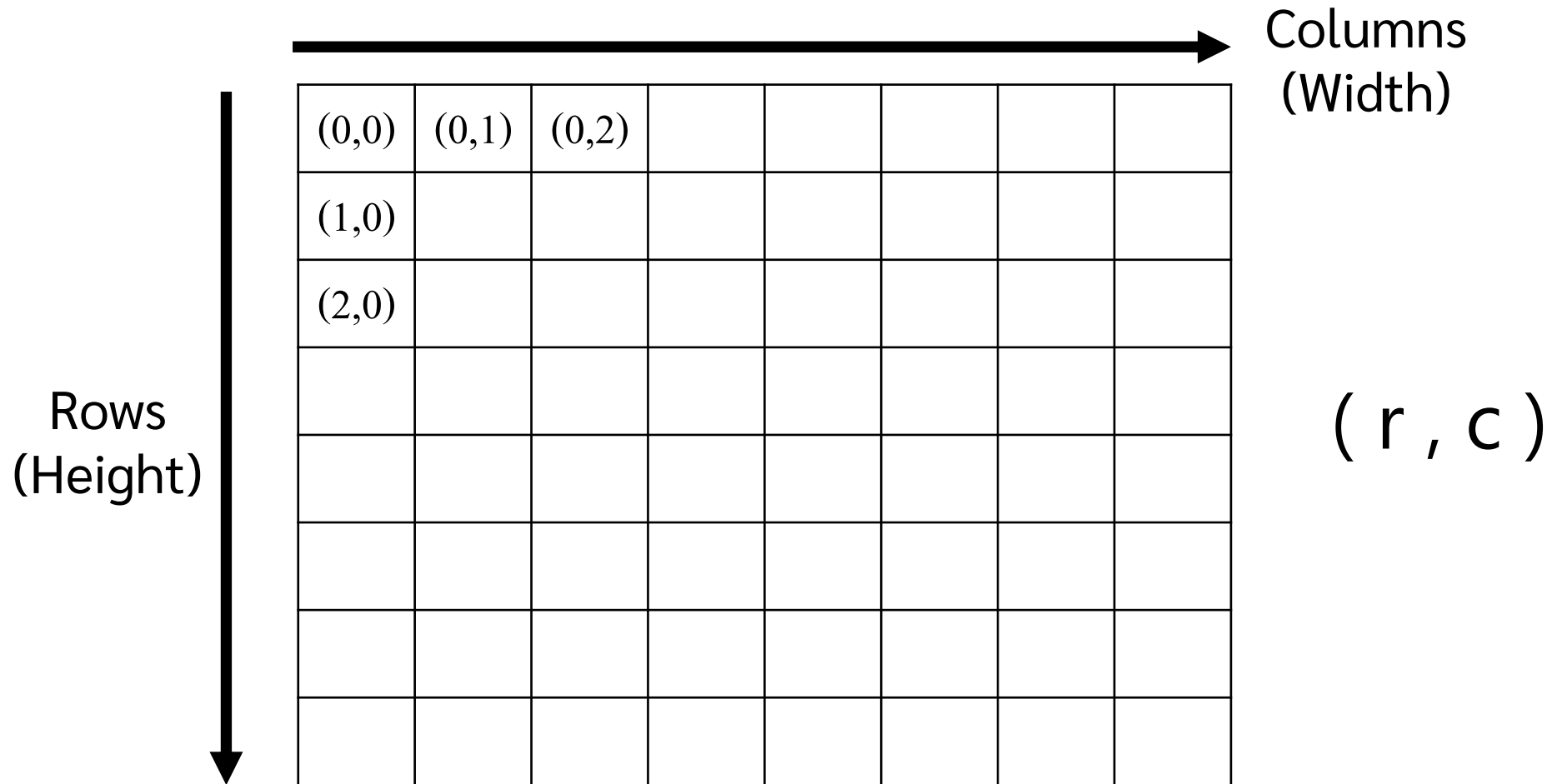
0

255

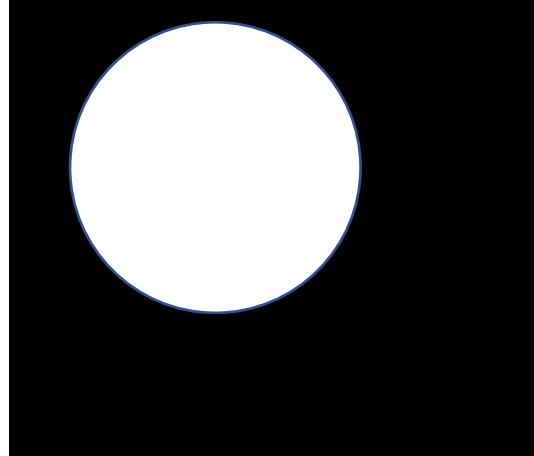
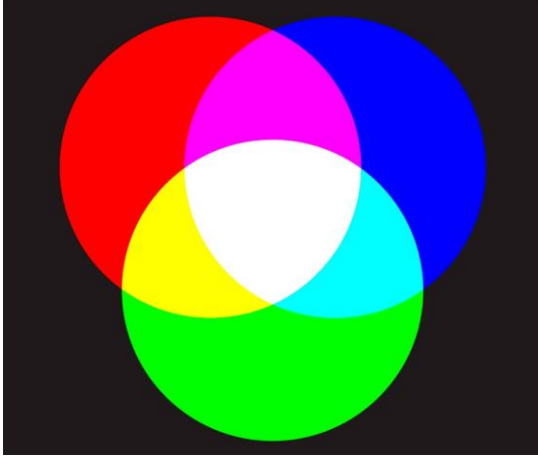
But, this is what computer see.

Digital Image → Array (N-dimensions)

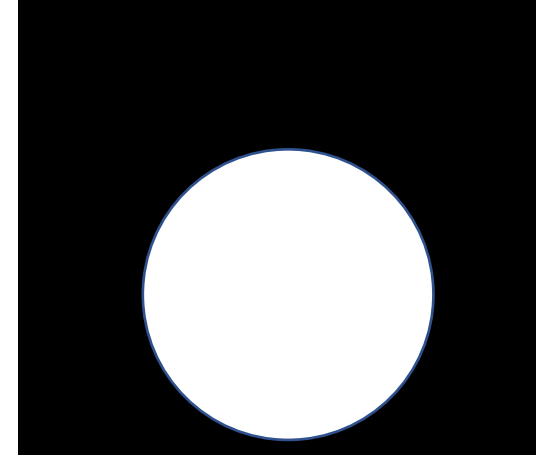
การจัดการ “ภาพ” ด้วย Python



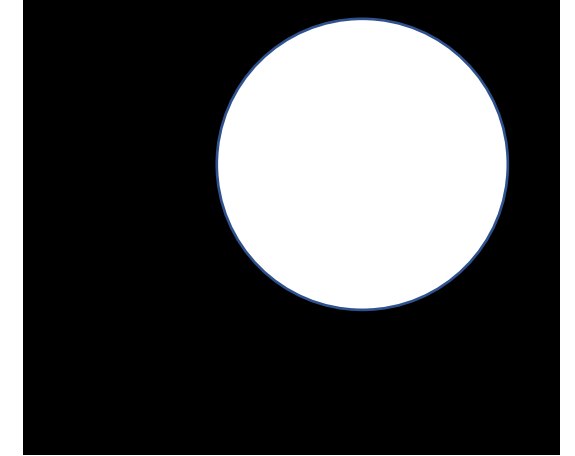
Digital Image (Color - RGB Model)



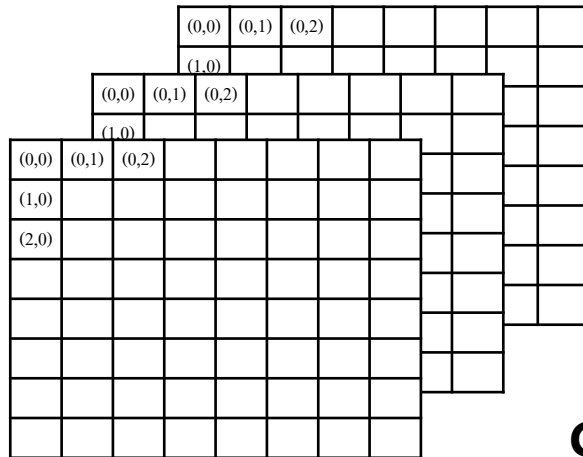
Red



Green



Blue



Channel : Red / Green / Blue

Python Implementation



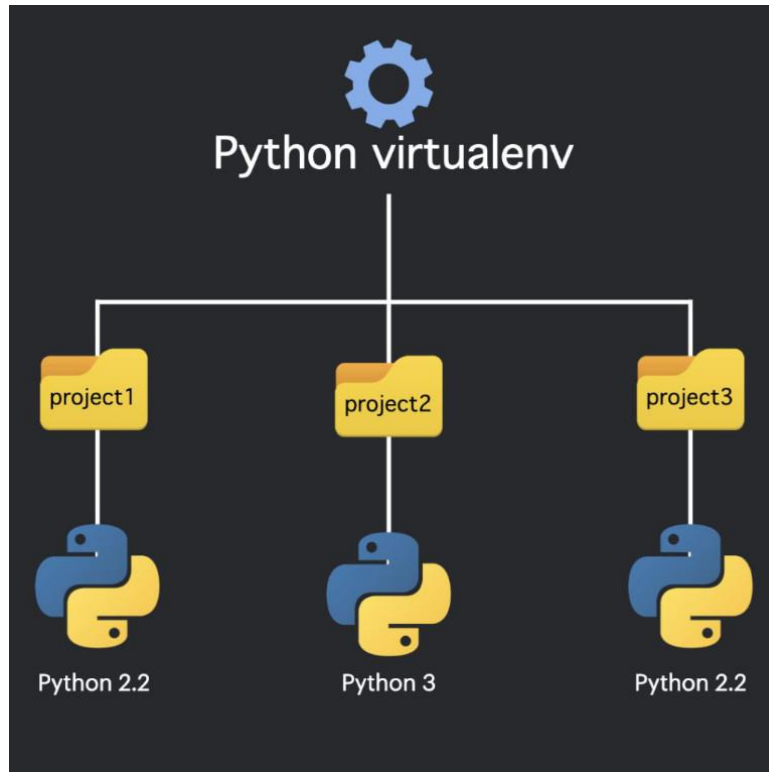
- 1.) Check Python version
- 2.) Install PIP
- 3.) Install library NumPy & Open CV

***** How to install PIP in macOS**

<https://www.geeksforgeeks.org/how-to-install-pip-in-macos>

Python Virtual Environment

“A virtual environment is a Python environment such that the Python interpreter, libraries and scripts installed into it are isolated from other”



1.) การสร้าง Virtual Environment

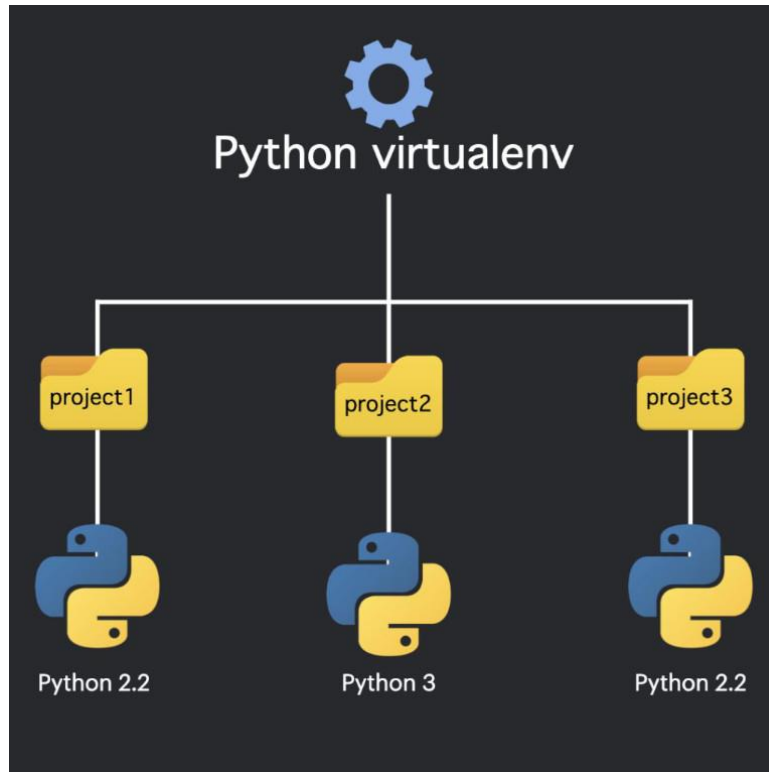
```
python -m venv <virtual-environment-name>
```

2.) การ Activate

[Run] env/Scripts/activate.bat

Python Virtual Environment

“A virtual environment is a Python environment such that the Python interpreter, libraries and scripts installed into it are isolated from other”



3.) การสร้าง File requirements.txt สำหรับเก็บ List

```
pip freeze > requirements.txt
```

4.) การ Install Package จาก requirements.txt

```
pip install -r requirements.txt
```

Other recommended management approaches

1. ใช้ `pyenv` + `pyenv-virtualenv` (สำหรับจัดการหลาย Python versions + env)

- `pyenv` ช่วยให้ติดตั้ง Python หลายเวอร์ชันในเครื่องเดียวกัน เช่น Python 3.8, 3.10, 3.12 ฯลฯ
- `pyenv-virtualenv` ช่วยสร้าง virtual environment แต่ละโปรเจกต์ แยก Python version และ packages ได้ชัดเจน
- รองรับ Mac/Linux ดีมาก และ Windows (ผ่าน WSL หรือ Git Bash)

bash

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ตัวอย่างการติดตั้ง Python 3.10

```
pyenv install 3.10.13
```

สร้าง environment

```
pyenv virtualenv 3.10.13 myproject-env
```

ใช้ environment นี้

```
pyenv activate myproject-env
```

Other recommended management approaches

2. ใช้ `conda` หรือ `mamba` (เหมาะกับ Data Science / Windows user)

- ใช้ Anaconda หรือ Miniconda เพื่อจัดการ environment + Python version
- `mamba` เป็นเวอร์ชันเร็วกว่า `conda` (แนะนำ)
- เหมาะกับคนที่ทำงานกับพวก NumPy, SciPy, TensorFlow, etc.

bash

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สร้าง environment ใหม่พร้อม Python 3.9

```
conda create -n myenv python=3.9
```

เข้า environment

```
conda activate myenv
```

ติดตั้ง library ตามโปรเจกต์

```
conda install numpy pandas
```

Read / Write / Show image

Example #1

```
Demo02_Read-Write-Show.py - C:\Users\Krisada\...
File Edit Format Run Options Window Help
1 import cv2 as cv
2
3 # Read the image
4 img = cv.imread("pic/B.bmp")
5
6 print("Data type of image (OpenCV): ", type(img))
7
8 # Display the image
9 cv.imshow("Example1 - imshow", img)
10
11 # Save the image
12 cv.imwrite("out.png", img)
13
14 # Wait for a key press
15 cv.waitKey(0)
16
17 # Clean up
18 cv.destroyAllWindows()
19
Ln: 19 Col: 0
```

```
*IDLE Shell 3.9.1*
File Edit Shell Debug Options Window Help
Python 3.9.1 (tags/v3.9.1:1e5d33e, Dec 7 2020, 16:33:24) [MSC v.1928 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Krisada\Desktop\DIP 2020\Code\Python\Demo 02\Demo02_Read-Write-Show.py
Data type of image (OpenCV): <class 'numpy.ndarray'>

Example1 - i...

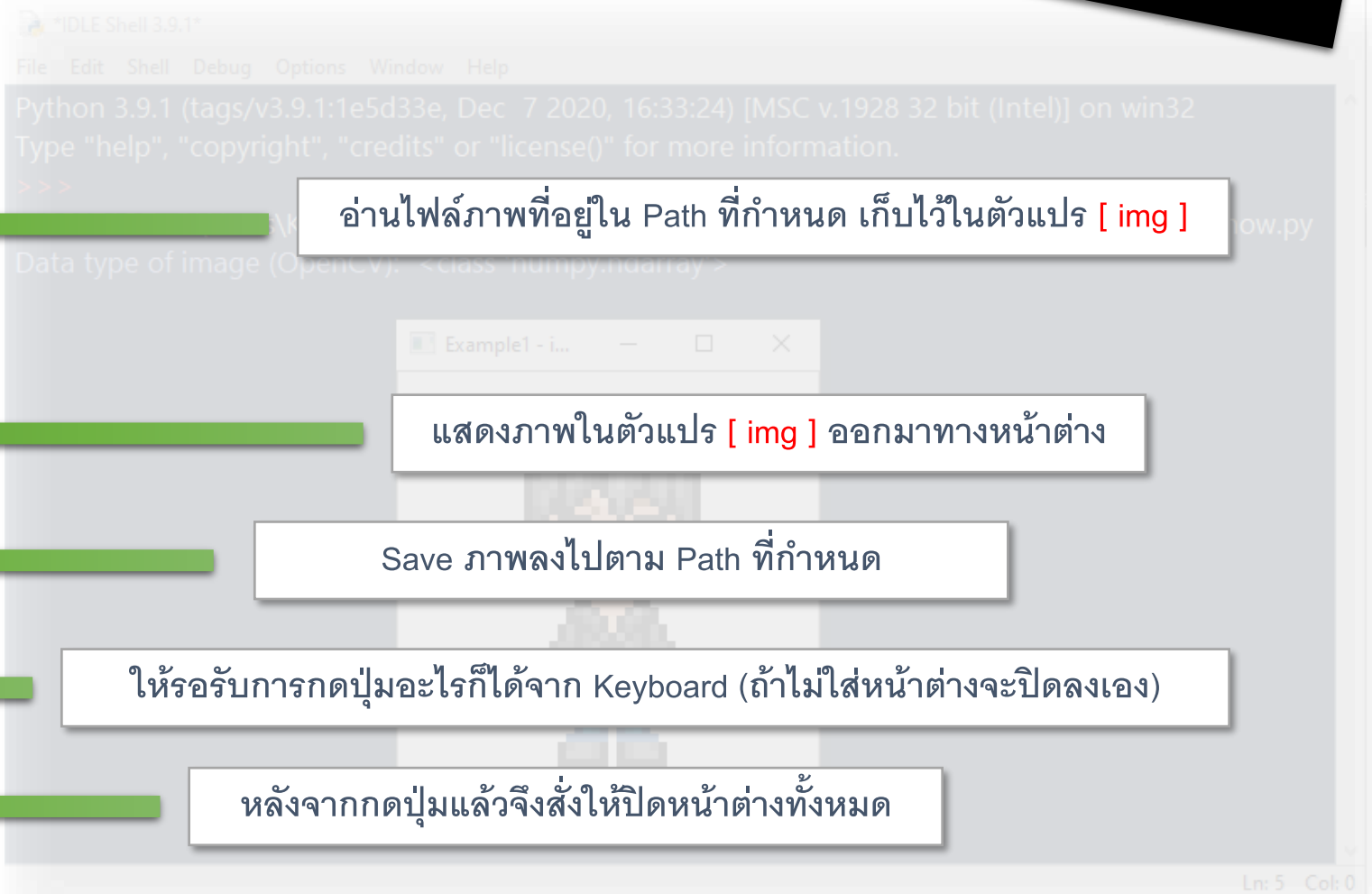
```

Ln: 5 Col: 0

Read / Write / Show image

Example #1

```
Demo02_Read-Write-Show.py - C:\Users\Krisada\...
File Edit Format Run Options Window Help
1 import cv2 as cv
2
3 # Read the image
4 img = cv.imread("pic/B.bmp")
5
6 print("Data type of image (OpenCV): ", type(img))
7
8 # Display the image
9 cv.imshow("Example1 - imshow", img)
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11 # Save the image
12 cv.imwrite("out.png", img)
13
14 # Wait for a key press
15 cv.waitKey(0)
16
17 # Clean up
18 cv.destroyAllWindows()
19
Ln: 19 Col: 0
```



อ่านไฟล์ภาพที่อยู่ใน Path ที่กำหนด เก็บไว้ในตัวแปร [img]

แสดงภาพในตัวแปร [img] ออกมาทางหน้าต่าง

Save ภาพลงไปตาม Path ที่กำหนด

ให้รอรับการกดปุ่มอะไรก็ได้จาก Keyboard (ถ้าไม่ใส่หน้าต่างจะปิดลงเอง)

หลังจากกดปุ่มแล้วจึงสั่งให้ปิดหน้าต่างทั้งหมด

Access image attributes

Example #2

```
Demo02_02_Grayscale.py - C:\Users\Krisada\Desktop\DIP 2020\Cod...
File Edit Format Run Options Window Help
1 import cv2 as cv
2
3 # Read the image
4 img = cv.imread("pic/C.jpg", cv.IMREAD_GRAYSCALE)
5
6 print("Data type of image (OpenCV): ", type(img))
7 print("number of dimension: ", img.ndim)
8 print("size of each dimension: ", img.shape)
9 print("image height: ", img.shape[0])
10 print("image width: ", img.shape[1])
11
12 # Display the image
13 cv.namedWindow('Example1 - imshow', cv.WINDOW_NORMAL)
14 cv.imshow('Example1 - imshow',img)
15
16 # Wait for a key press
17 cv.waitKey(0)
18
19 # Clean up
20 cv.destroyAllWindows()
21
Ln: 21 Col: 0
```

```
*IDLE Shell 3.9.1*
File Edit Shell Debug Options Window Help
Python 3.9.1 (tags/v3.9.1:1e5d33e, Dec 7 2020, 16:33:24) [MSC v.1928 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Krisada\Desktop\DIP 2020\Code\Python\Demo 02\Demo02_02_Grayscale.
py
Data type of image (OpenCV): <class 'numpy.ndarray'>
number of dimension: 2
size of each dimension: (1440, 1080)
image height: 1440
image width: 1080
Ln: 10 Col: 0
```

File Edit

1 impo
2
3 # Re
4 img
5
6 print
7 print
8 print
9 print
10 print
11
12 # Dis
13 cv.na
14 cv.im
15
16 # Wa
17 cv.wa
18
19 # Cle
20 cv.de
21

ร่ม

ไม่ได้ถูกสร้างขึ้นมาให้ฝนหยุดตก แต่
มันถูกสร้างขึ้นมาให้เราเดินต่อไปได้ท่ามกลางสายฝน



กำลังใจ

ก็คงไม่ต่างจากร่ม เพราะมันไม่สามารถหยุดปัญหาหรือ
อุปสรรคที่จะเข้ามาได้ แต่
มันเป็นสิ่งที่จะพาเราก้าวผ่านสิ่งต่างๆ
เพื่อนำเราไปสู่จุดหมายปลายทางแห่งความสำเร็จได้



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Happiness

www.facebook.com/ClubToHappiness

Ln: 21 Col: 0

ribun

Example1 - imshow

ร่ม

ไม่ได้ถูกสร้างขึ้นมาให้ฝนหยุดตก แต่
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Ln: 10 Col: 0

ole #2

it (Intel)] on win32
mo02_02_Grayscale.

Ln: 10 Col: 0

Fundamental Operation

Page 43

Access / Modify each pixel

Example #3

การสร้าง Array ด้วย NumPy

`numpy.empty(shape, dtype)`

It creates an uninitialized array of specified shape and dtype.

`numpy.zeros(shape, dtype)`

Returns a new array of specified size, filled with zeros.

`numpy.ones(shape, dtype)`

Returns a new array of specified size and type, filled with ones.

Access / Modify each pixel

Example #3

```
Demo02_03_DrawRect.py - C:\Users\Krisada\...
File Edit Format Run Options Window Help
1 import numpy as np
2 import cv2 as cv
3
4 # Create empty image
5 img = np.zeros([200, 300], dtype = np.uint8)
6
7 for y in range( 75, 125 ):
8     for x in range( 50, 250 ):
9         img[y,x] = 255
10
11 # Display the image
12 cv.imshow('Drawing', img)
13
14 # Wait for a key press
15 cv.waitKey(0)
16
17 # Clean up
18 cv.destroyAllWindows()
19
```

Ln: 19 Col: 0

#	Data Types	Description
1	bool_	Boolean (True or False) stored as a byte
2	int8	Byte (-128 to 127)
3	int16	Integer (-32768 to 32767)
4	int32	Integer (-2147483648 to 2147483647)
5	int64	Integer (-9223372036854775808 to 9223372036854775807)
6	uint8	Unsigned integer (0 to 255)
7	uint16	Unsigned integer (0 to 65535)
8	uint32	Unsigned integer (0 to 4294967295)
9	uint64	Unsigned integer (0 to 18446744073709551615)
10	float_	Shorthand for float64
11	float32	Single precision float: sign bit, 8 bits exponent, 23 bits mantissa
12	float64	Double precision float: sign bit, 11 bits exponent, 52 bits mantissa
13	complex	Shorthand for complex128
14	complex64	Complex number, represented by two 32-bit floats (real and imaginary components)
15	complex128	Complex number, represented by two 64-bit floats (real and imaginary components)

Access / Modify each pixel

Example #3

```
Demo02_03_DrawRect.py - C:\Users\Krisada\...
File Edit Format Run Options Window Help
1 import numpy as np
2 import cv2 as cv
3
4 # Create empty image
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12 cv.imshow('Drawing', img)
13
14 # Wait for a key press
15 cv.waitKey(0)
16
17 # Clean up
18 cv.destroyAllWindows()
19
Ln: 19 Col: 0
```

```
*IDLE Shell 3.9.1*
File Edit Shell Debug Options Window Help
Python 3.9.1 (tags/v3.9.1:1e5d33e, Dec 7 2020, 16:33:24) [MSC v.1928 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Krisada\Desktop\DIP 2020\Code\Python\Demo 02\Demo02_03_DrawRect.py

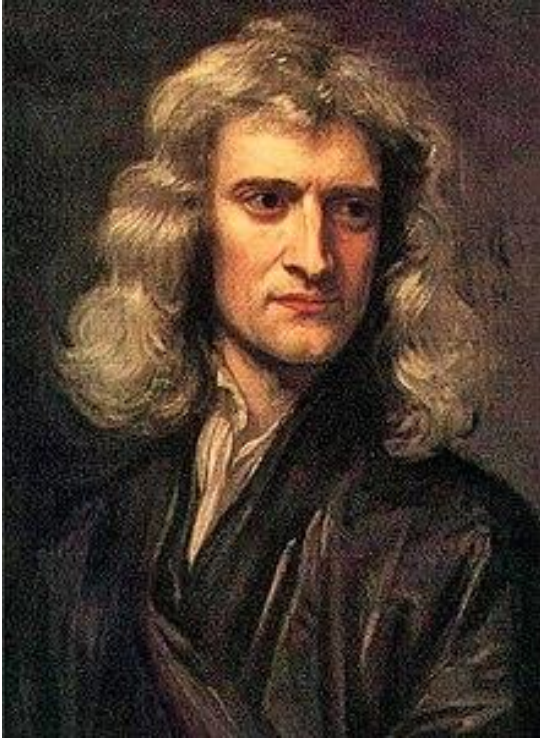
Drawing
Ln: 5 Col: 0
```

Access / Modify each pixel

Question ?

ถ้าเราจะวาดรูปวงกลม 1 รูป จะต้องทำอย่างไร ?

What is color?



Isaac Newton

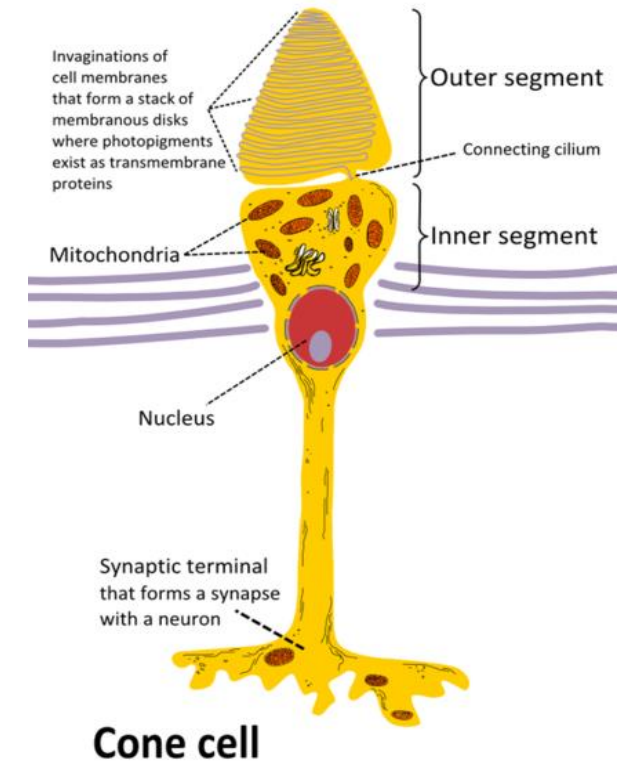
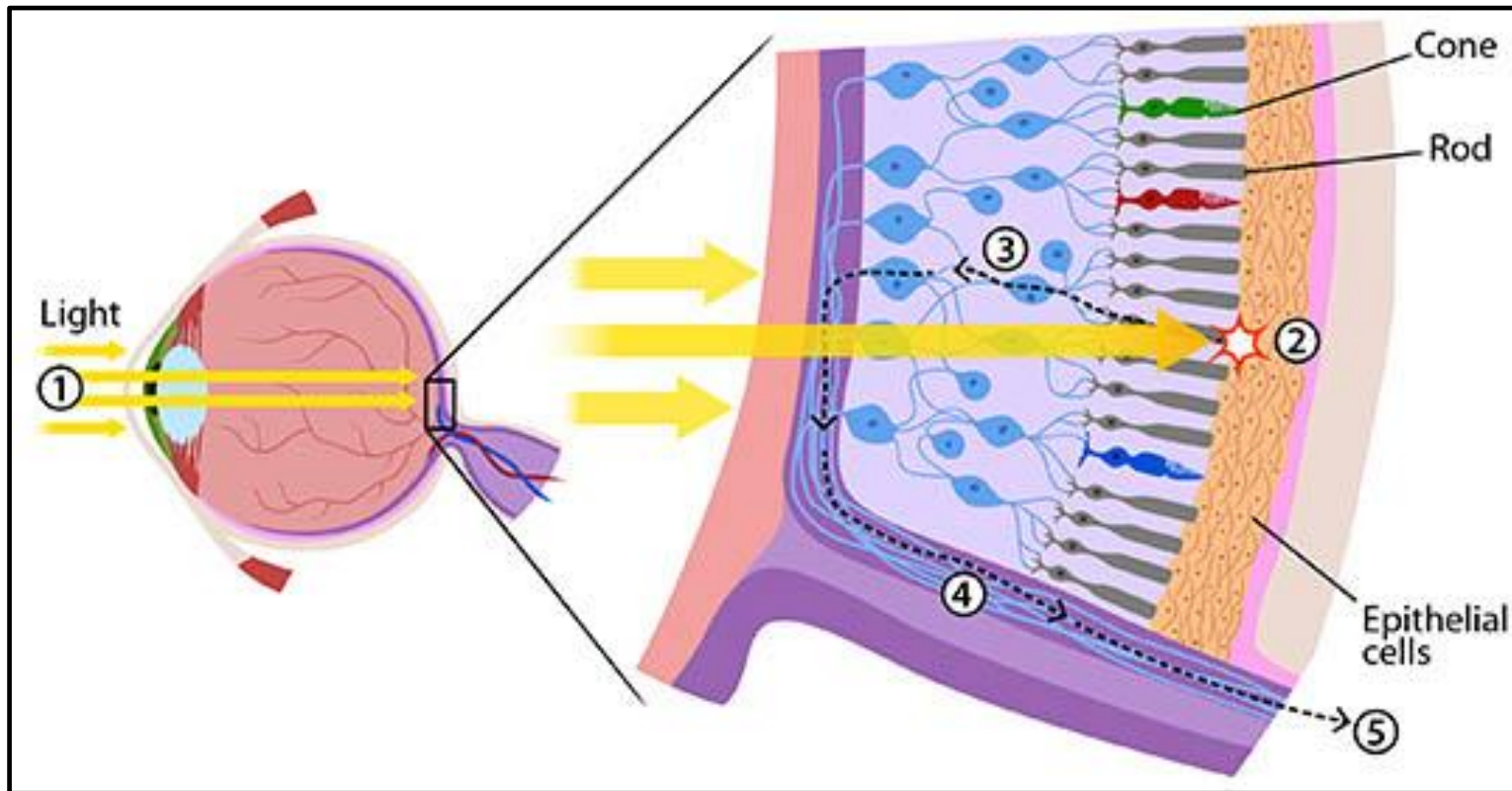


Color is the characteristic of visual perception.

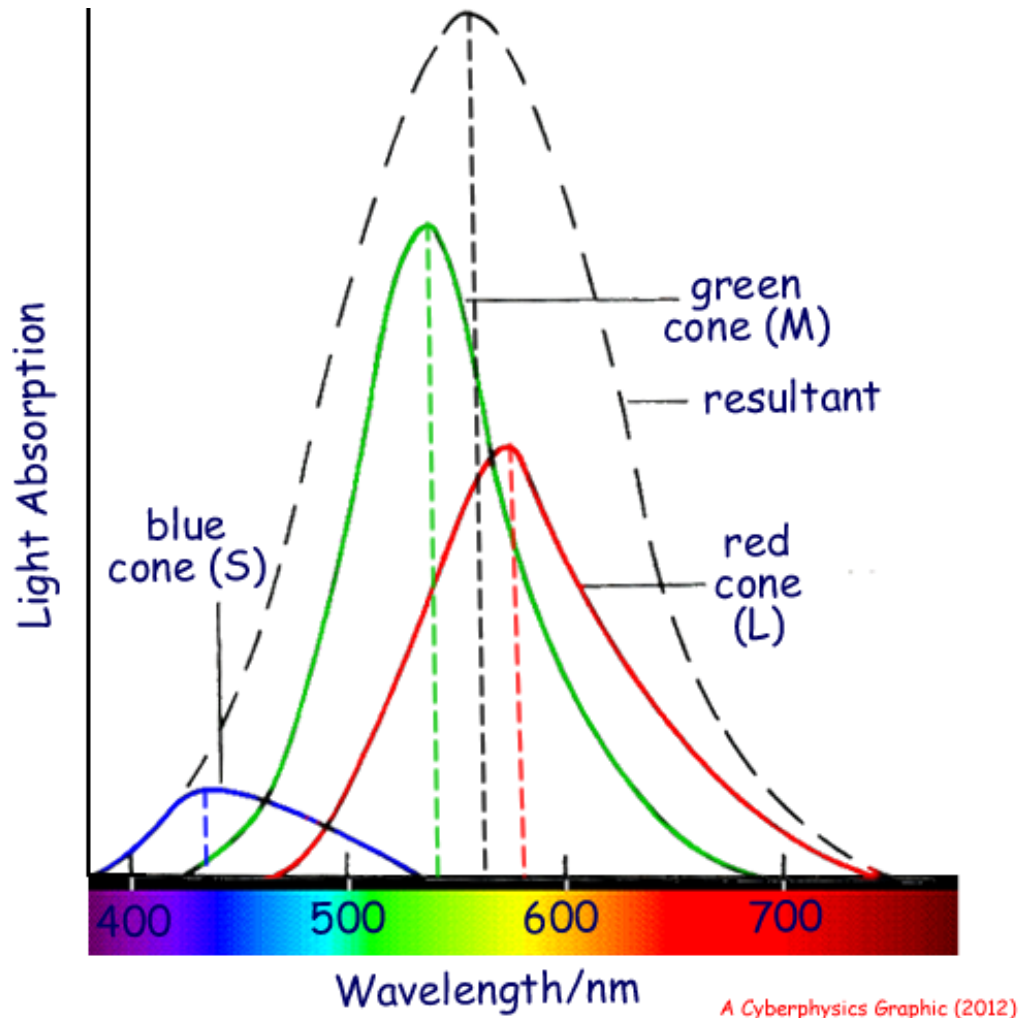
This perception of color derives from the stimulation of photoreceptor cells (in particular cone cells in the human eye and other vertebrate eyes) by electromagnetic radiation (in the visible spectrum in the case of humans).

Physiology of color perception

Perception of color begins with specialized retinal cells containing pigments with **different spectral sensitivities**, known as “**cone cells**”.



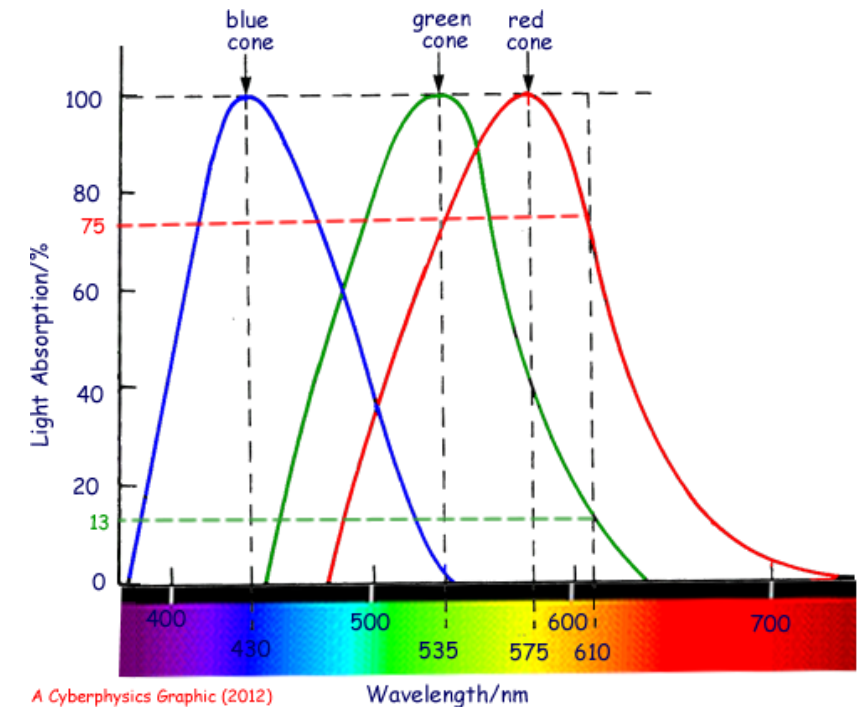
Physiology of color perception



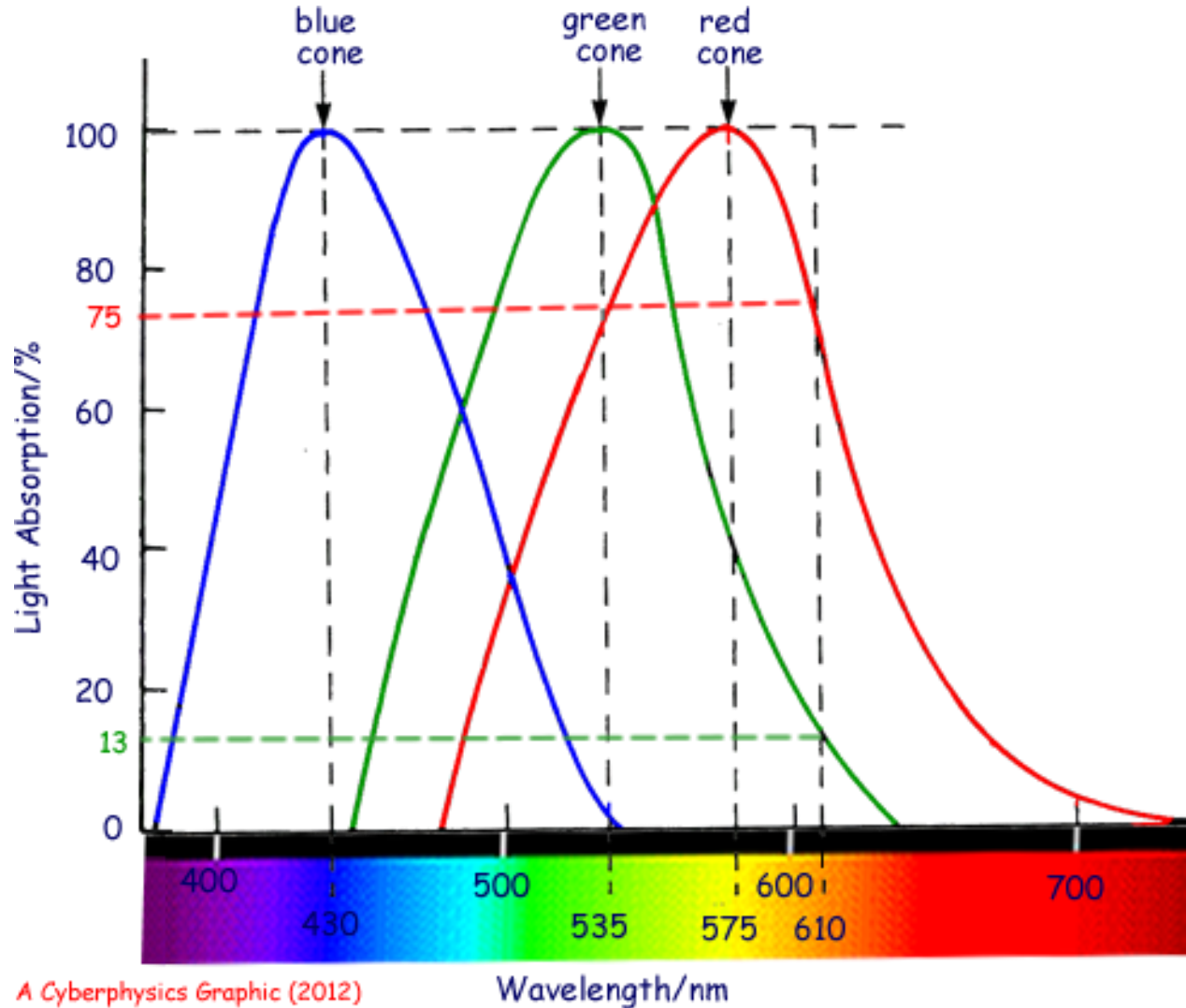
Normalization



Cone Type	Range (nm)
S (Blue)	400 – 500
M (Green)	450 – 630
L (Red)	500 – 700



Physiology of color perception



Example:

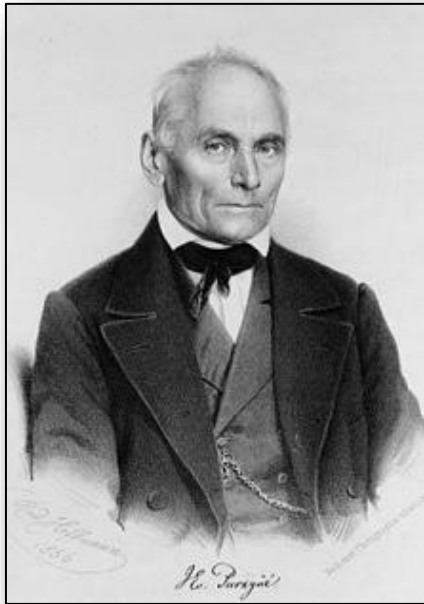
Monochromatic light of wavelength 610 nm

red cones	stimulus value	75%
green cones	stimulus value	13%
blue cones	stimulus value	0%

The brain then interprets the stimulus ratios 75:13:0 as an orange red.

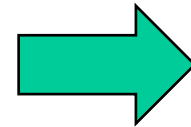
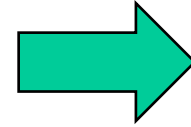
Physiology of color perception

ในที่สว่างเราจะเห็นว่าแสงสีแดงดูสว่างกว่าสีเขียวและน้ำเงิน

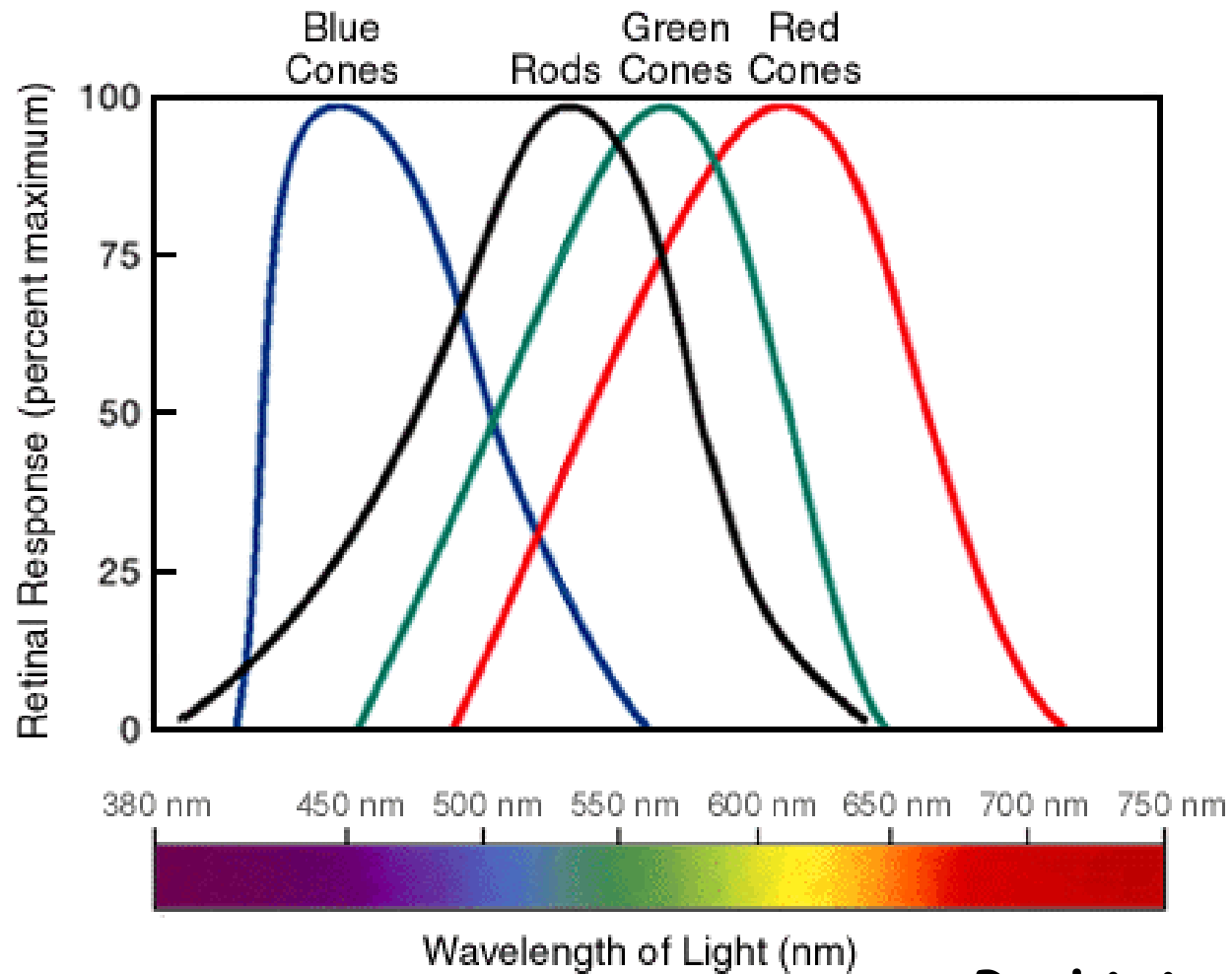


Putkinje effect

ในที่มืด เราเห็นสีเขียวเป็นสีขาวสว่าง ขณะที่สีแดงเห็นเป็นสีดำ



Physiology of color perception



Putkinje effect



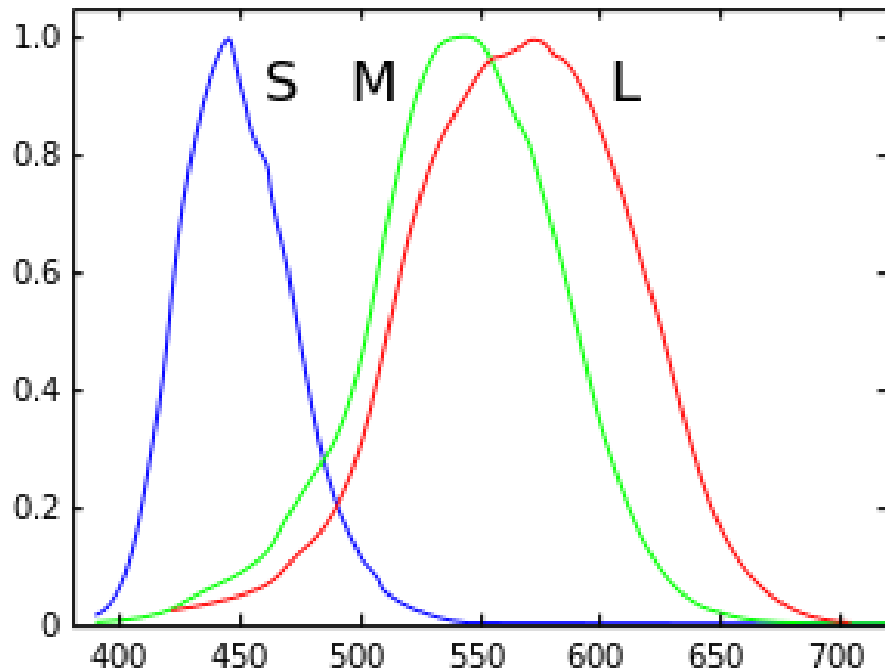
Cones

Rod

Color Space



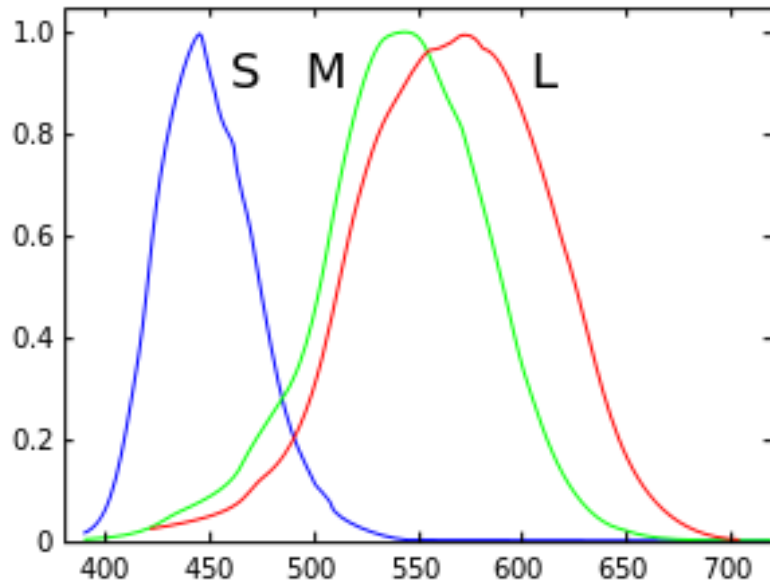
International Commission on Illumination
Commission Internationale de l'Eclairage
Internationale Beleuchtungskommission



หาวิธี Represent สีให้สะดวกขึ้น
ตอนนี้ S, M, L มีช่วงการรับสีแตกต่างกัน
ทำให้บอกได้ยากกว่าเป็นสีอะไรเมื่อรู้ช่วงของ
คลื่นแสง เช่น 400nm → สี ?

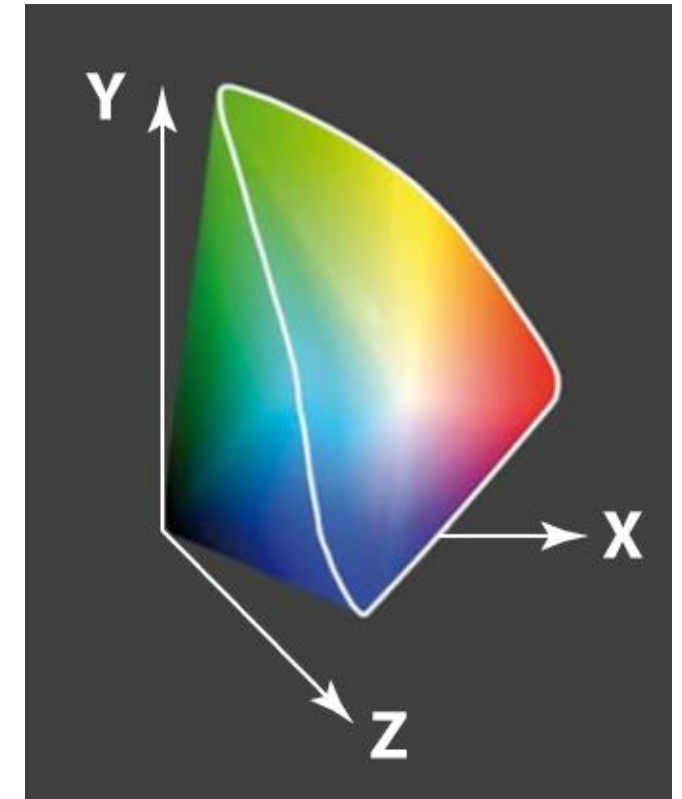
long-wavelength (L), medium-wavelength (M),
and short-wavelength (S) light receptors.

Color Space



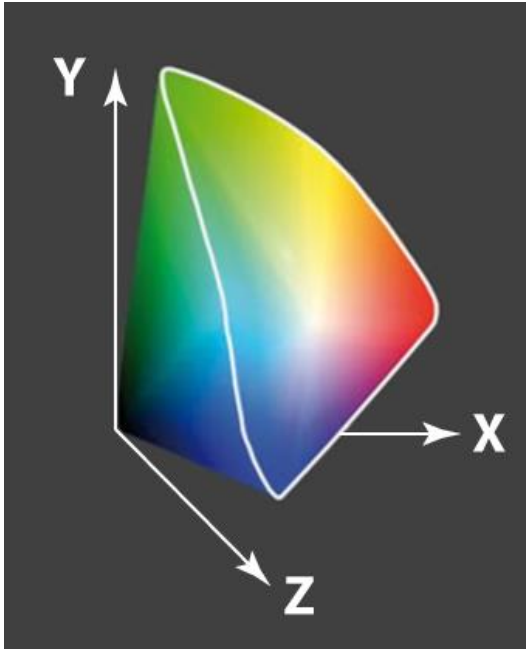
long-wavelength (L), medium-wavelength (M),
and short-wavelength (S) light receptors.

$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} = \begin{bmatrix} 0.7328 & 0.4296 & -0.1624 \\ -0.7036 & 1.6975 & 0.0061 \\ 0.0030 & 0.0136 & 0.9834 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$



From a physiological point of view, the LMS color space describes a more fundamental level of human visual response, so it makes more sense to define XYZ by LMS, rather than the other way around

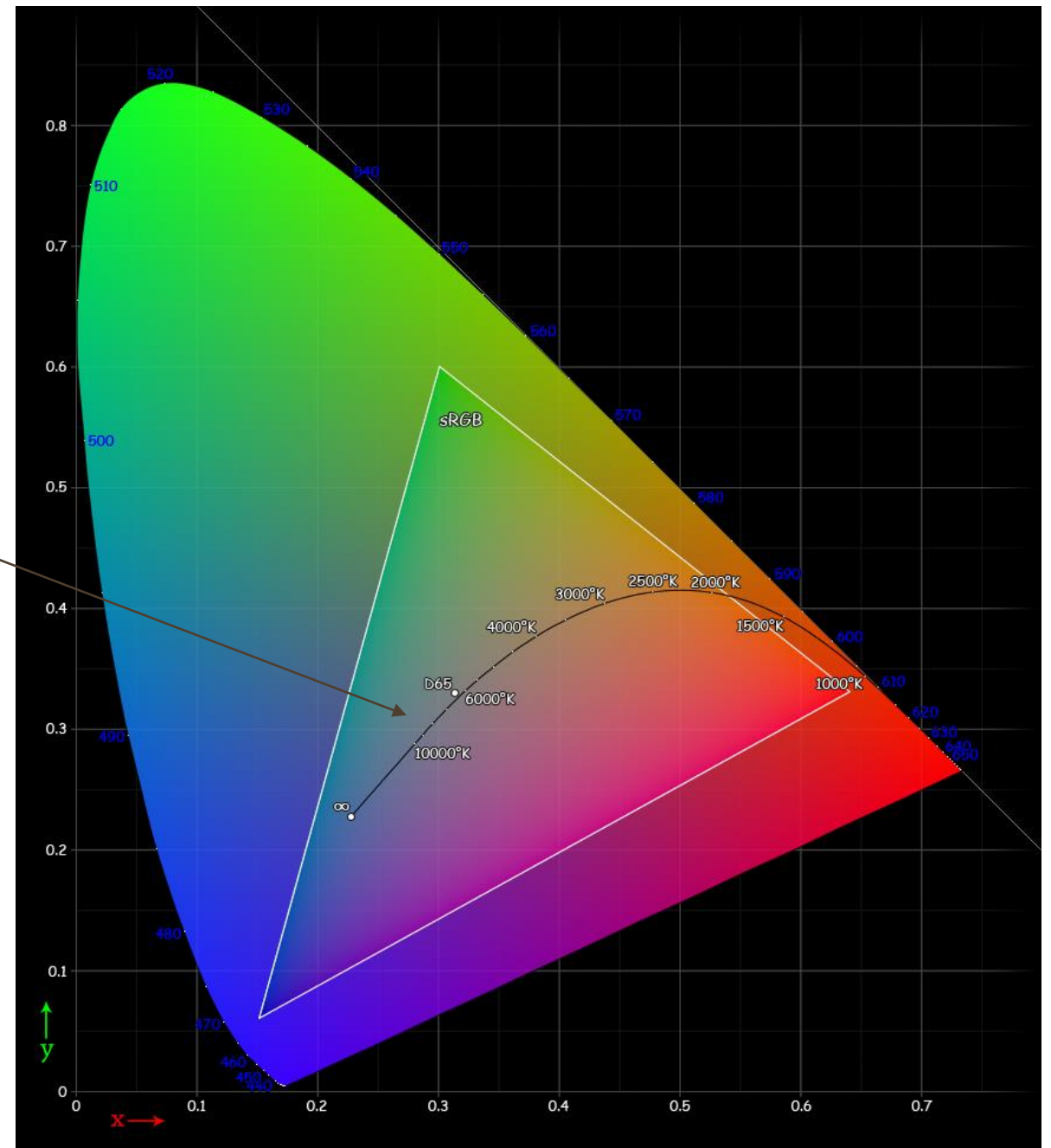
Color Space



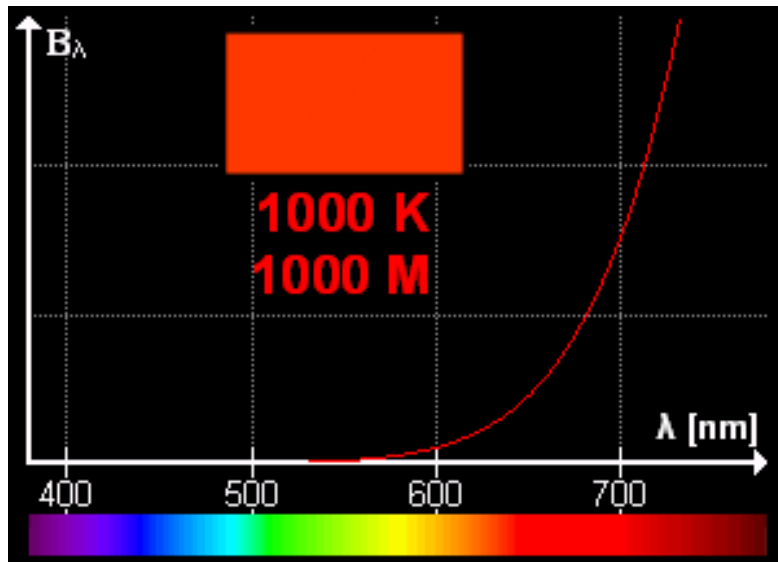
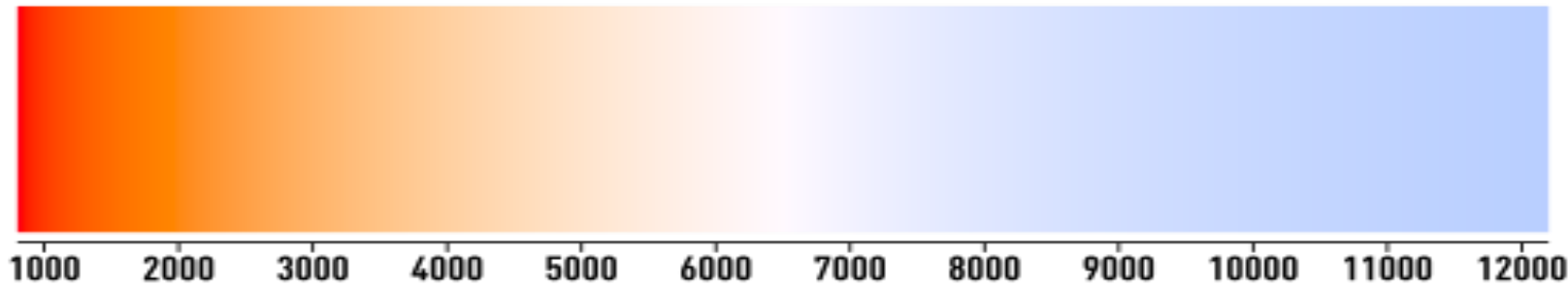
White?
(มีสีขาวอยู่หรือไม่?)

The origin, $(S,M,L) = (0,0,0)$ or $(x, y, z) = (0,0,0)$, corresponds to black.

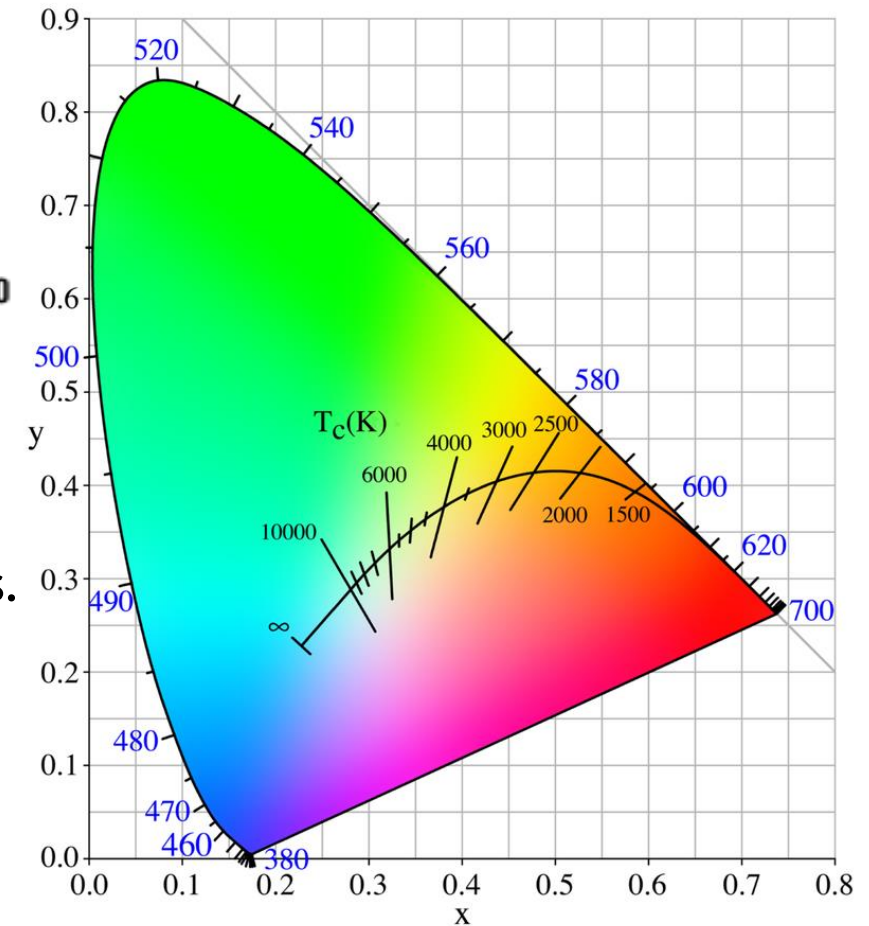
White has no definite position in this diagram.



Color Space : Color temperature



The black-body radiance (B_λ) vs. wavelength (λ) curves for the visible spectrum.



Color Space : Color temperature

Temperature	Source
1700 K	Match flame, low pressure sodium lamps (LPS/SOX)
1850 K	Candle flame, sunset/sunrise
2400 K	Standard incandescent lamps
2550 K	Soft white incandescent lamps
2700 K	"Soft white" compact fluorescent and LED lamps
3000 K	Warm white compact fluorescent and LED lamps
3200 K	Studio lamps, photofloods, etc.
3350 K	Studio "CP" light
5000 K	Horizon daylight
5000 K	Tubular fluorescent lamps or cool white/daylight compact fluorescent lamps (CFL)
5500 – 6000 K	Vertical daylight, electronic flash
6200 K	Xenon short-arc lamp ^[3]
6500 K	Daylight, overcast
6500 – 9500 K	LCD or CRT screen
15,000 – 27,000 K	Clear blue poleward sky
<i>These temperatures are merely characteristic; there may be considerable variation</i>	

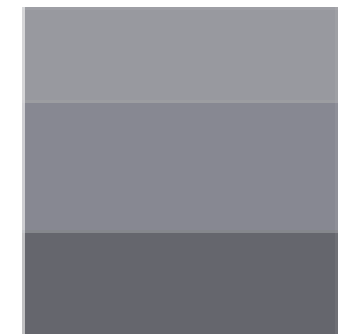


กำแพงบ้านสีขาว → สีนํ้าเงิน

เพราะเหตุนี้เลยต้องมีแวนตัดแสง “สีฟ้า”



"Warm" gray



"Cool" gray

Color Space : Color temperature

WHITE BALANCE PRESETS

Your guide to what they do and when to use them

AUTO A simple failsafe mode for snapshots, but the White Balance may vary from one shot to the next, and you may not get the colours you expect.

Incandescent This is the closest match for regular domestic lighting, and will correct the colour much more effectively than auto White Balance.

Fluorescent This comes in many different types and current D-SLRs offer no fewer than seven alternatives, so some trial and error may be needed.

Direct Sunlight Calibrated to give neutral colours under midday sun – and you can use it as a fixed standard for recording

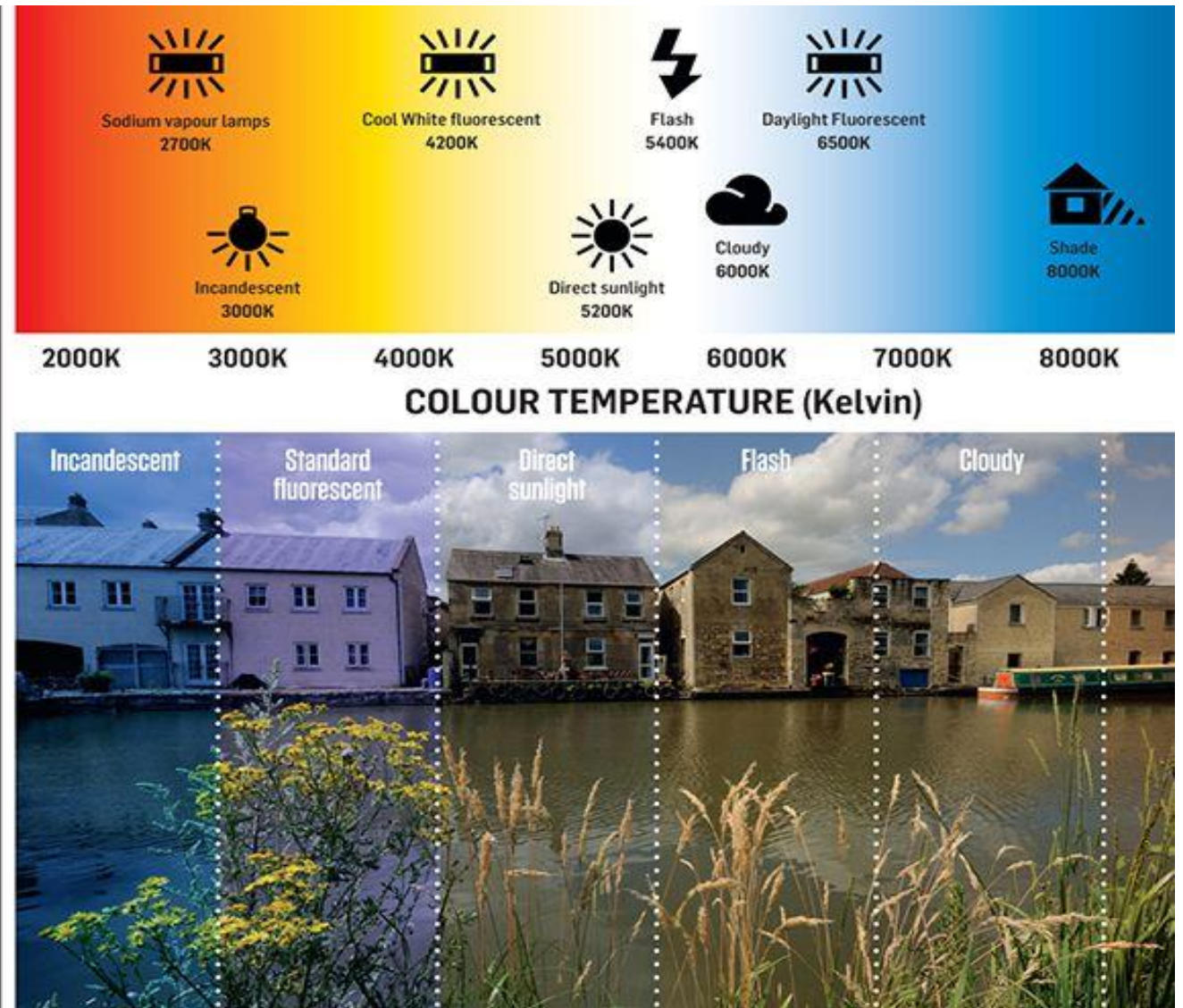
sunlight, and using this preset can prevent skin tones turning 'cold'.

Cloudy Light has a cooler tone under a cloudy sky, and this preset will warm up the colours. It's good for portraits but can be too much for landscapes.

Shade This is designed for open shade under a blue sky. This preset will give you more natural-looking skin tones.

K More advanced D-SLRs let you set the White Balance colour temperature manually – useful with some studio lighting.

PRE Sometimes it's impossible to predict the colour of the lighting and the



Color Space : Color temperature



Cool White



Daylight



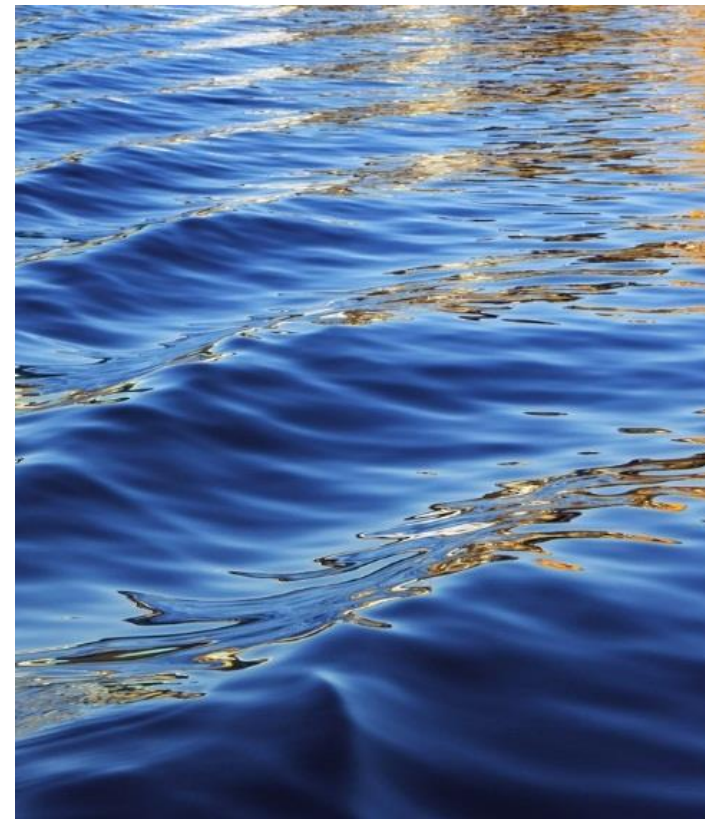
Warm White





Color Models

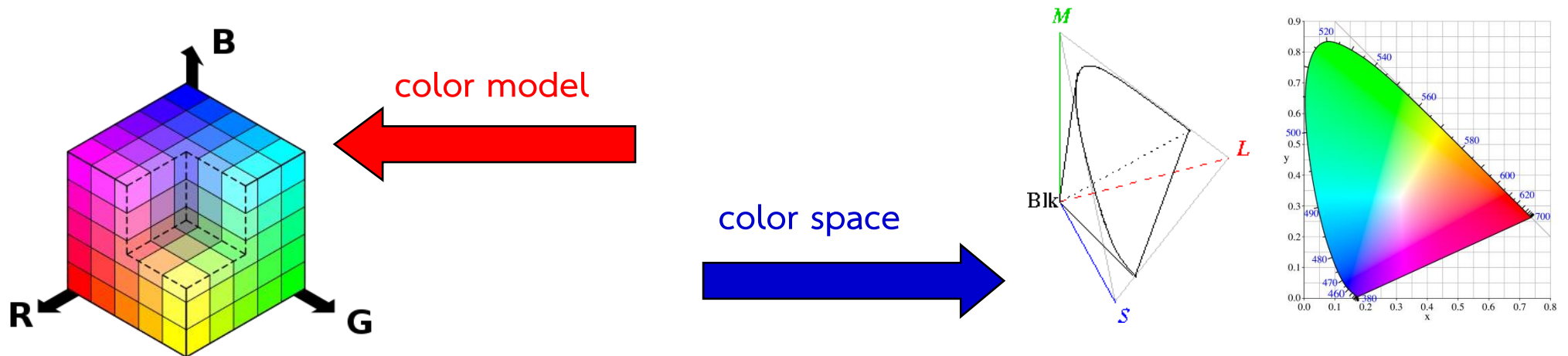
Describing color with abstract mathematical



Color Models

A **color model** is an abstract mathematical model **describing the way colors can be represented as tuples of numbers**.

When this model is associated with a precise description of how the components are to be interpreted (viewing conditions, etc.), **the resulting set of colors is called “color space”**.



Color Models

Additive & Subtractive Color Models

- ☐ RGB Color Model
- ☐ CMYK Color Model

Cylindrical Coordinate Color Models

- ☐ HSL Color Model
- ☐ HSV Color Model
- ☐ HSI Color Model

Color Models

The RGB color model is an **additive color model** in which red, green and blue light are added together in various ways to reproduce a broad array of colors.

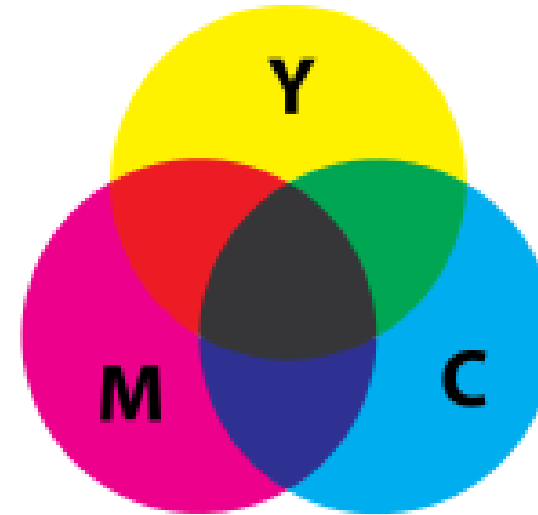
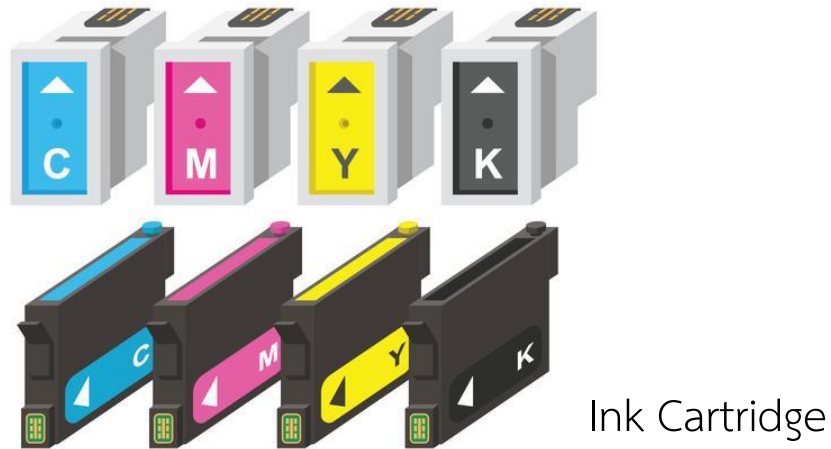


Additive & Subtractive Color Models

Cylindrical Coordinate Color Models

Color Models

The CMYK color model is a **subtractive color model**, used in color printing, and is also used to describe the printing process itself. CMYK refers to the four inks used in some color printing: **C**yan, **m**agenta, **y**ellow, and black.

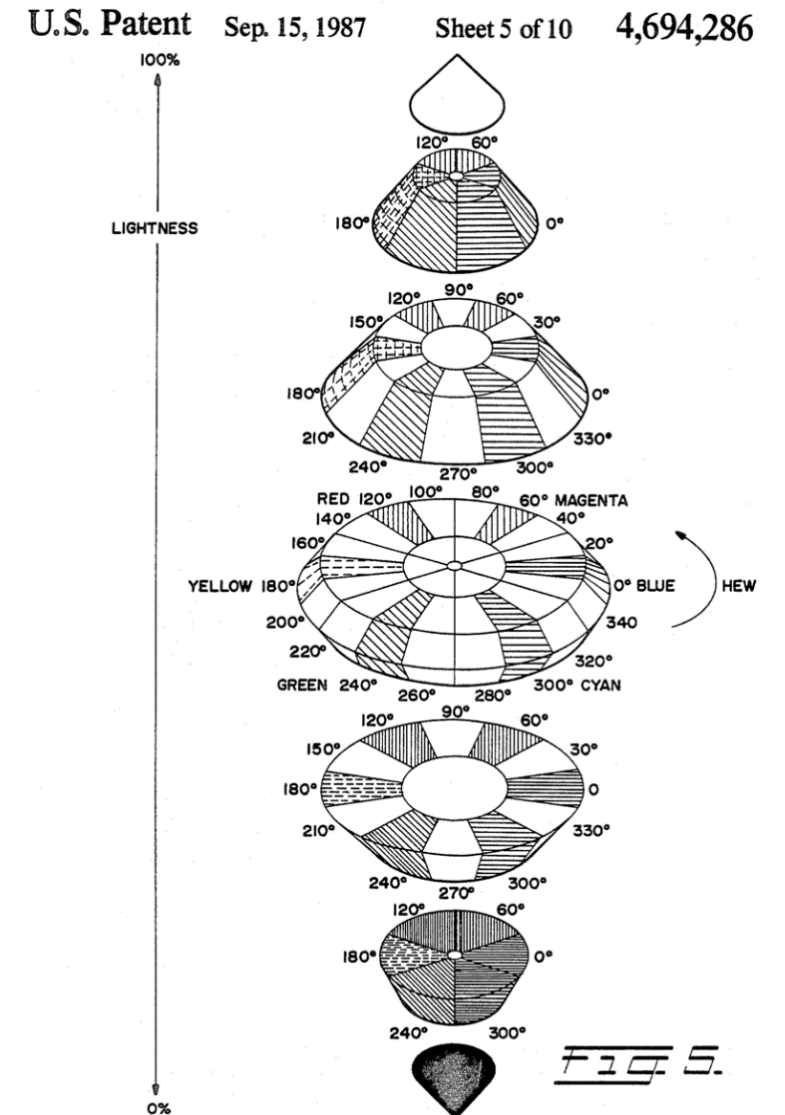


Additive & Subtractive Color Models

Cylindrical Coordinate Color Models

Color Models

The **HSL color space** was invented in 1938 by Georges Valensi as a method to **add color encoding to existing monochrome broadcasts**, without modification the luminance signal.



Additive & Subtractive Color Models

Cylindrical Coordinate Color Models

Color Models

Hue

(degree)



0

60

120

180

240

300

360

Saturation
(HSL)

1.00	0.75	0.50	0.25	0.00
(255,0,0)	(224,32,32)	(192,64,64)	(160,96,96)	(128,128,128)
(255,128,0)	(224,128,32)	(192,128,64)	(160,128,96)	(128,128,128)
(255,255,0)	(224,224,32)	(192,192,64)	(160,160,96)	(128,128,128)

Additive & Subtractive Color Models

Cylindrical Coordinate Color Models

Color Models



Intensity (HSI)



Value (HSV)



Lightness (HSL)

HSI

≠

HSV

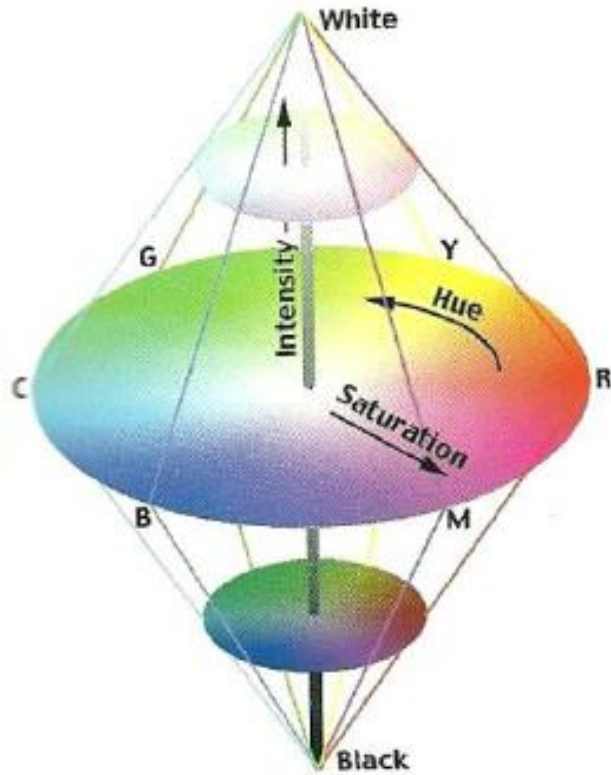
≠

HSL

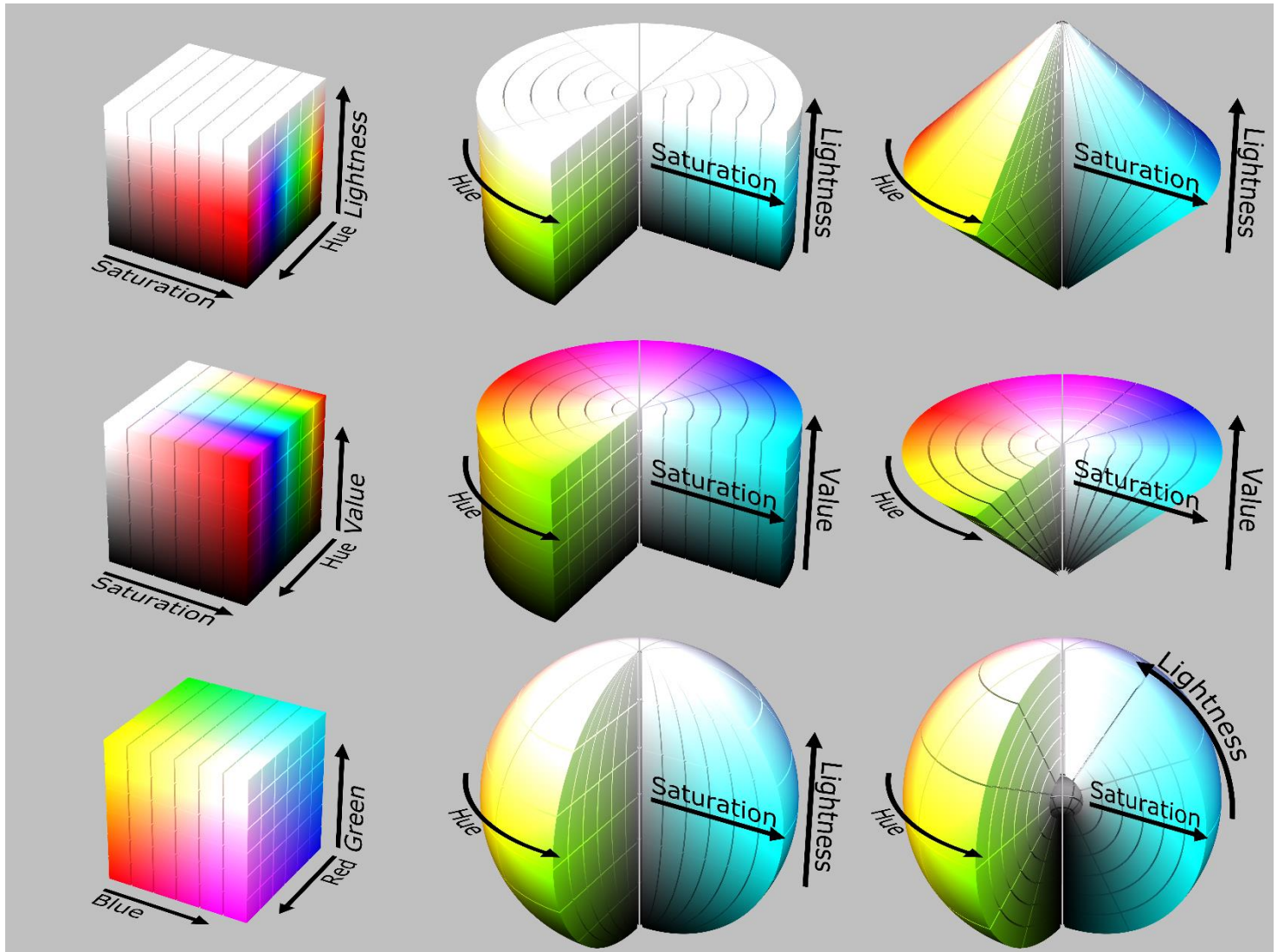
Additive & Subtractive Color Models

Cylindrical Coordinate Color Models

Color Models



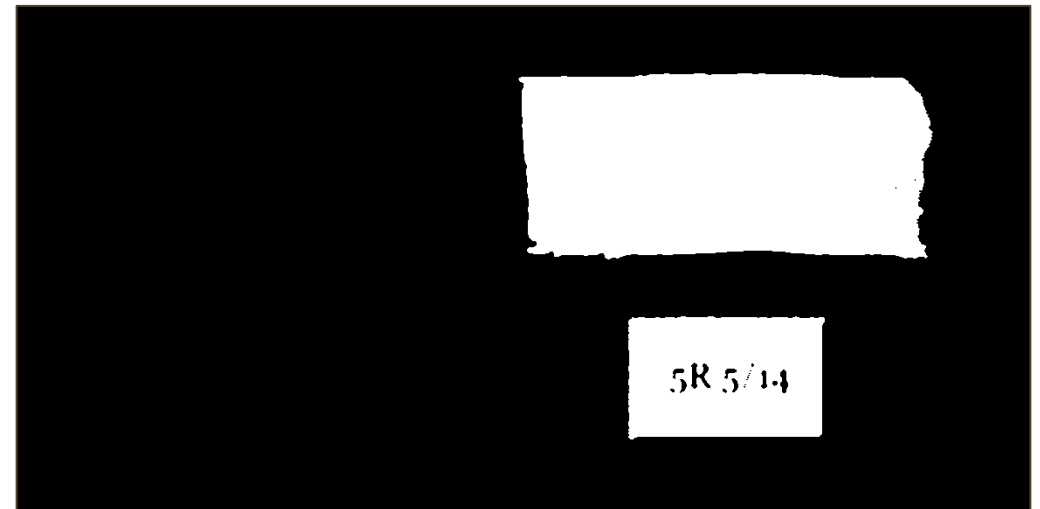
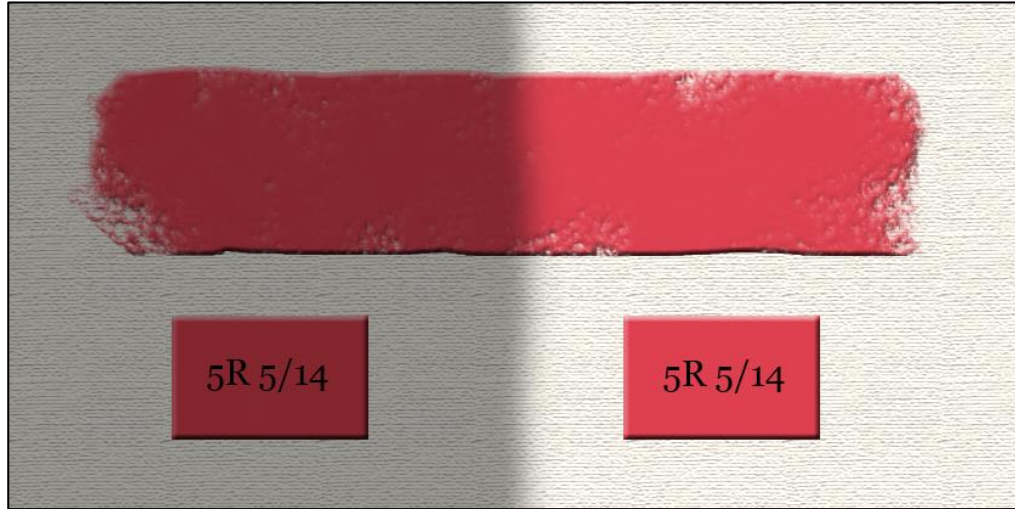
HSI Model



Additive & Subtractive Color Models

Cylindrical Coordinate Color Models

Which color model matches your problem?



Example: Play with Hue

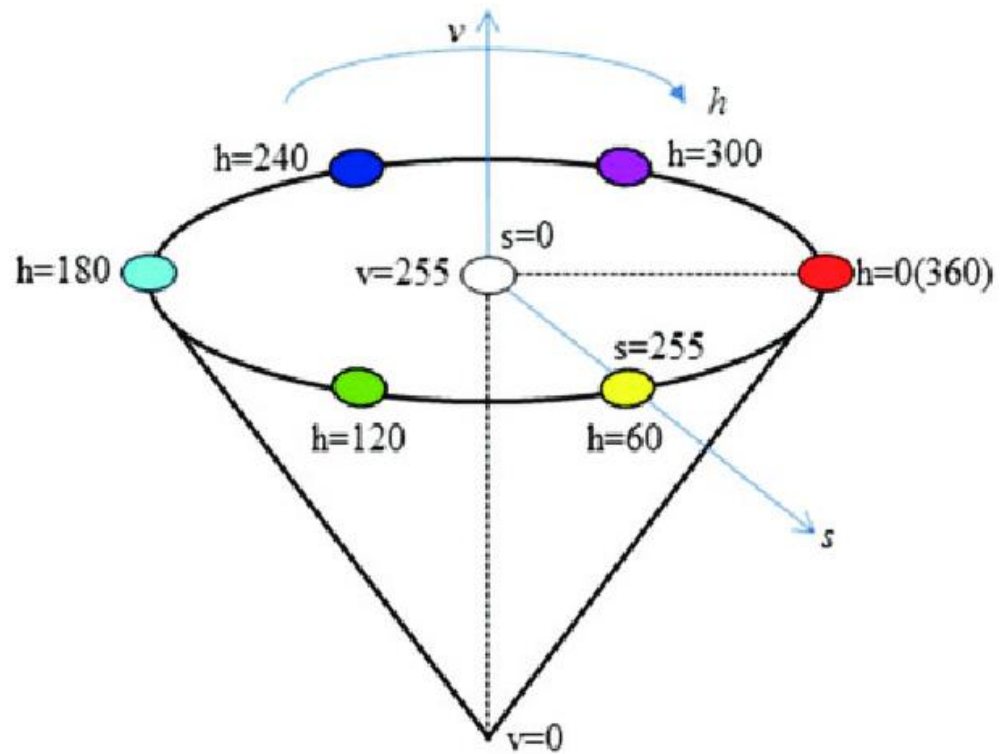




Image Acquisition

How we capture the image from the World

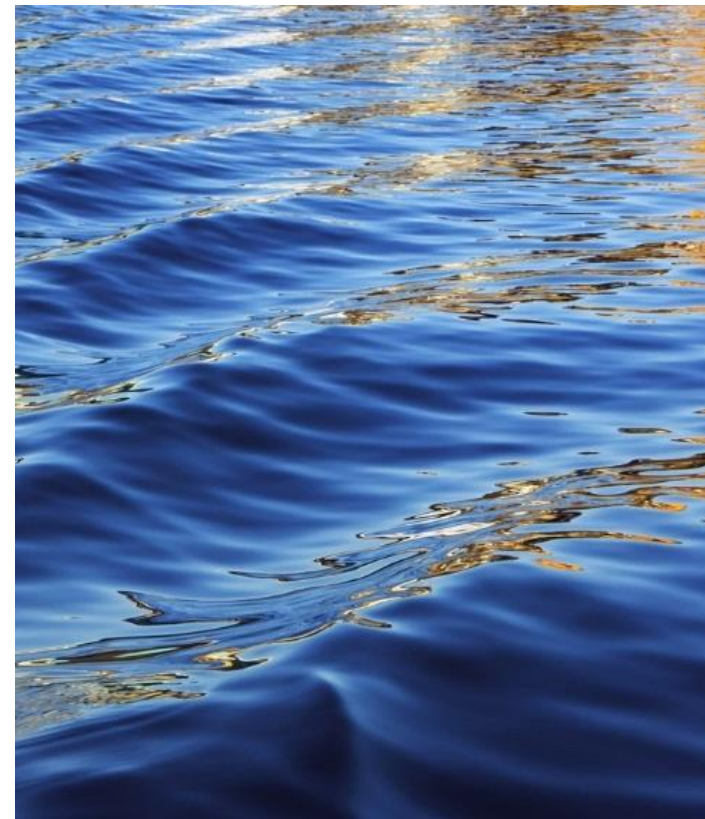


Image Acquisition (Real world $\rightarrow$ Image (Film))

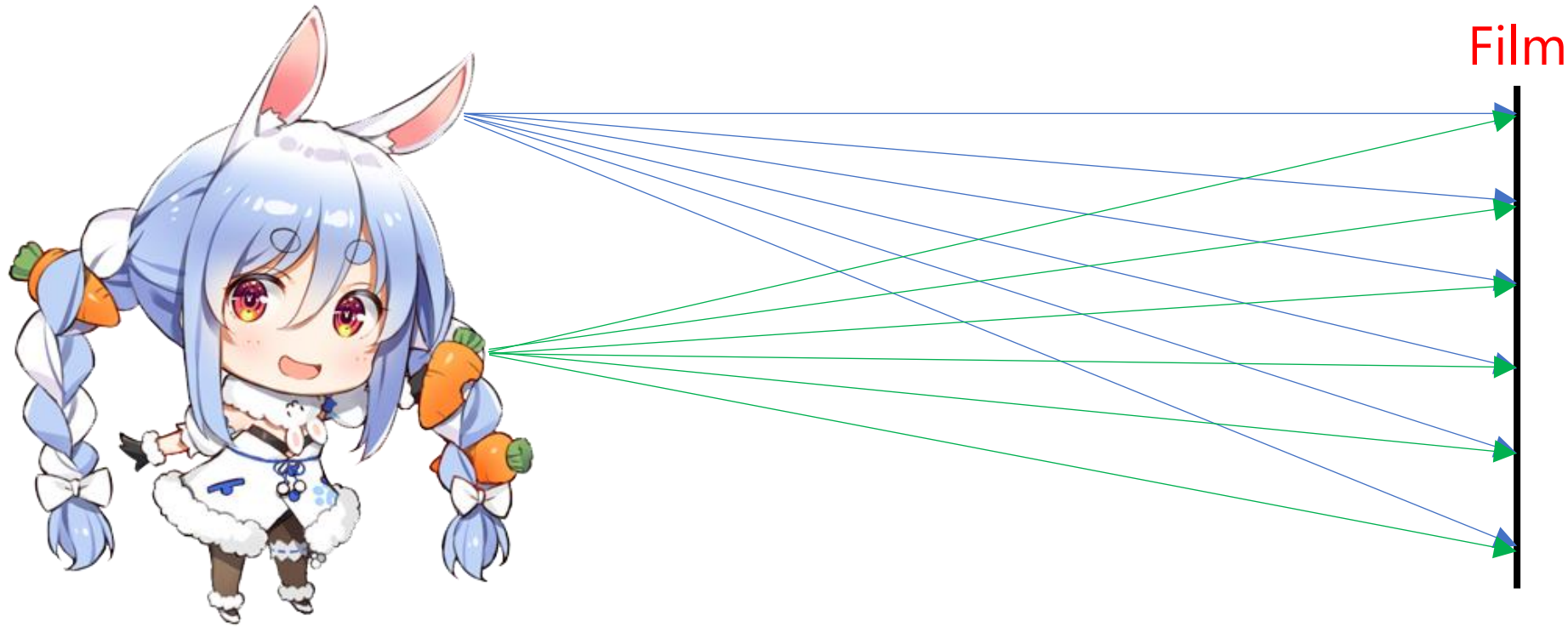


Image Acquisition (Pinhole Camera)

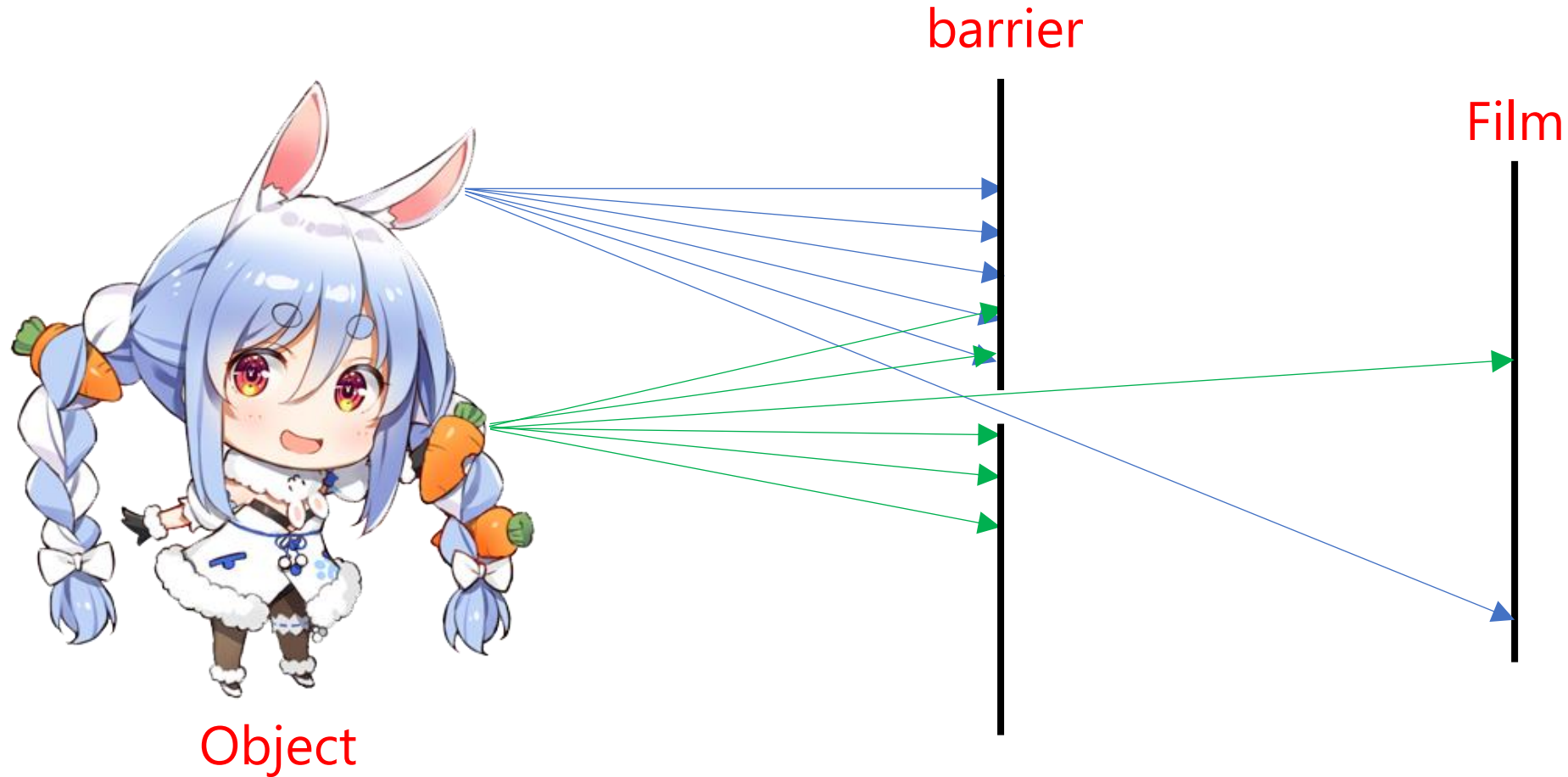
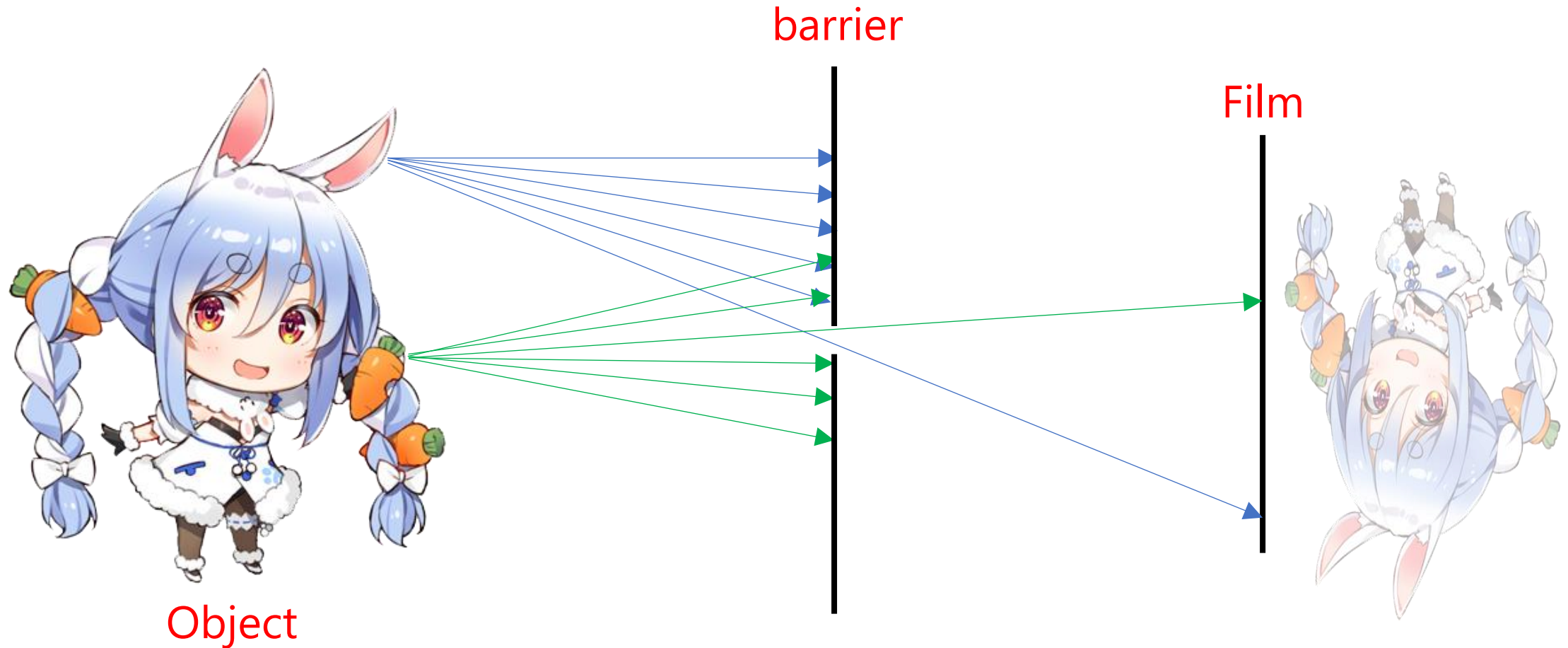
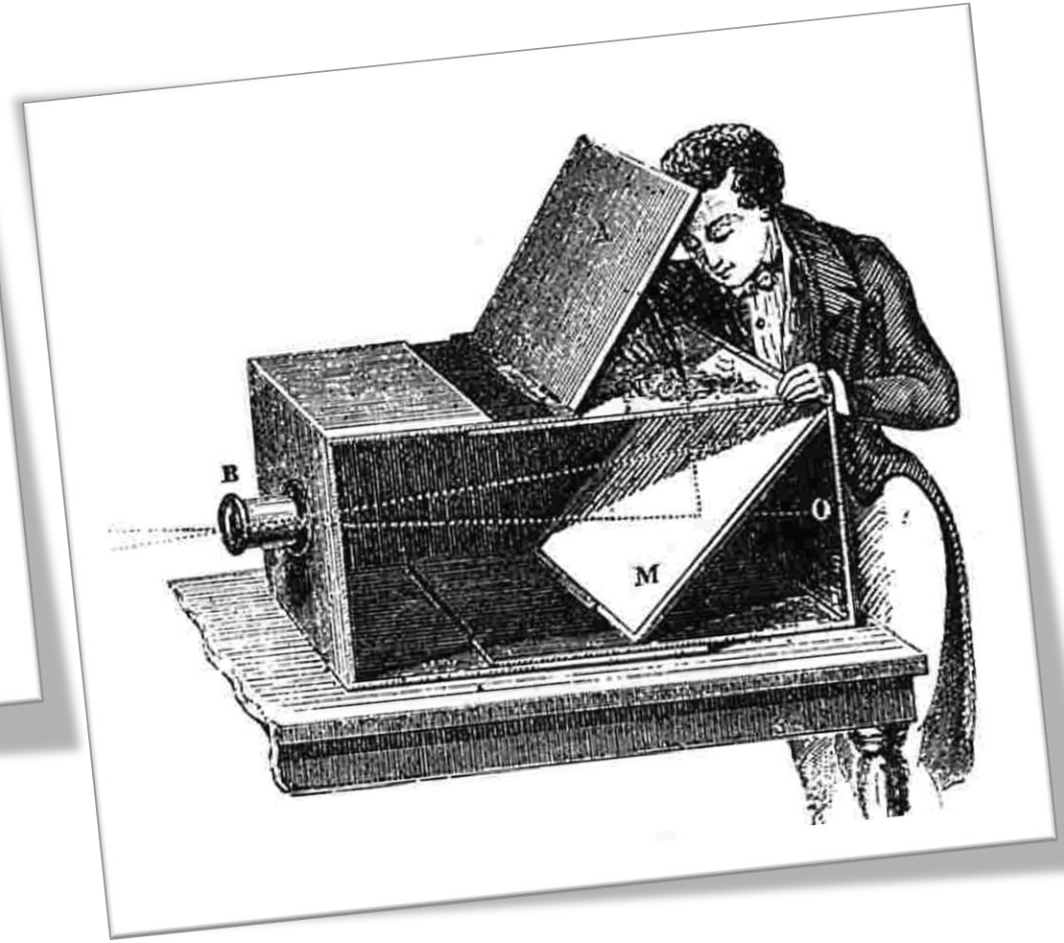
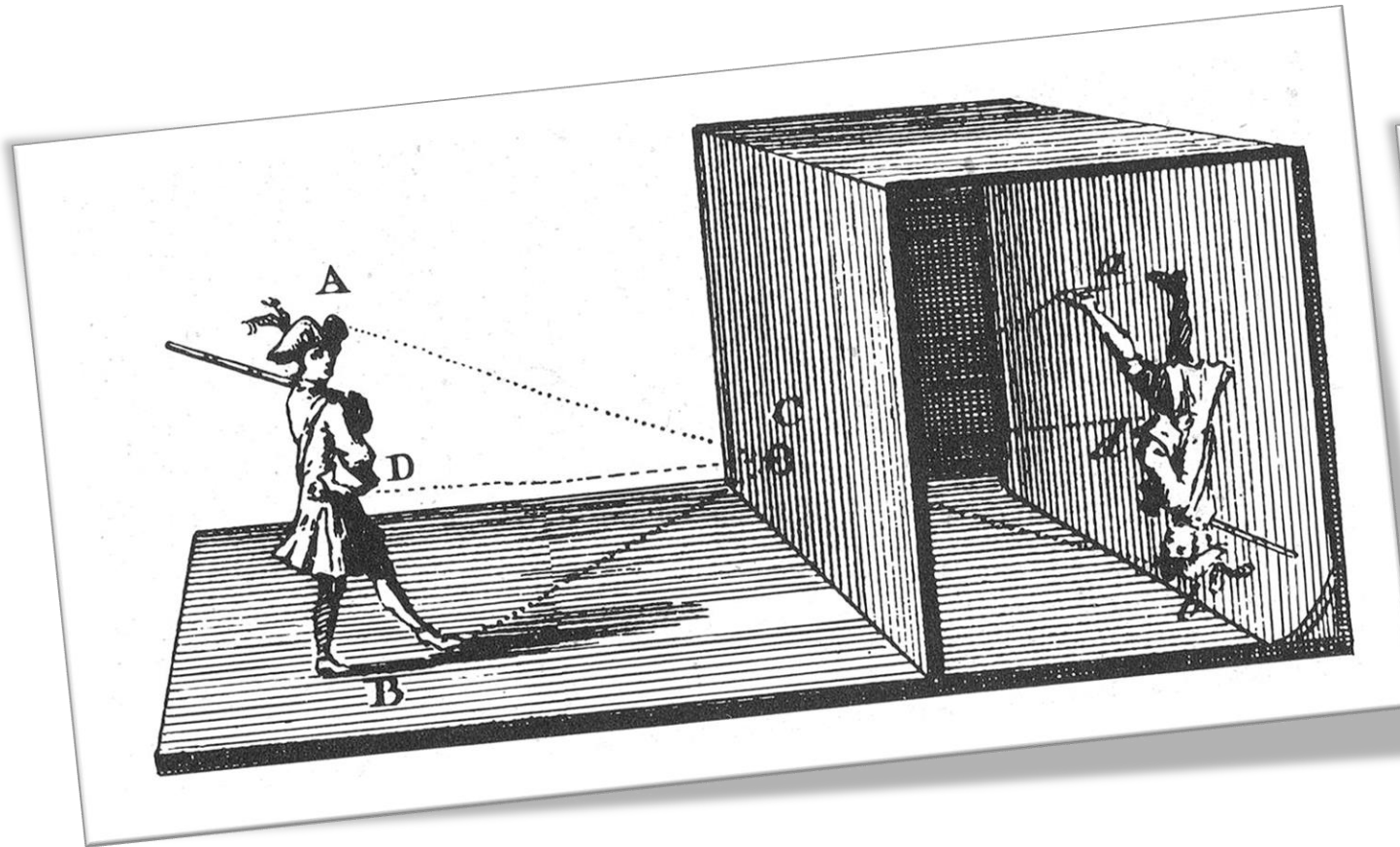


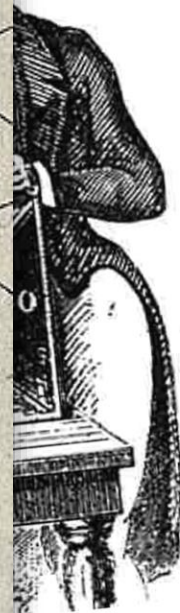
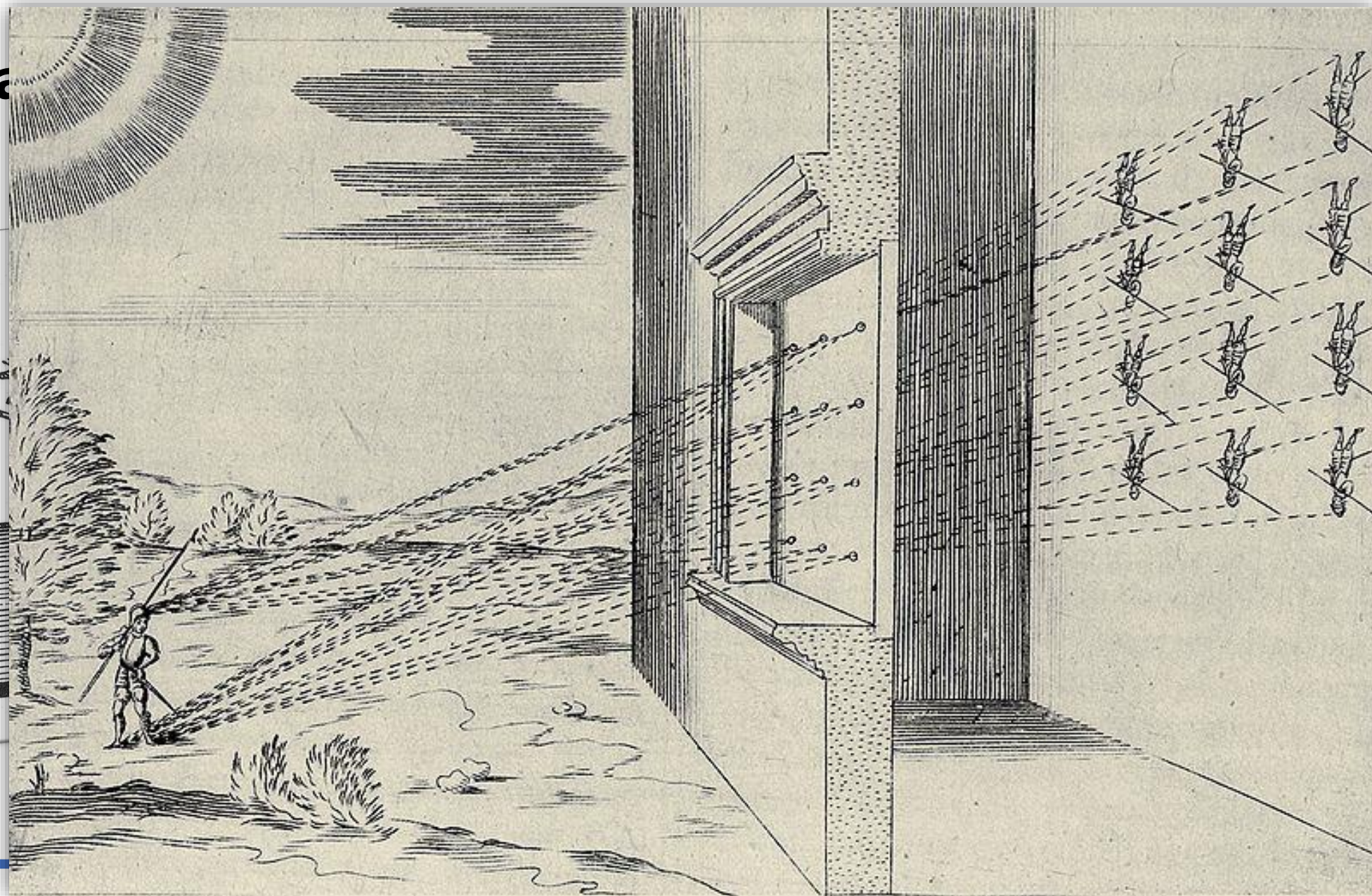
Image Acquisition (Pinhole Camera)



Camera obscura (the beginnings of photograph)



Ca



Niepce "View from the Window at Le Gras", 1826



1826
8h exp

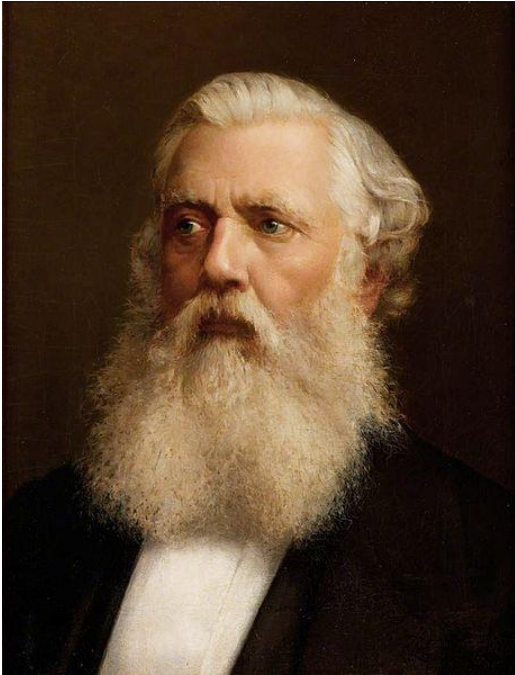


Louis Daguerre

Lens focus light
Better chemical → 10 – 12 min



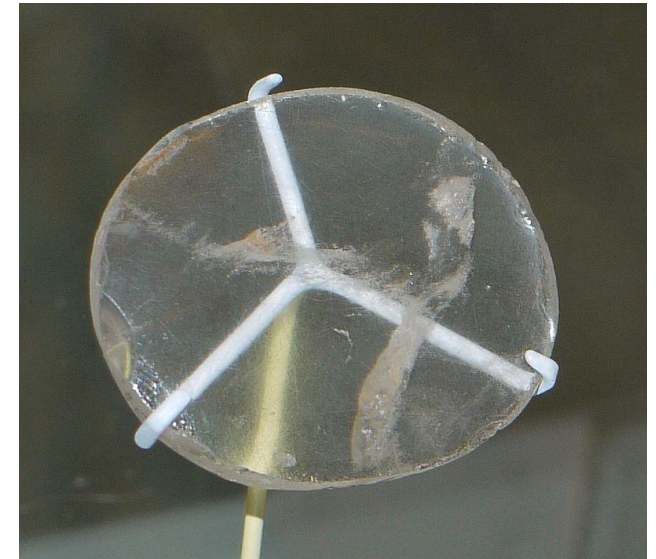
Lens (???)



Sir Austen Henry Layard
English traveller



Assyrian palace reliefs
ancient Assyrian states (Iraq)



Nimrud lens

Alhazen Hasan Ibn al-Haytham

ابن الهيثم



อิบนุ ฮัยษัม (Ibn al-Haytham)
หรือที่รู้จักในชื่อละตินว่า Alhazen

- ยุค: ศตวรรษที่ 10-11 (ประมาณ ค.ศ. 965 – 1040)
- ผลงานเด่น: เขียนหนังสือชื่อ "Kitab al-Manazir" (Book of Optics)
- สิ่งที่น่าสนใจ:
 - พิสูจน์ว่าแสงเดินทางเป็นเส้นตรง
 - อธิบายหลักการการสะท้อนและหักเหของแสง
 - ใช้เลนส์นูนเพื่อ ขยายภาพ
 - วางรากฐานของ กล้องถ่ายภาพ (Camera Obscura)

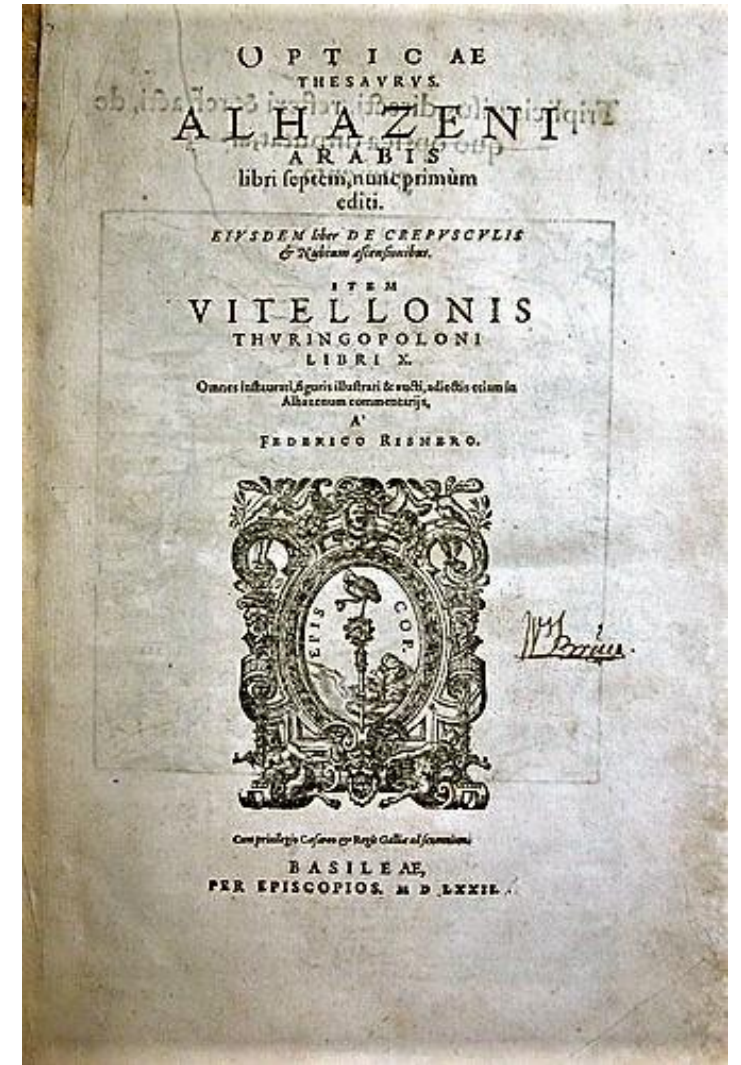


Image Acquisition (Pinhole Camera)

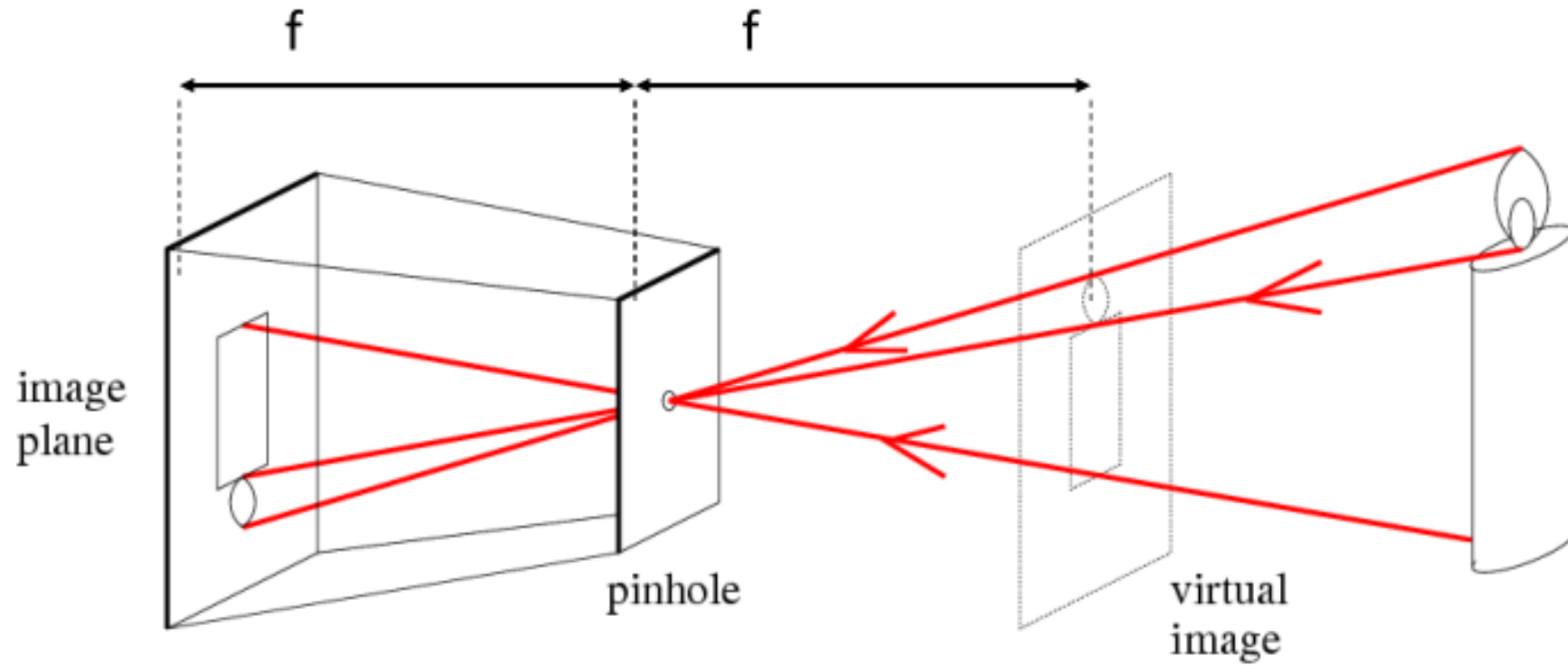
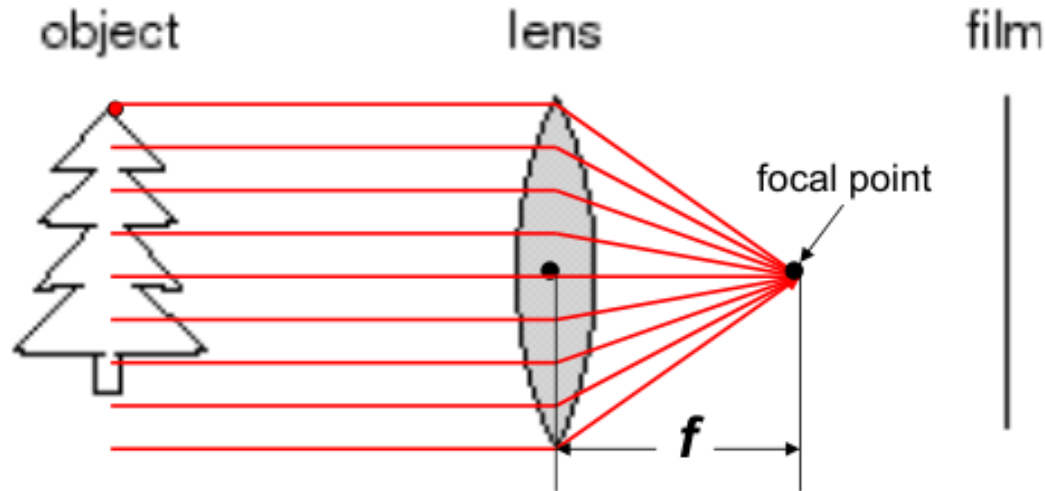


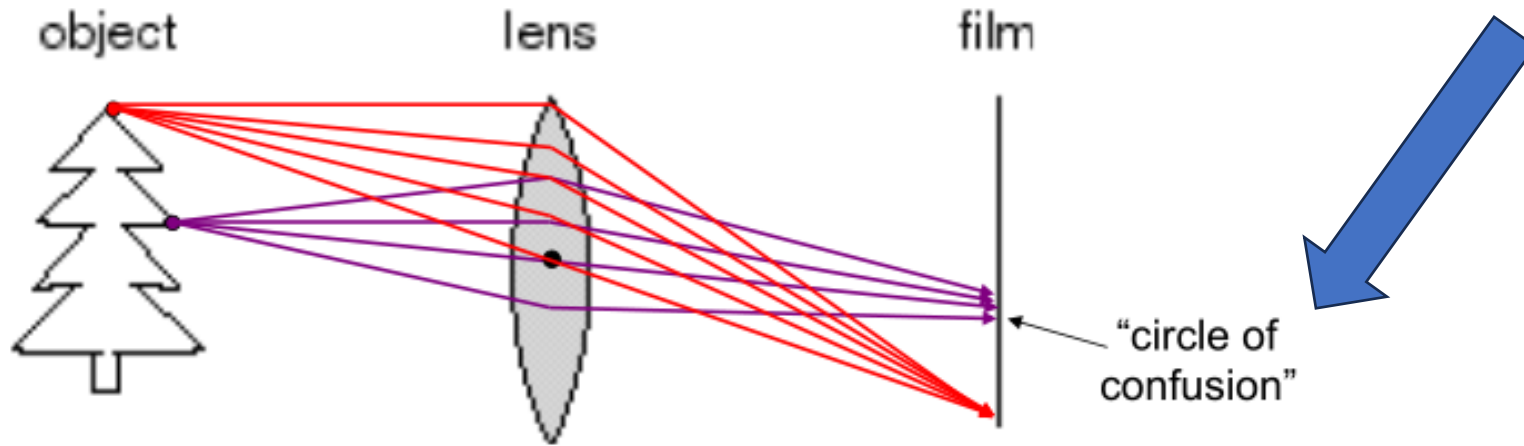
Image Acquisition (Lens)



A lens focuses light onto the film

- Rays passing through the center are not deviated
- All parallel rays converge to one point on a plane located at the *focal length* f

Image Acquisition (Lens)



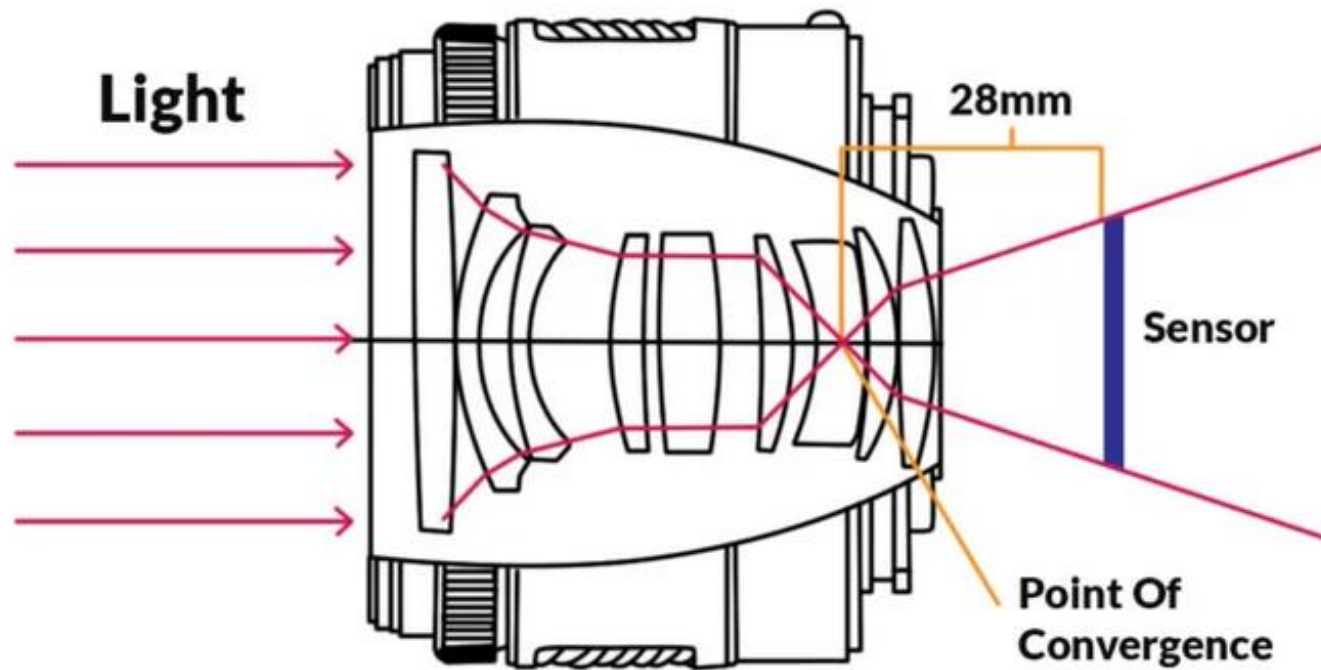
A lens focuses light onto the film

- There is a specific distance at which objects are “in focus”
[other points project to a “circle of confusion” in the image]

Image Acquisition (Lens)

Why do we use complex, multi-element designs and large lens housings if a single lens will also provide an image for much cheaper?

EXPERT
PHOTOGRAPHY





gns a

28m

Real Glass



Convergence

Approximate (Thin lens)

Image Acquisition (Lens)

Snell's law

$$n_1 \sin \alpha_1 = n_2 \sin \alpha_2$$

α_1 = incident angle
 α_2 = refraction angle
 n_i = index of refraction

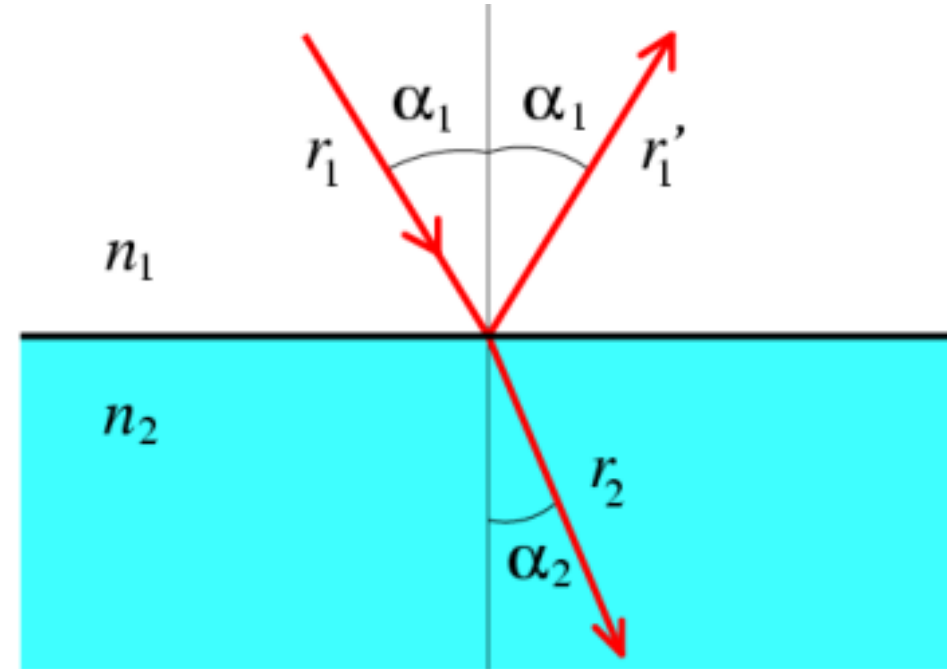
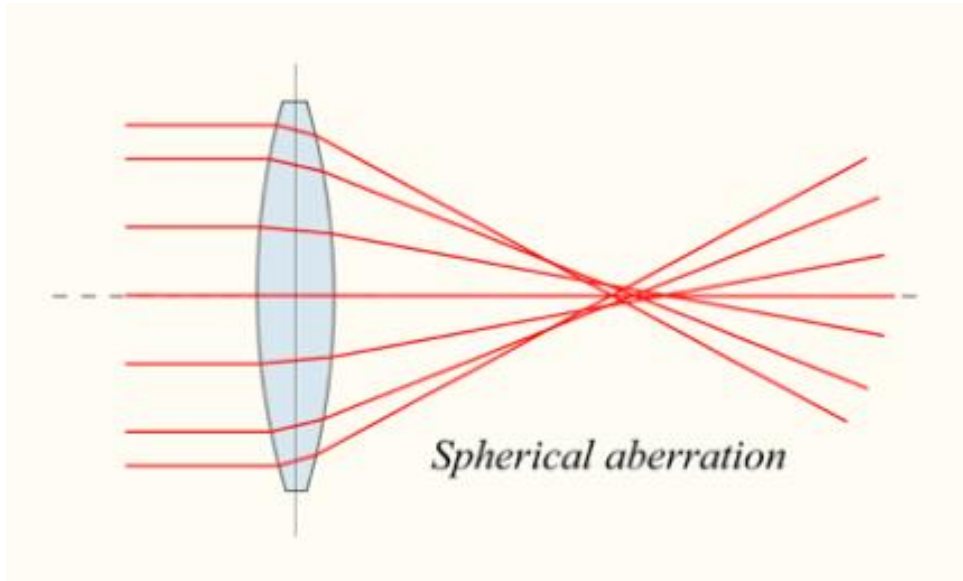
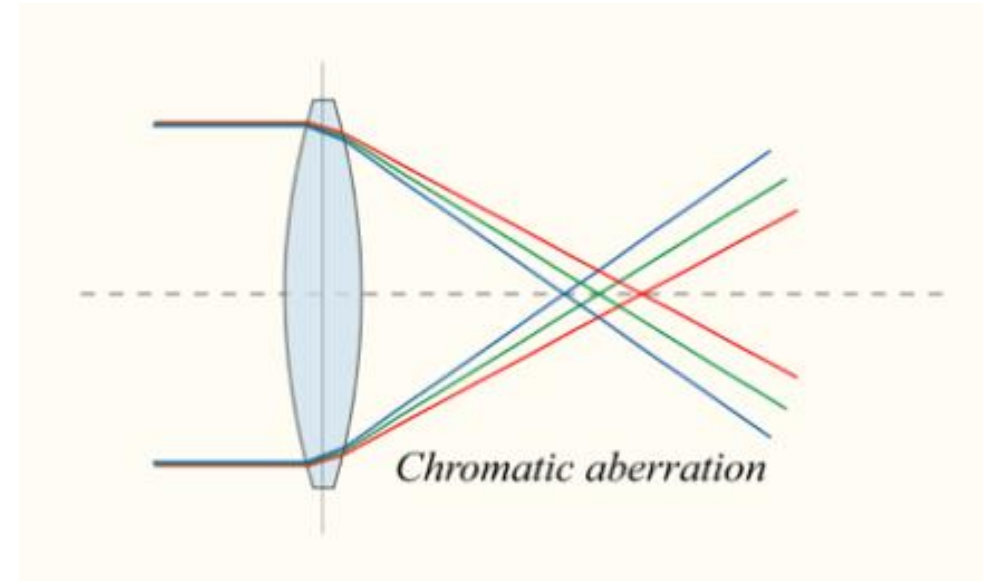


Image Acquisition (Aberrations (Abnormal))

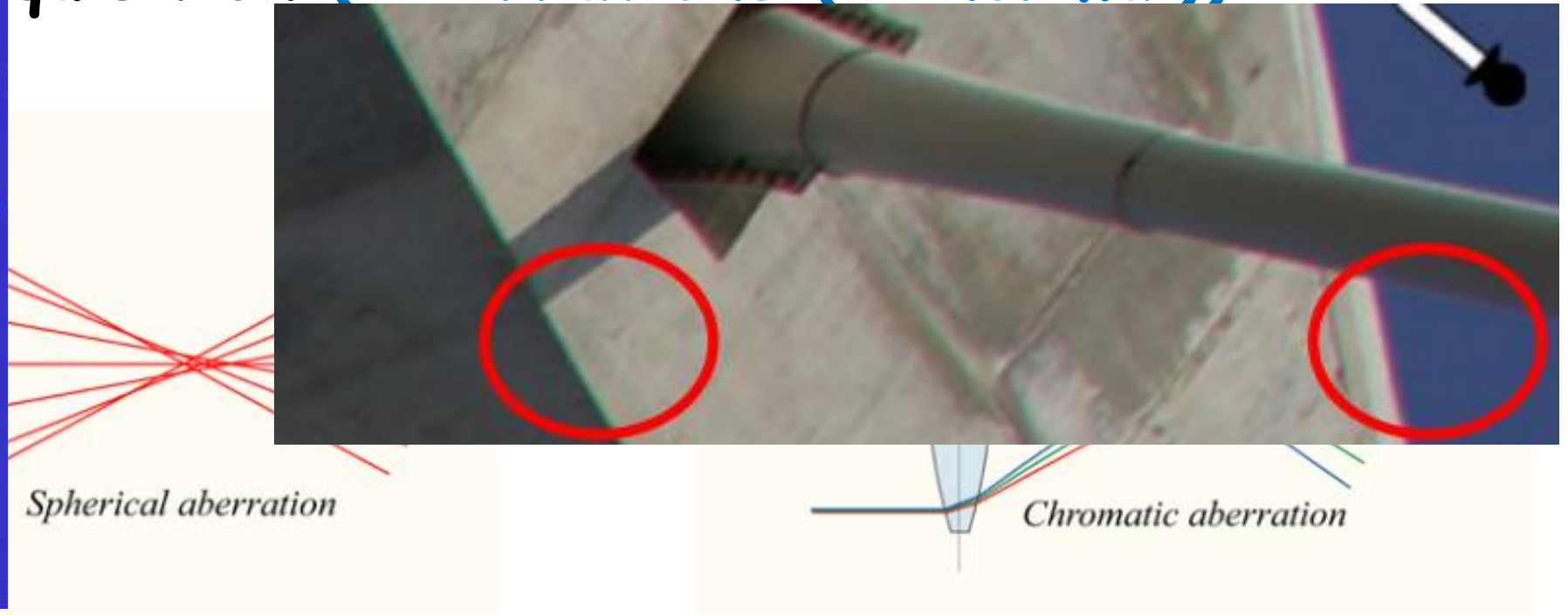
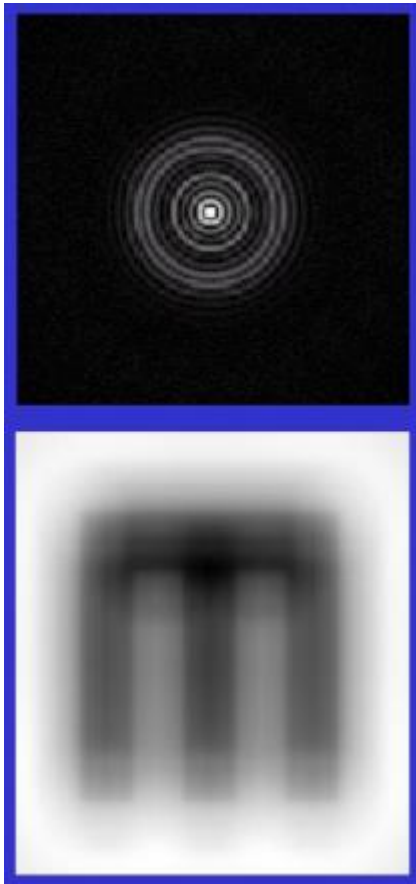


Rays farther from the optical axis focus closer
(แถวๆ ขอบเลนส์จะโฟกัสใกล้กว่า)



Lens has different refractive indices for different wavelengths: causes colour fringing

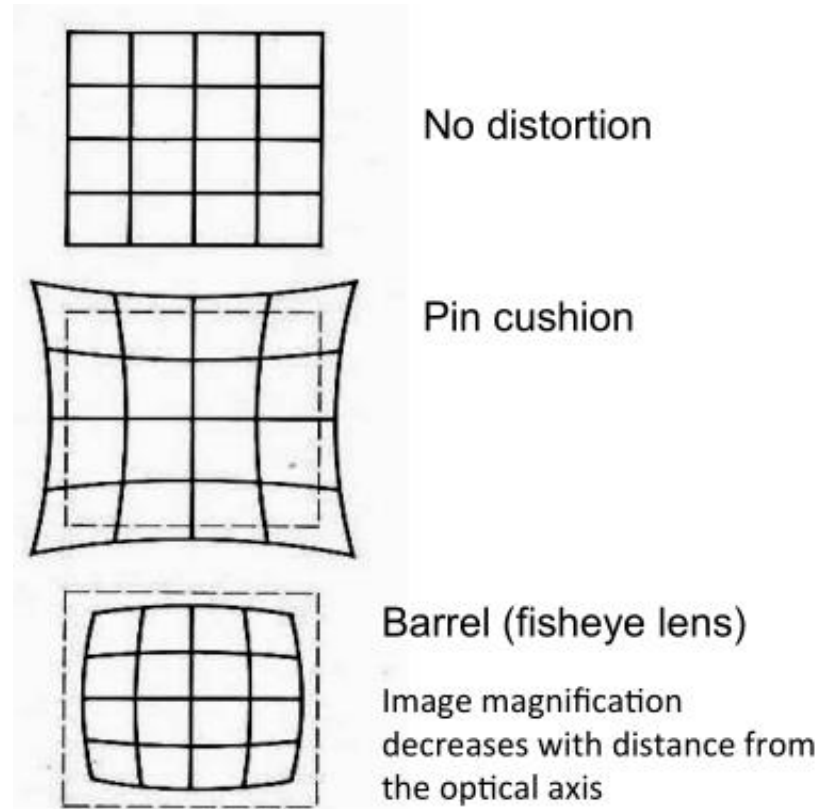
quisition (Aberrations (Abnormal))



Rays farther from the optical axis focus closer
(แถวๆ ขอบเลนส์จะโฟกัสใกล้กว่า)

Lens has different refractive indices for different
wavelengths causes colour fringing

Image Acquisition (Aberrations (Abnormal))

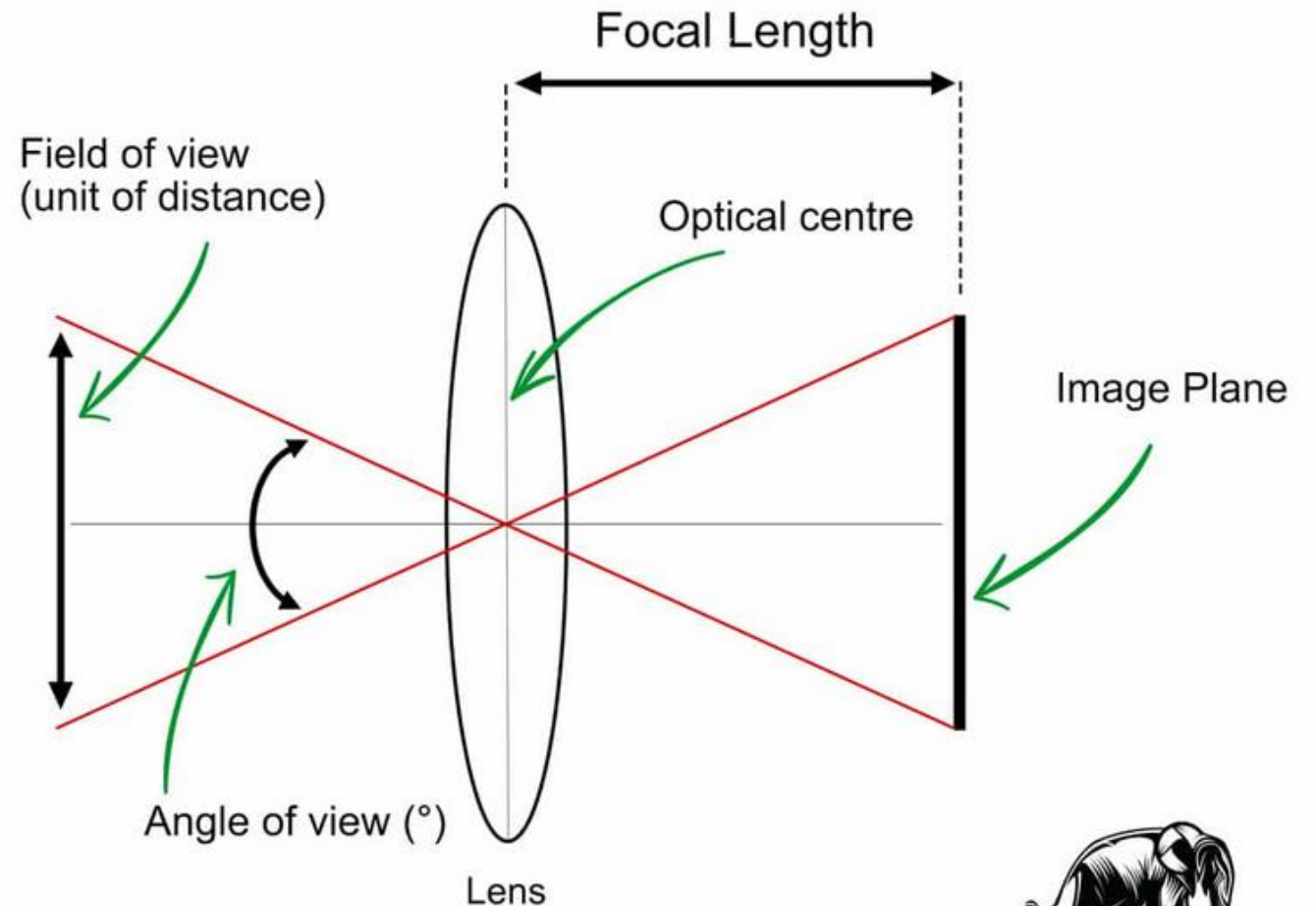
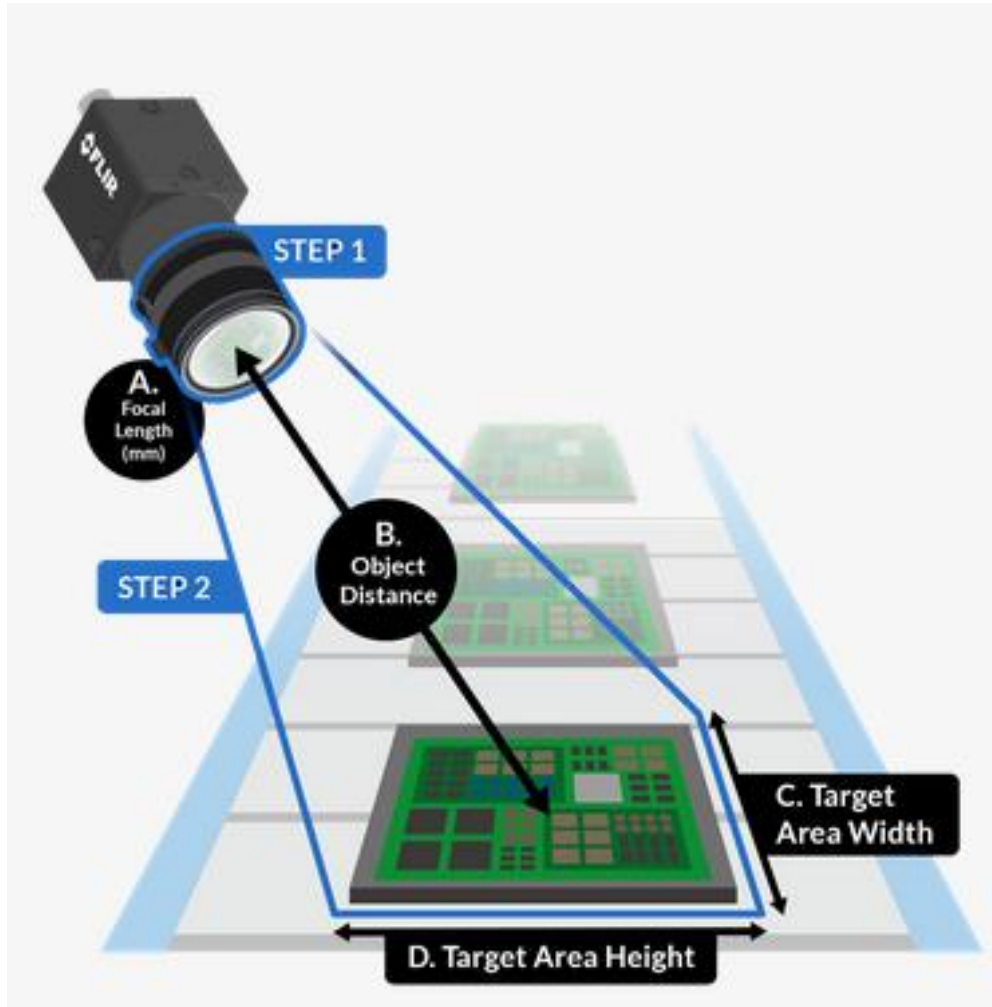


Deviations are most noticeable for rays that pass through the edge of the lens

How to choose camera? (for image processing project)

- [1] **Data type** (Still Image or Video) → **CCD** or **CMOS**
- [2] **Coverage area** (**Field of View** / **Size of Object**)
- [3] **Resolution of image** / **Frame rate** (**Detection** / **Identify**)
- [4] **Connectivity** / **Hardware Memory**
- [5] **Reasonable price**

Field of View (FOV)



Check your camera

Recommend to do it with calculator software such as Microsoft Excel, etc,.

Camera and Lens Settings

Results

Sensor Size: Full Frame / Nikon FX (1.00x)

Focal Length: 50 mm

Aperture: f/ 8 Custom AV

Distance: 10 Feet Meters

AF Test distance

Closeup of Face

Headshot

Head & Shoulder

Upper Body

Full Body

Angle and Field of View

	Hor.	Vert.	Diag
AoV	39.6°	27°	46.8°
FoV (US)	23' 7.00"	15' 9.00"	28' 5.00"
FoV (SI)	7.2 m	4.8 m	8.65 m
Method 1: Fixed AoV [See Notes]			
AoV	39.4°	26.9°	46.6°
FoV (US)	23' 6.00"	15' 8.00"	28' 3.00"
FoV (SI)	7.16 m	4.78 m	8.61 m
Method 2: AoV shifts w/ focus [See Notes]			

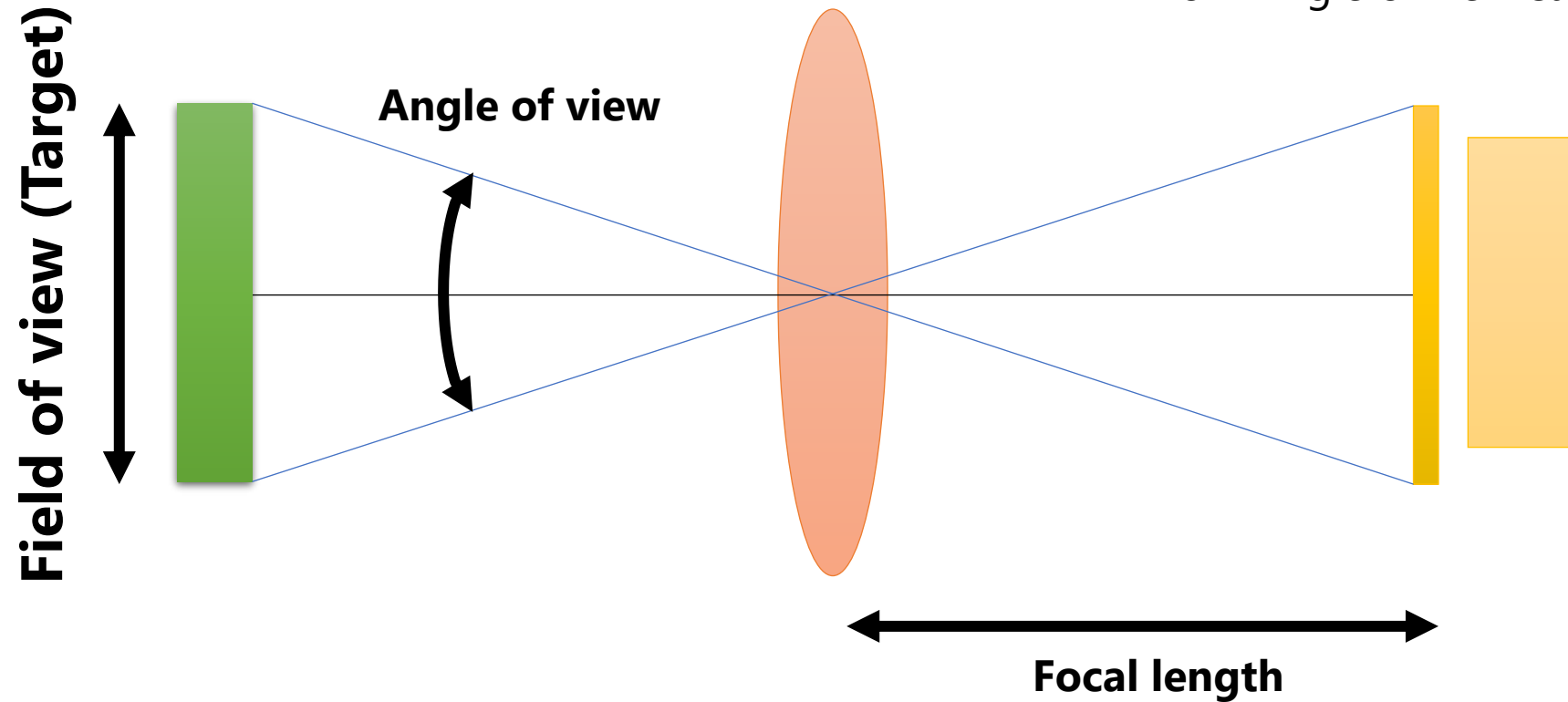
<https://www.pointsinfocus.com/tools/depth-of-field-and-equivalent-lens-calculator/>



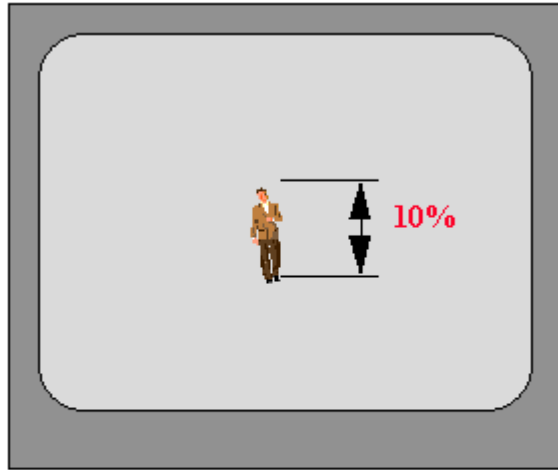
Check your camera

Find "**Angle of view**"
from Focal length & Sensor size

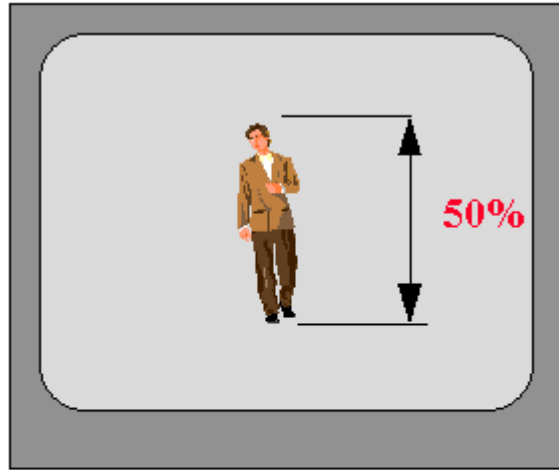
Find "**Field of view**"
From Angle of view & Distance to Obj.



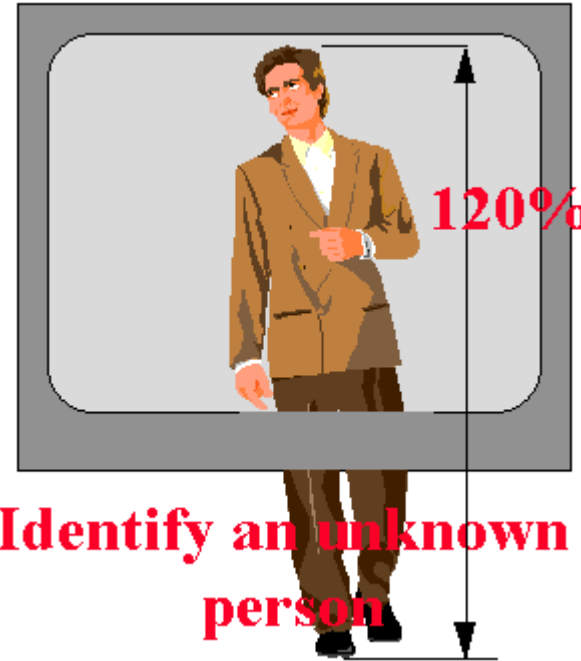
Choose Sensor & Focal Length with your GOAL



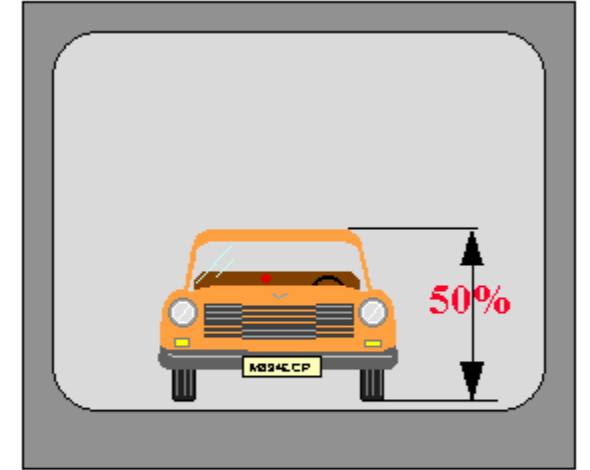
Detect a person



Recognise a known person



Identify an unknown person



Car Number Plate

เรื่องการคำนวณ Field of view (FOV)

[1] หา Angle of view (θ) จาก sensor size กับ Focal length (f)

$$\theta = 2 \tan^{-1} \left(\frac{\text{sensor size}}{2f} \right)$$

[2] หา Field of view (FOV) จาก Angle of view กับ Distance to Object (S_o)

$$\text{FOV} = 2 \left(S_o * \tan \left(\frac{\theta}{2} \right) \right)$$

เรื่องการคำนวณ Field of view (FOV)

ข้อแนะนำ: ควรระมัดระวังเรื่องการใช้หน่วยวัด (ให้ใช้หน่วยวัดเดียวกันจะได้ไม่สับสน)

$$\theta = 2 \tan^{-1} \left(\frac{\text{sensor size}}{2f} \right) \longrightarrow \theta = 2 \tan^{-1} \left(\frac{35\text{mm}}{2(50\text{mm})} \right)$$

$$\text{FOV} = 2 \left(S_o * \tan \left(\frac{\theta}{2} \right) \right) \longrightarrow \text{FOV} = 2 \left((5\text{m}) * \tan \left(\frac{\theta}{2} \right) \right)$$

Field of view (FOV) จะได้ผลลัพธ์เป็นหน่วย เมตร (m) ตามหน่วยของระยะทางวัตถุถึงเลนส์

Sensor

	CCD	CMOS
Resolution	Up to 100+ Megapixels	Up to 100+ Megapixels
Frame rate	Low frame rates	High frame rates
Noise figure	Low noise floor	High noise floor
Sensitive of intensity	More sensitive at low intensity	Less sensitive at low intensity
Shutter type	Global	Rolling
Skew	No	Yes

Sensor

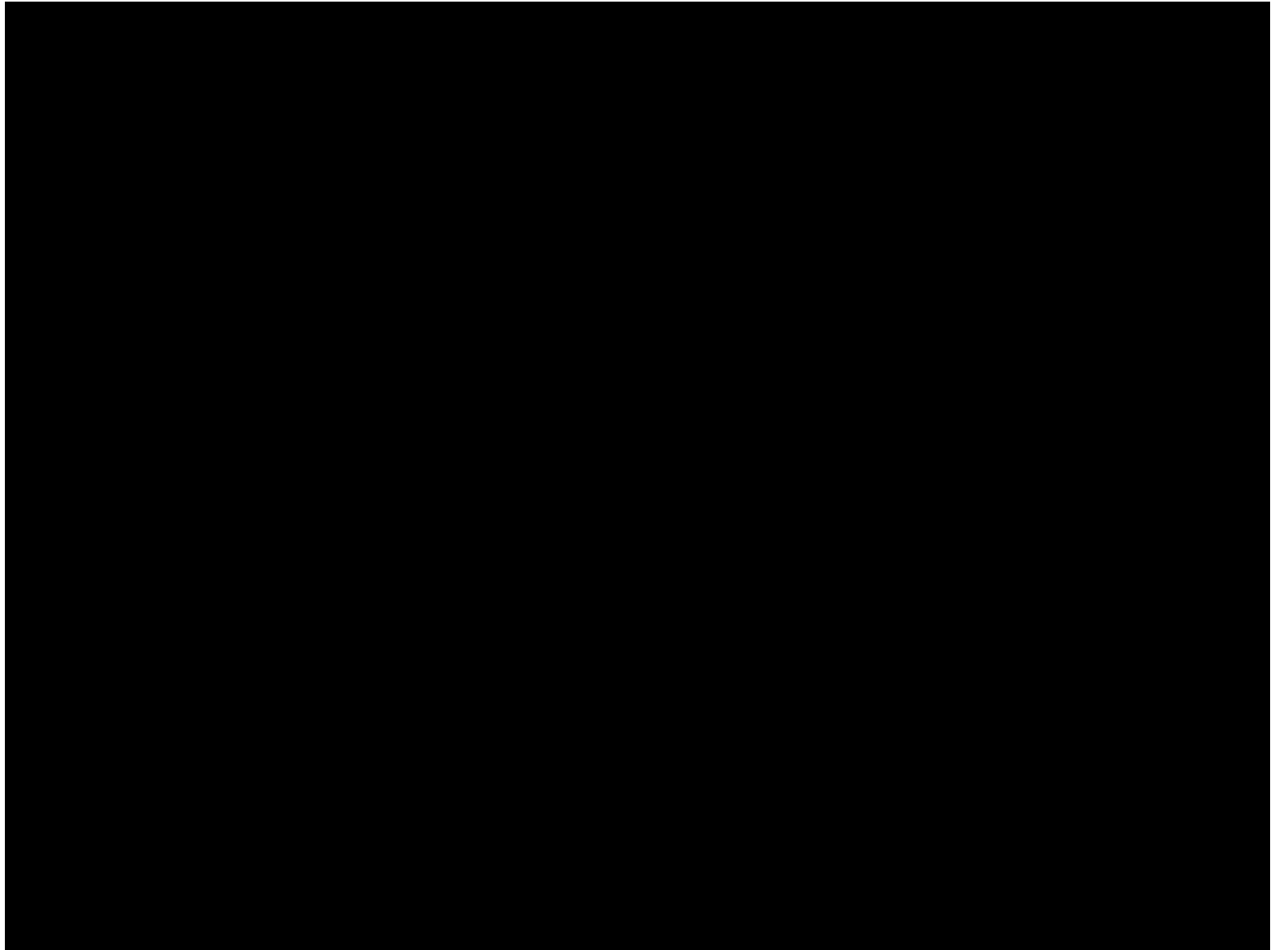
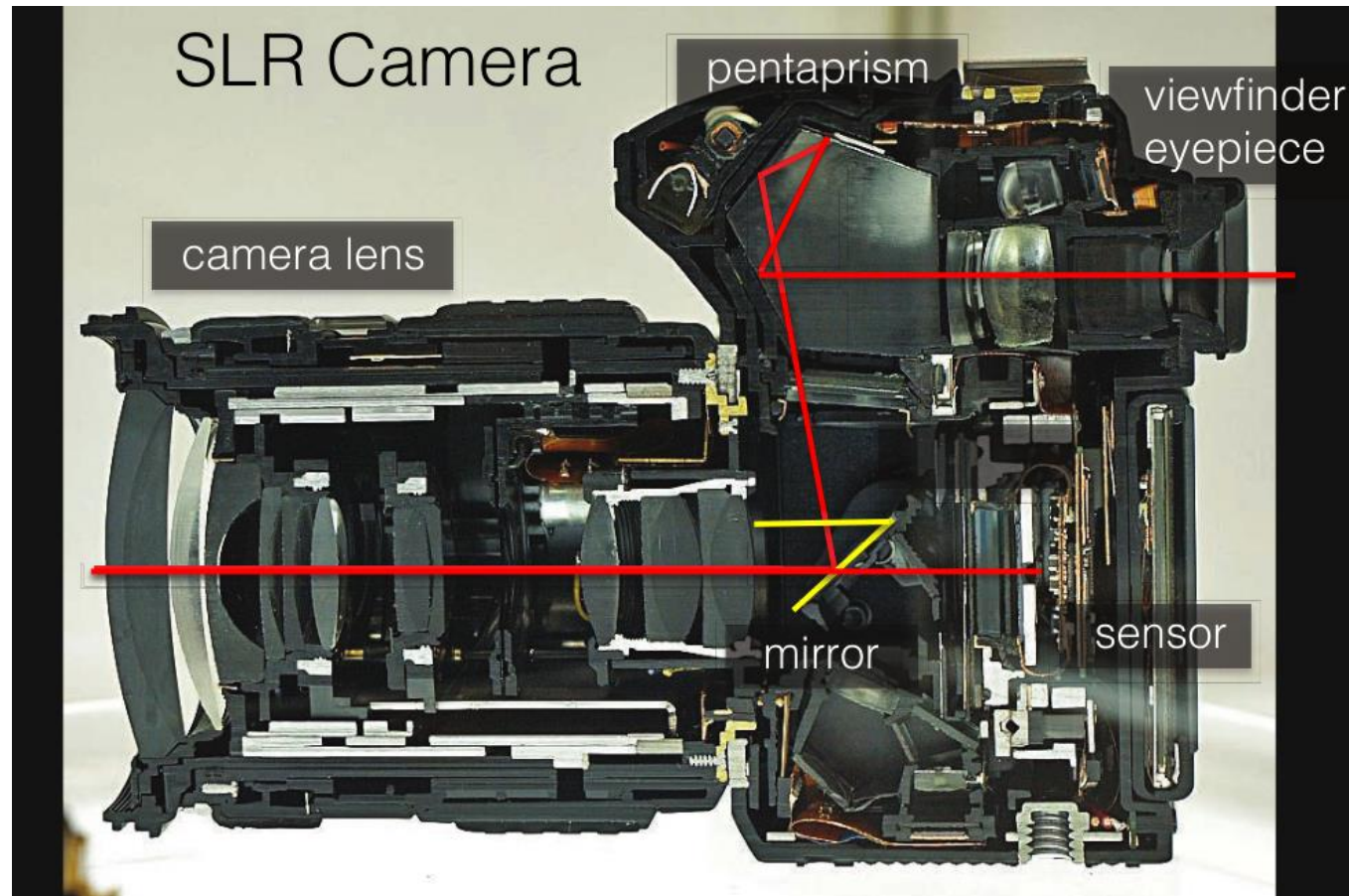


Image Acquisition

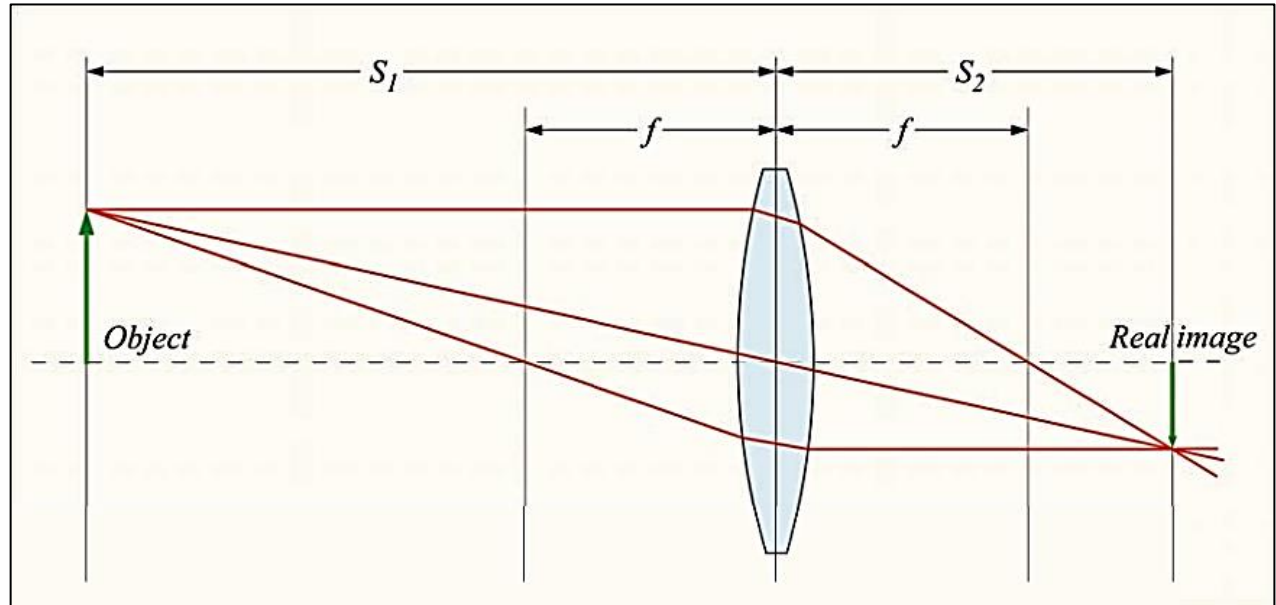
Inside Camera



Some important part inside the Single-lens reflex (SLR) camera

Image Acquisition

Inside Camera: Lens



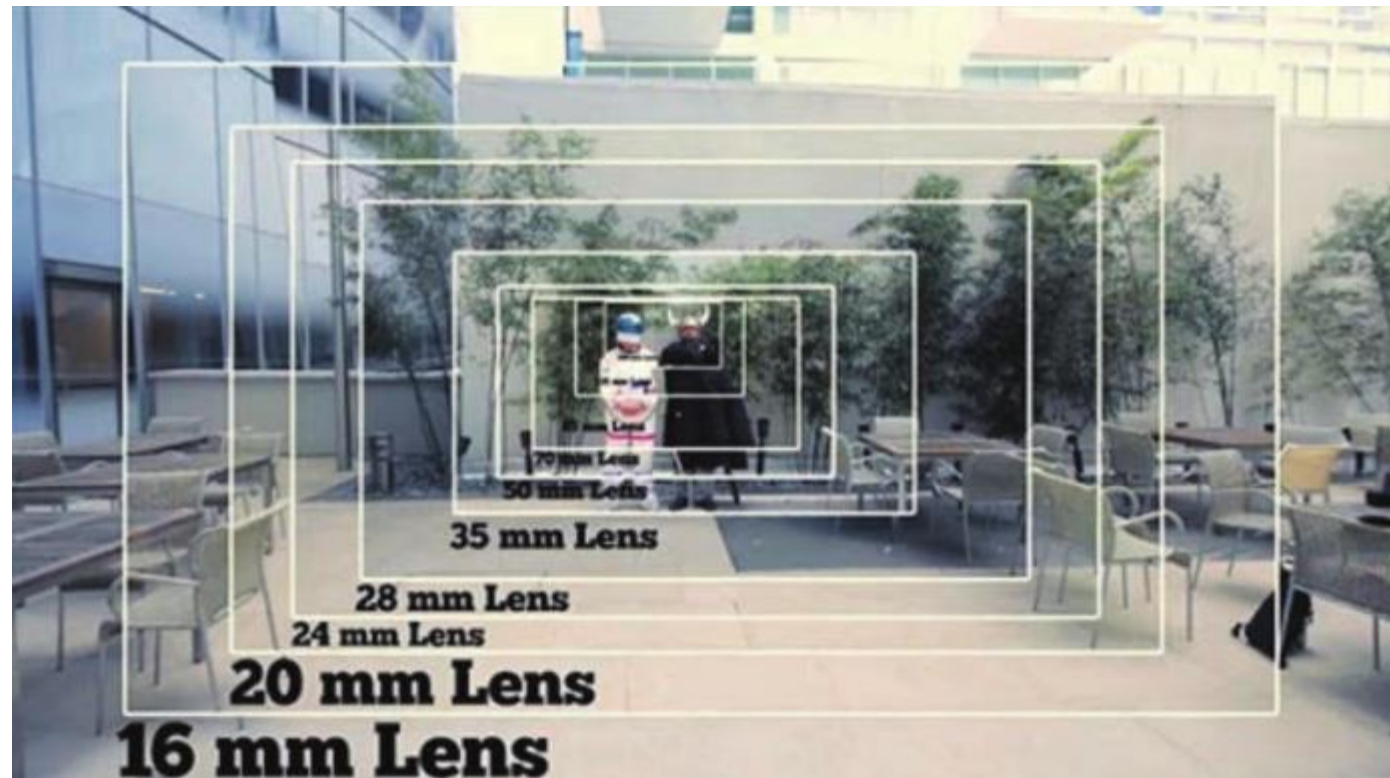
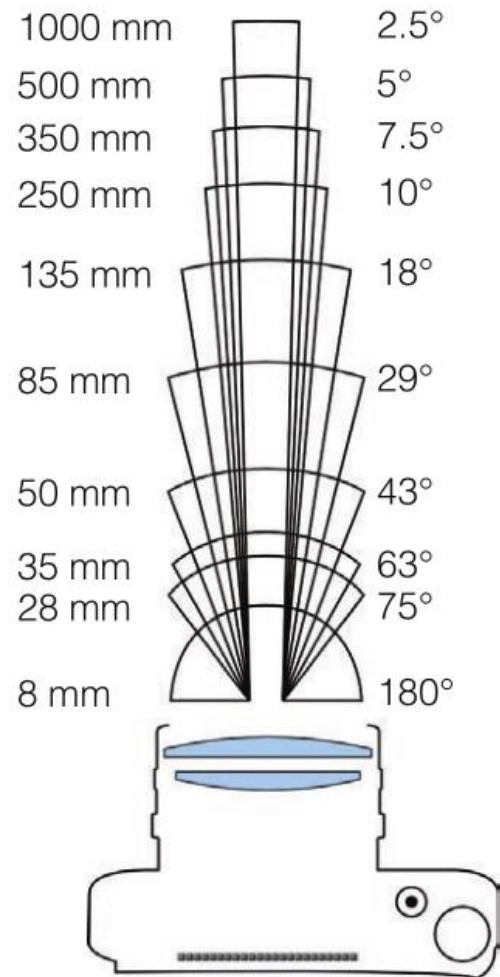
$$\frac{1}{f} = \frac{1}{S_1} + \frac{1}{S_2}$$

Lens maker's equation

$$M = -\frac{S_2}{S_1} = \frac{f}{f - S_1}$$

Magnification

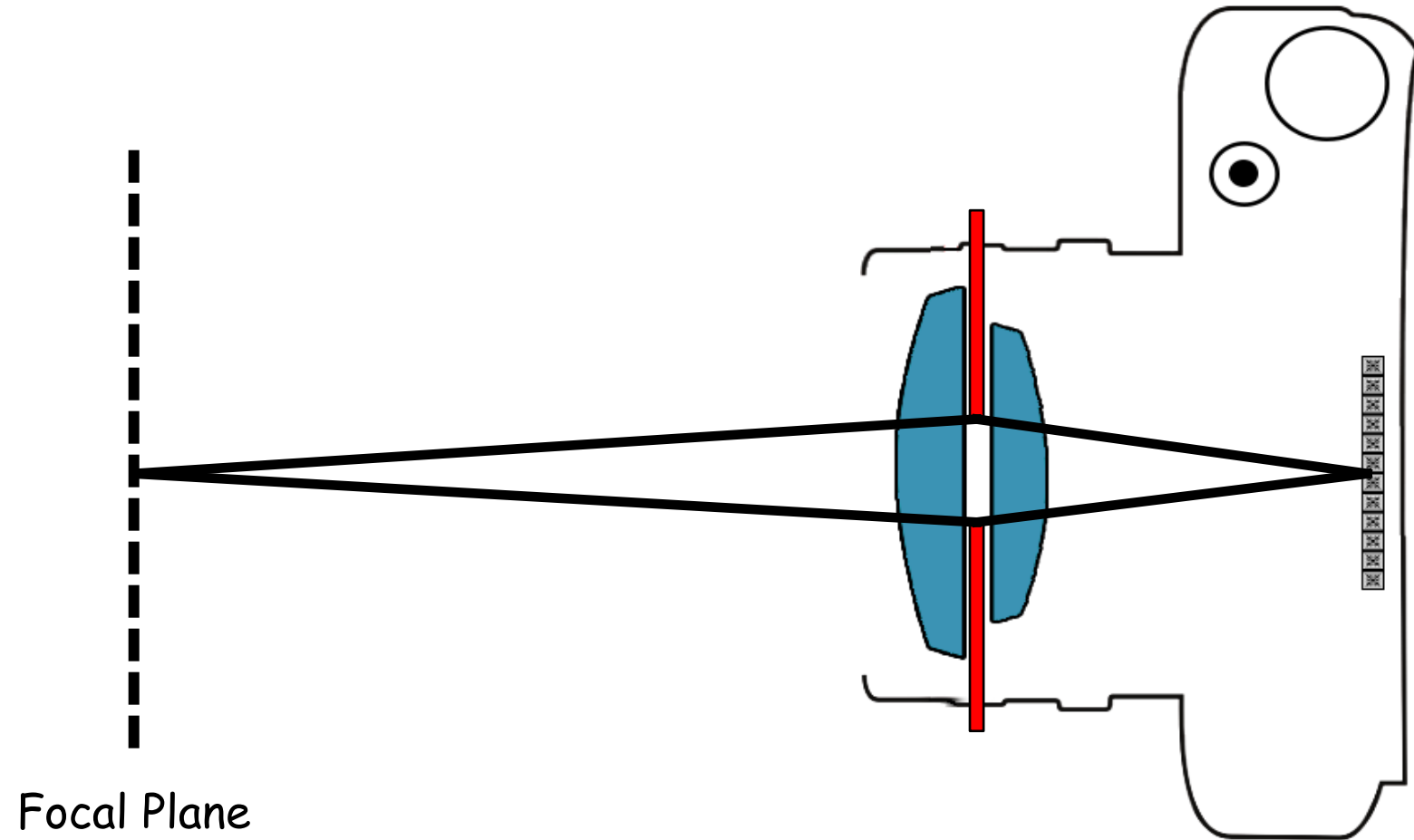
Field of View (FOV)



Focal Length VS Field of View

Image Acquisition

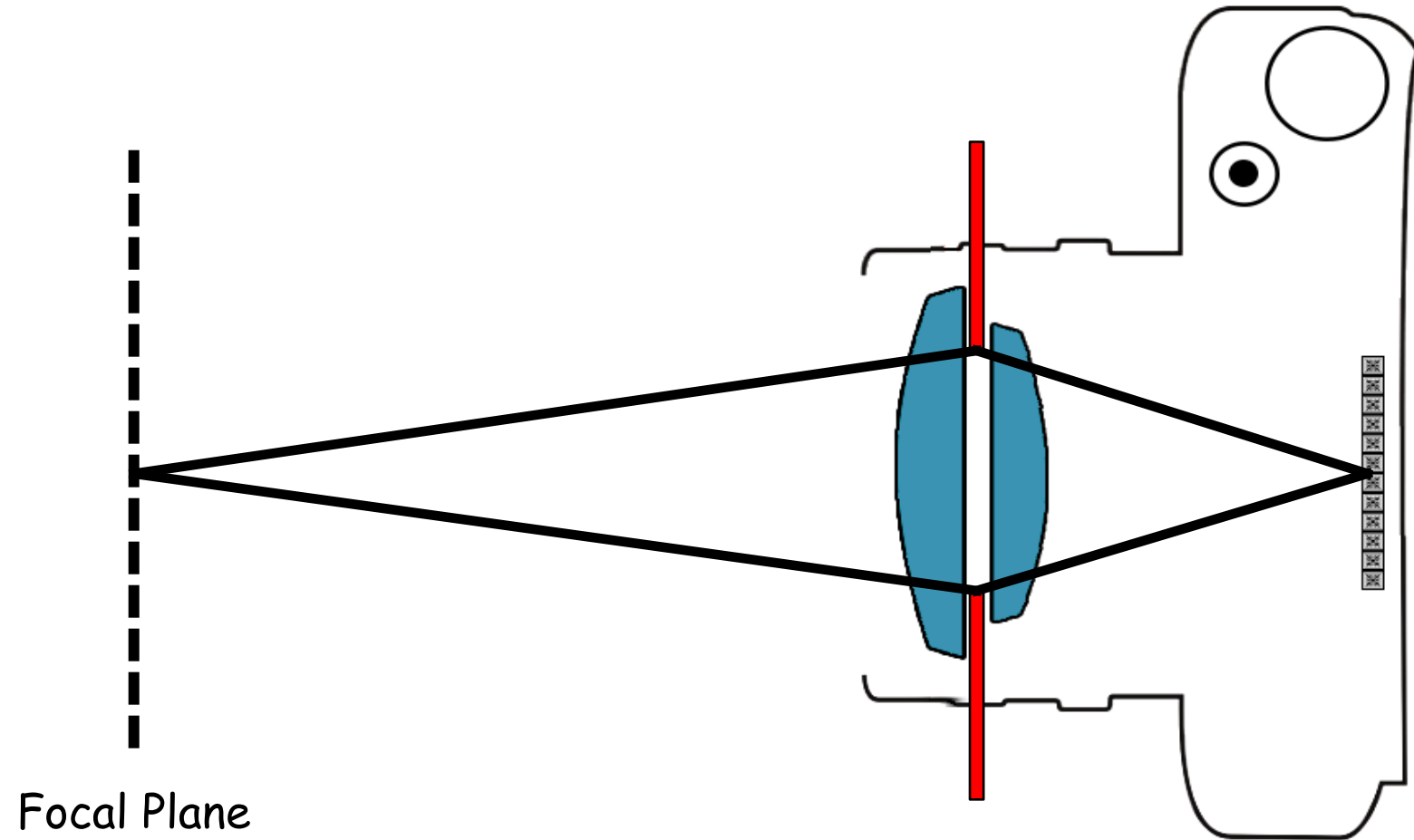
Inside Camera: **Aperture**



Aperture control "amount of light"

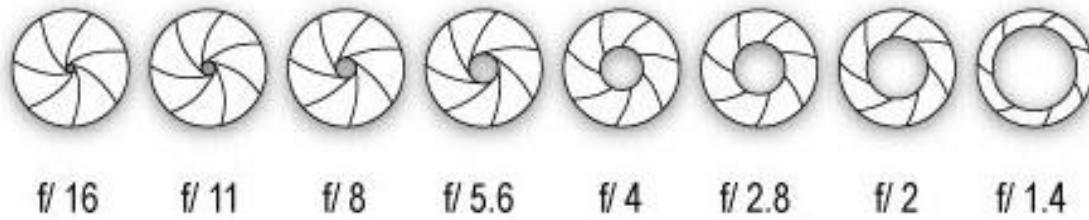
Image Acquisition

Inside Camera: **Aperture**



Aperture control "amount of light"

Inside Camera: Aperture

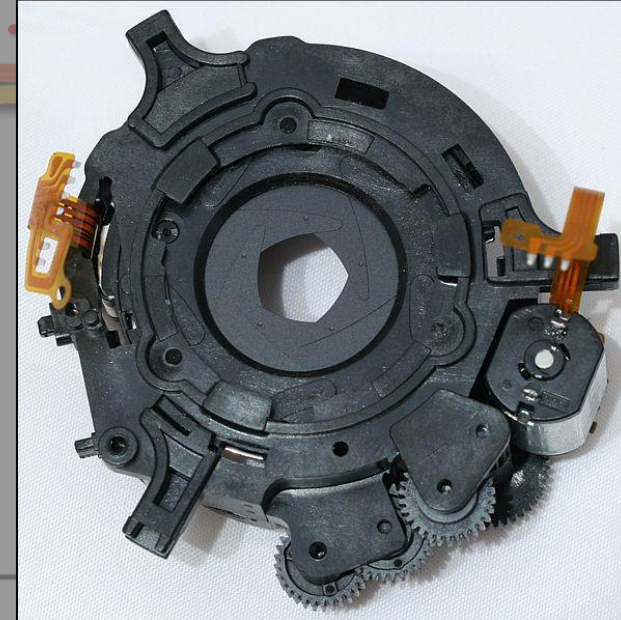


Unit: f-number

$$N = f / D$$

f: focal length

D: diameter



Focal Plane

Apert



Image Acquisition

Inside Camera: Aperture

Out of Focus "Blur"

Focal Plane

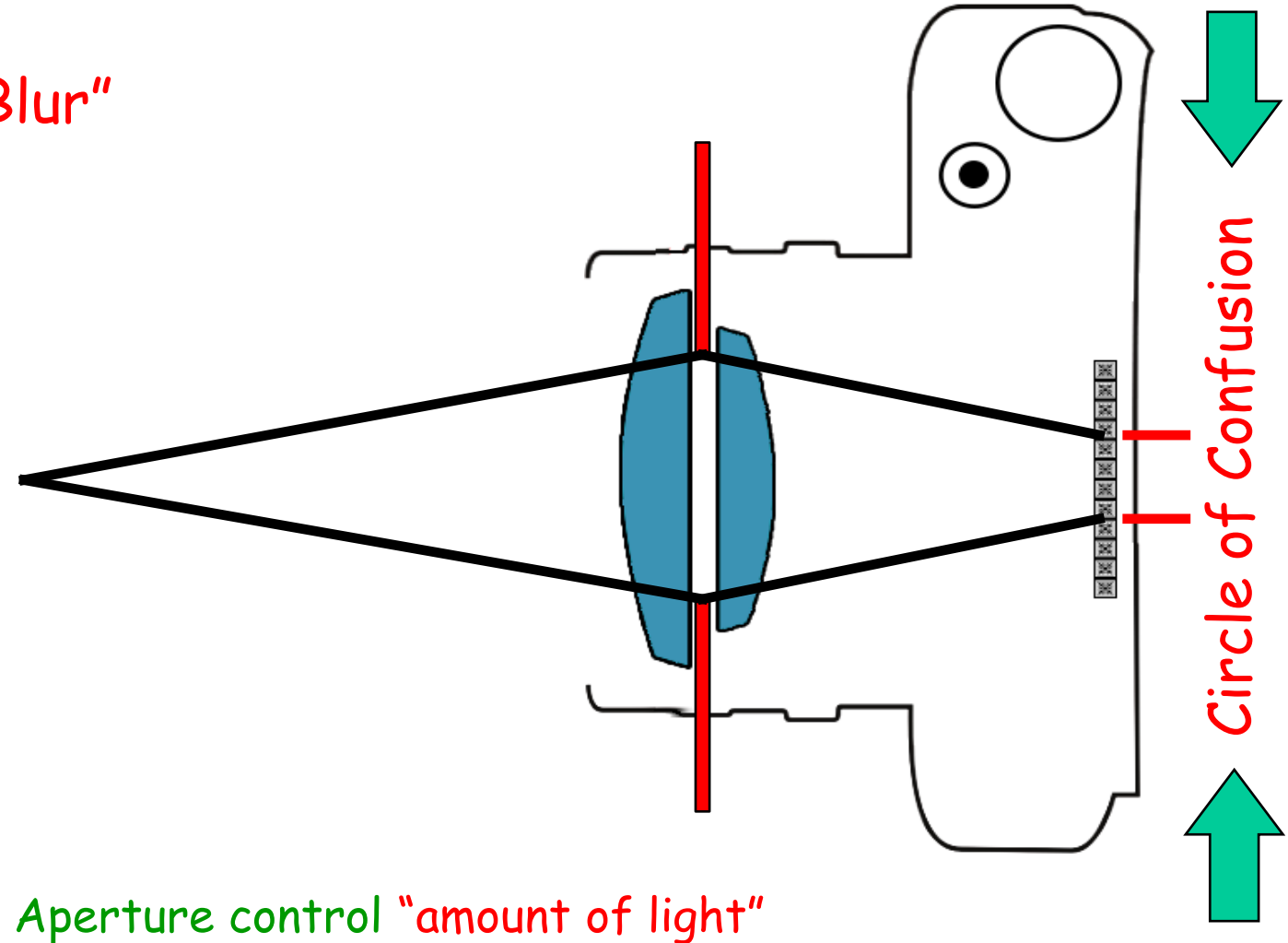


Image Acquisition

Inside Camera: Aperture

Out of Focus "Blur"

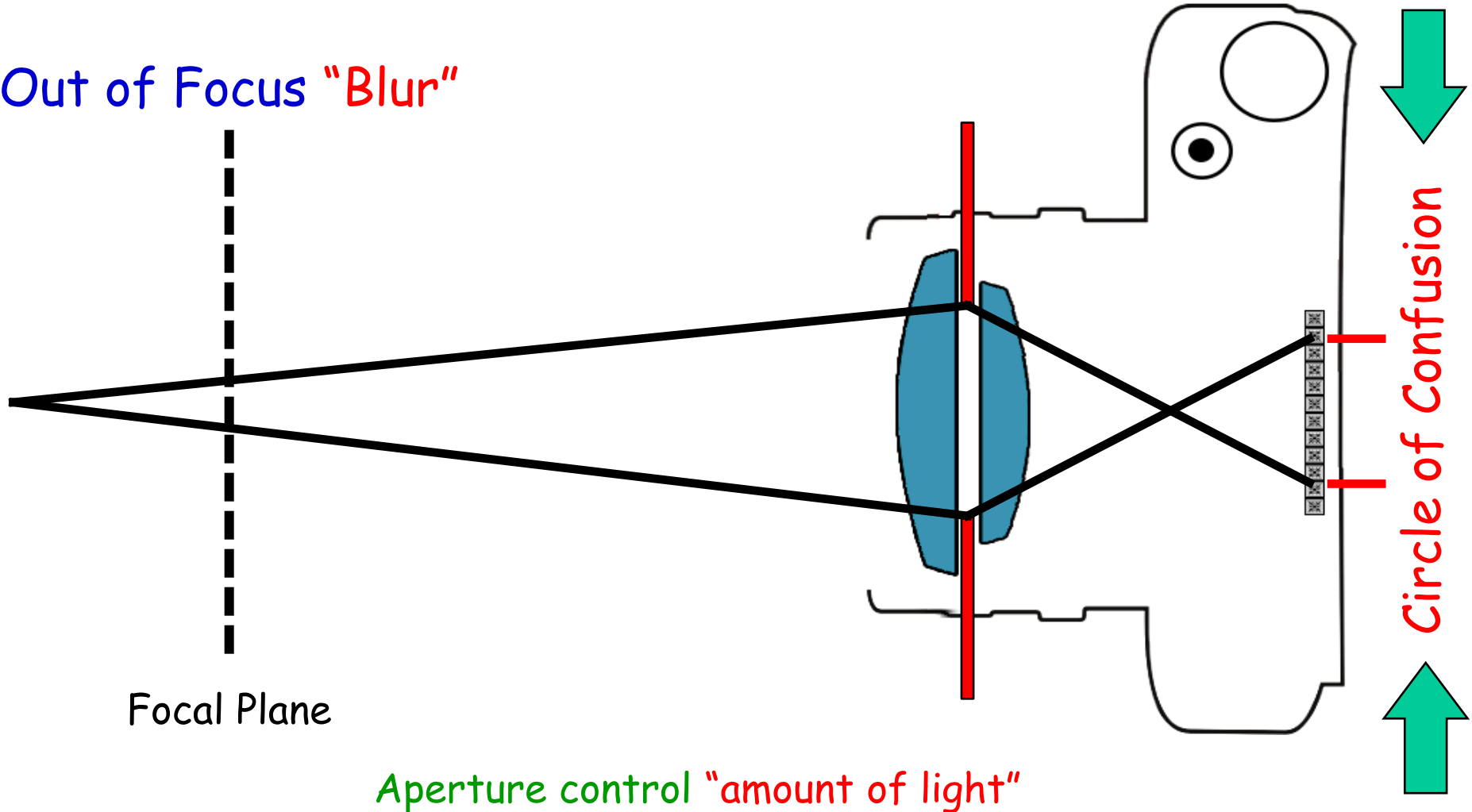


Image Acquisition

Inside Camera: Aperture

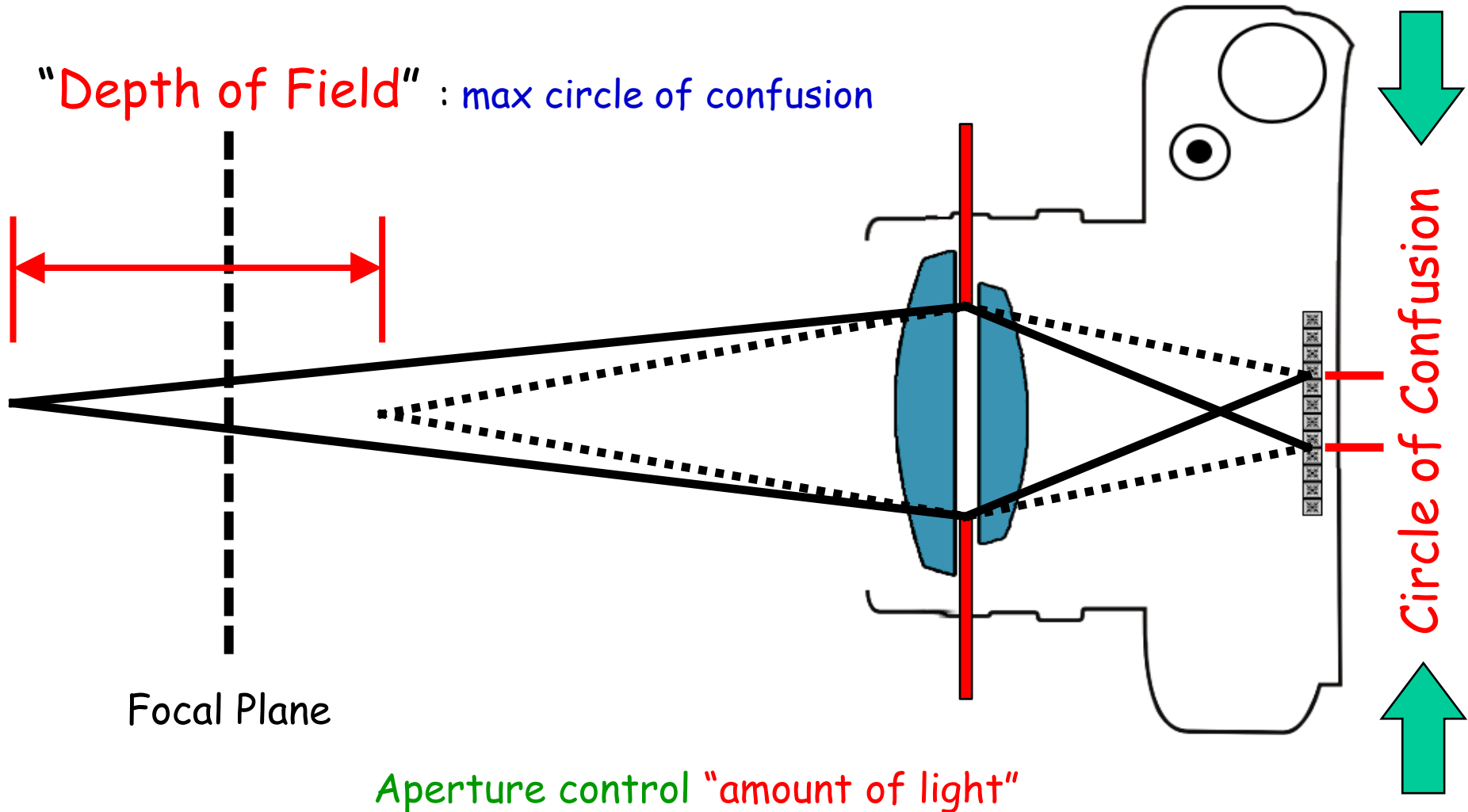


Image Acquisition

Inside



aperture....f 1.8
shutter.....1/500
ISO.....100
distance...~3ft

aperture....f 4
shutter.....1/125
ISO.....100
distance...~3ft

aperture....f 8
shutter.....1/40
ISO.....125
distance...~3ft



Circle of Confusion



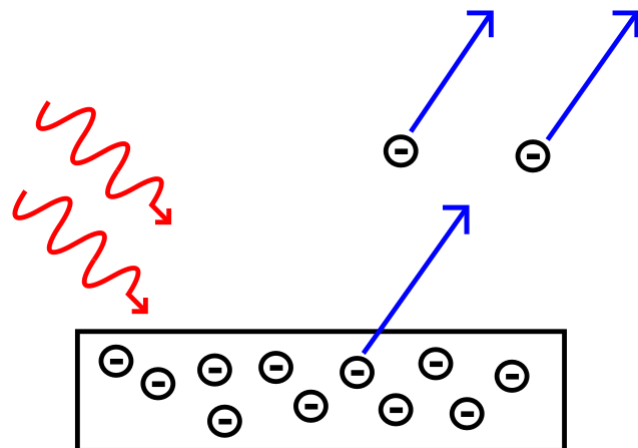
Aperture control "amount of light"

Image Acquisition

Inside Camera: **Sensor**

Micro lens: focus light on photodiode

Color Filter: select color channel



Photoelectric effect Energy transfer from “Light” to “Electron”

Anatomy of the Active Pixel Sensor Photodiode

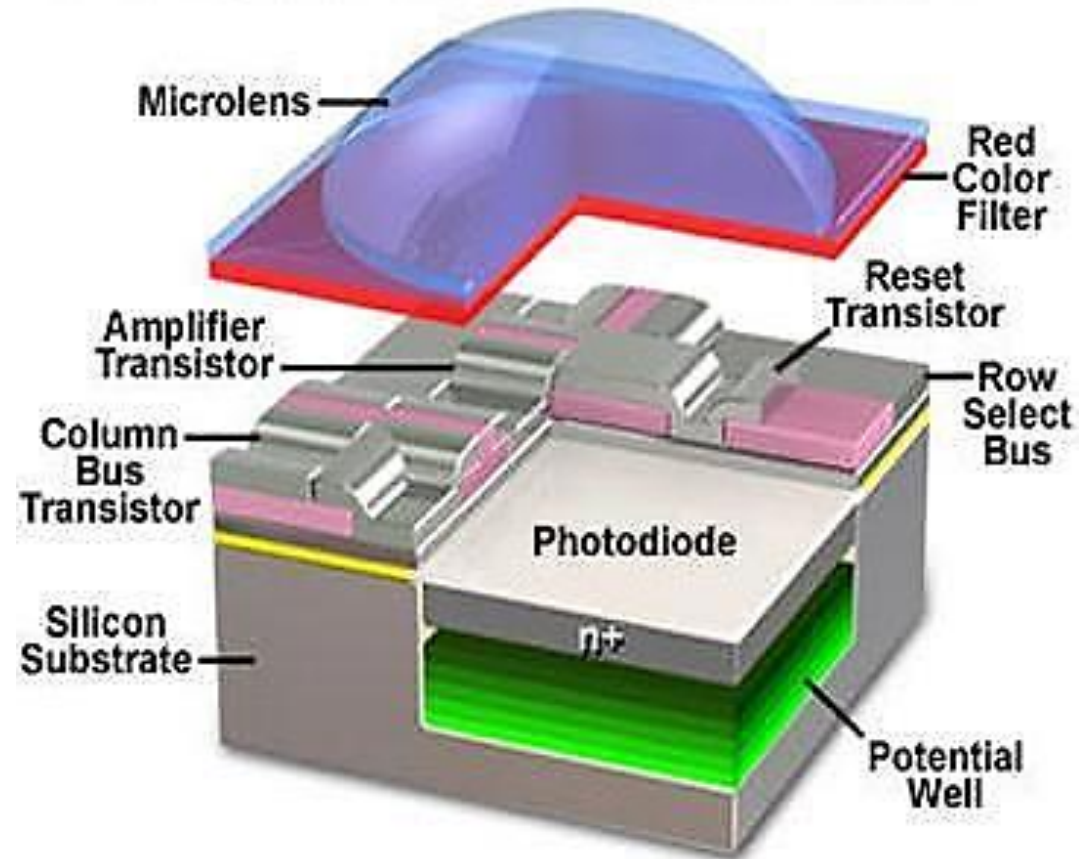


Image Acquisition

He used **twice** as many **green** elements as **red** or **blue**

to mimic the physiology of the human eye.

(The eye is most sensitive to green light (555 nm))

Inside Camera: Sensor

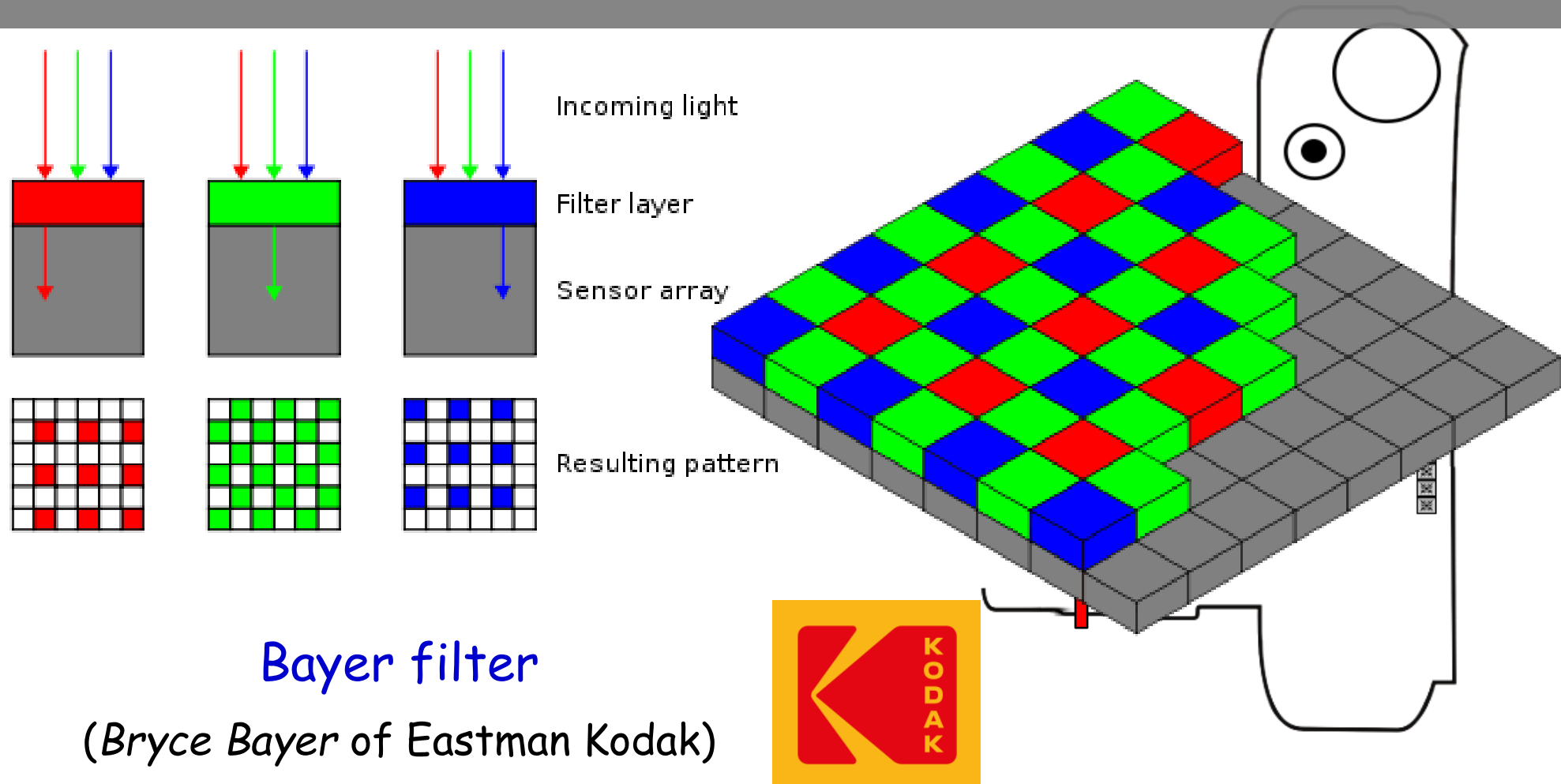
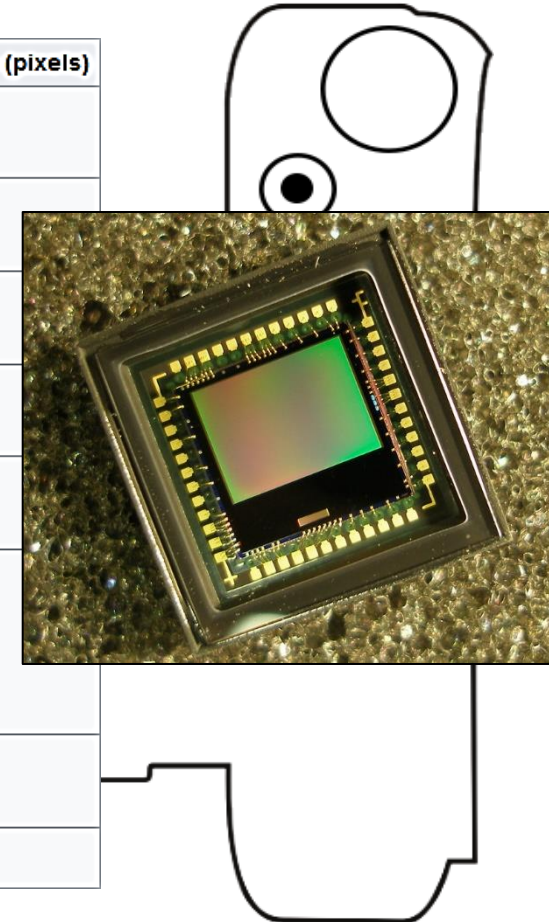


Image Acquisition

Inside Camera: Sensor

Image	Name	Description	Pattern size (pixels)
	Bayer filter	Very common RGB filter. With one blue, one red, and two green.	2×2
	RGBE filter	Bayer-like with one of the green filters modified to "emerald"; used in a few Sony cameras.	2×2
	CYYM filter	One cyan, two yellow, and one magenta; used in a few cameras of Kodak.	2×2
	CYGM filter	One cyan, one yellow, one green, and one magenta; used in a few cameras.	2×2
	RGBW Bayer	Traditional RGBW similar to Bayer and RGBE patterns.	2×2
	RGBW #1	Three example RGBW filters from Kodak, with 50% white. (See <i>Bayer filter#Modifications</i>)	4×4
	RGBW #2		
	RGBW #3		2×4
	X-Trans	Fujifilm-specific RGB matrix filter, with a large pattern, studied for diminishing Moiré effect.	6×6



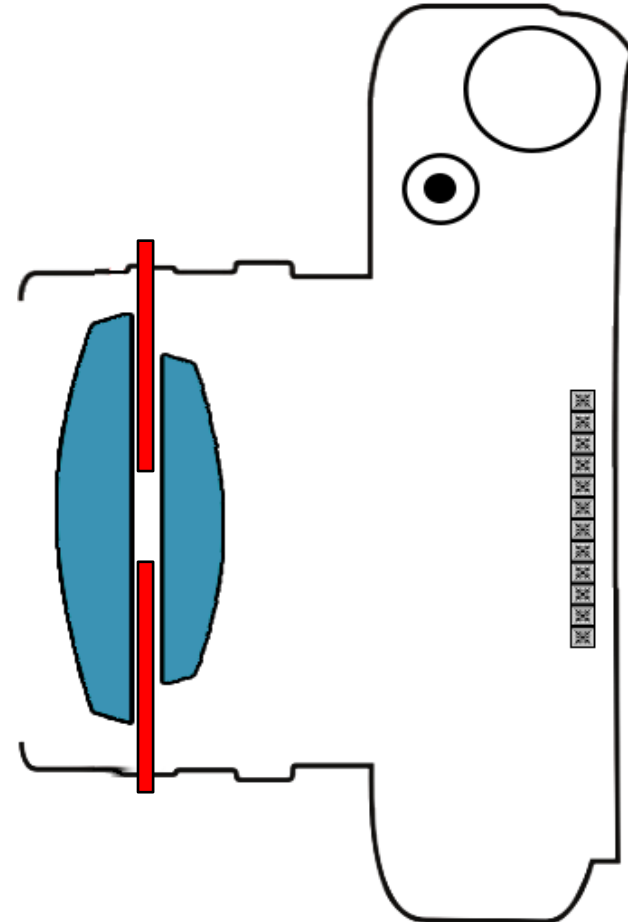
List of color filter array

Image Acquisition

Inside Camera: **Sensor**



Buckets (Sensor)

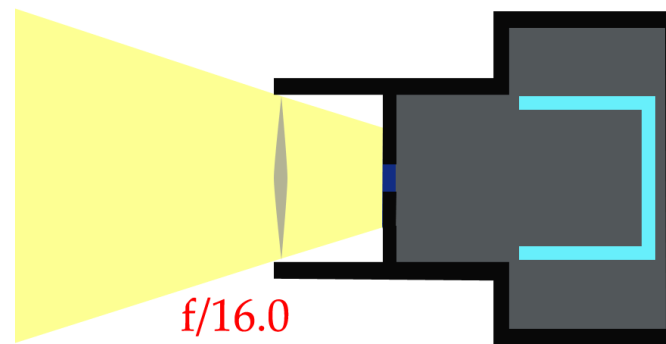
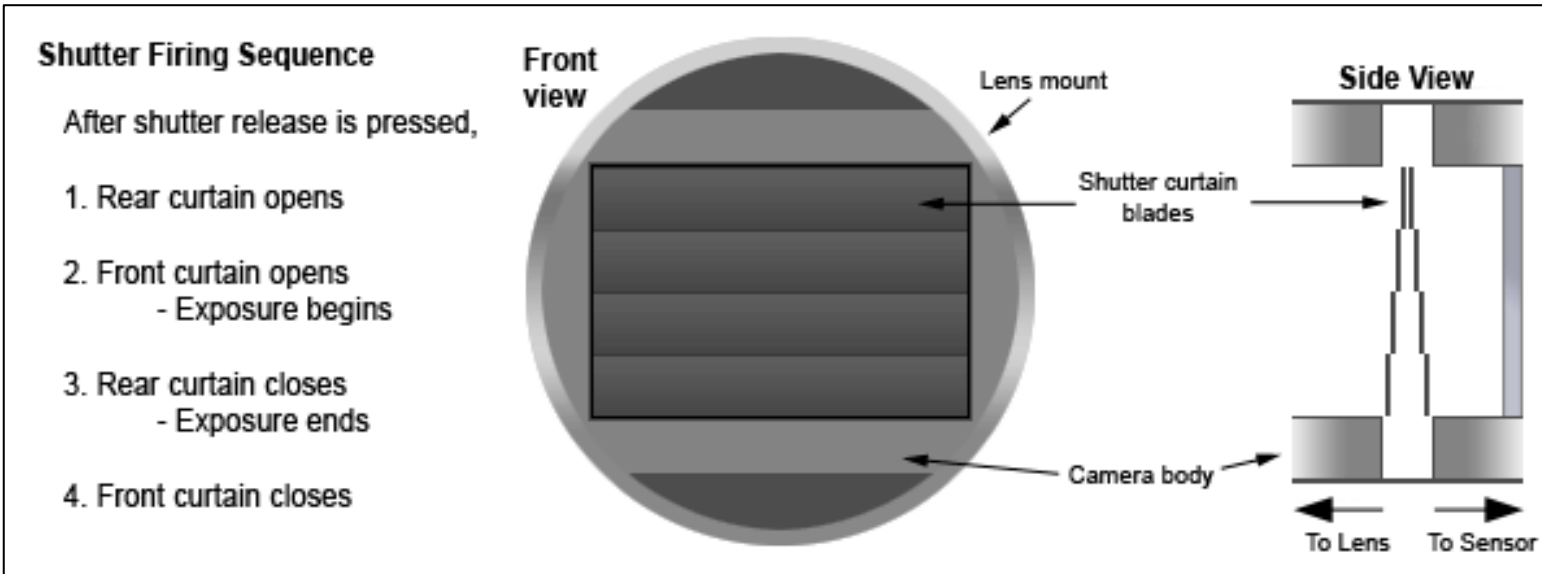


Note: Great concept idea from Stanford Univ.

“Collect Photons like a Bucket”

Image Acquisition

Mechanical: Exposure (Shutter Speed)



Exposure: amount of light per unit area

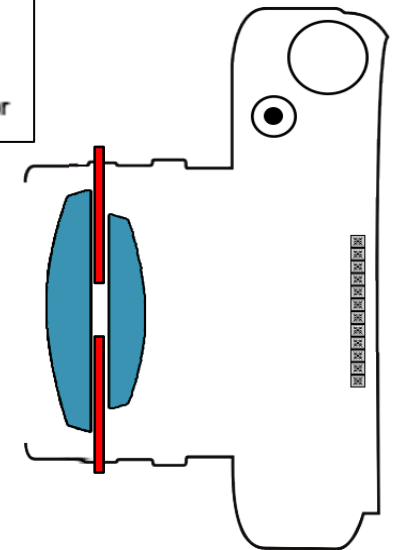


Image Acquisition

Mechanical: Exposure (Shutter Speed)



1/4 sec, f/3.3



2 sec, f/6.3

f/16.0

Exposure: amount of light per unit area

Image Acquisition

Digital: ISO (Film Speed)

Sensitivity of sensor to light
(Digital Gain)

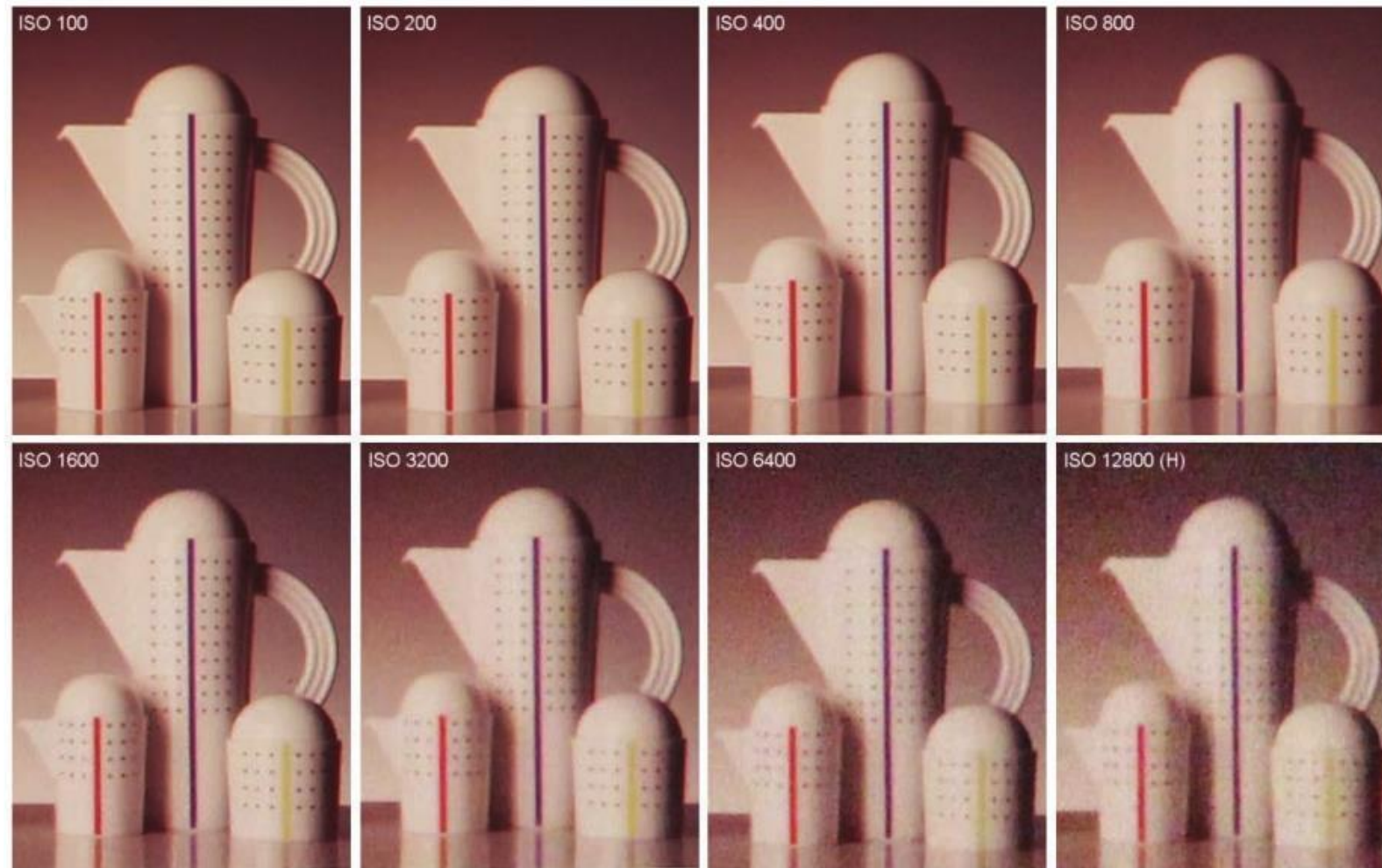


Image Acquisition

Camera Guide

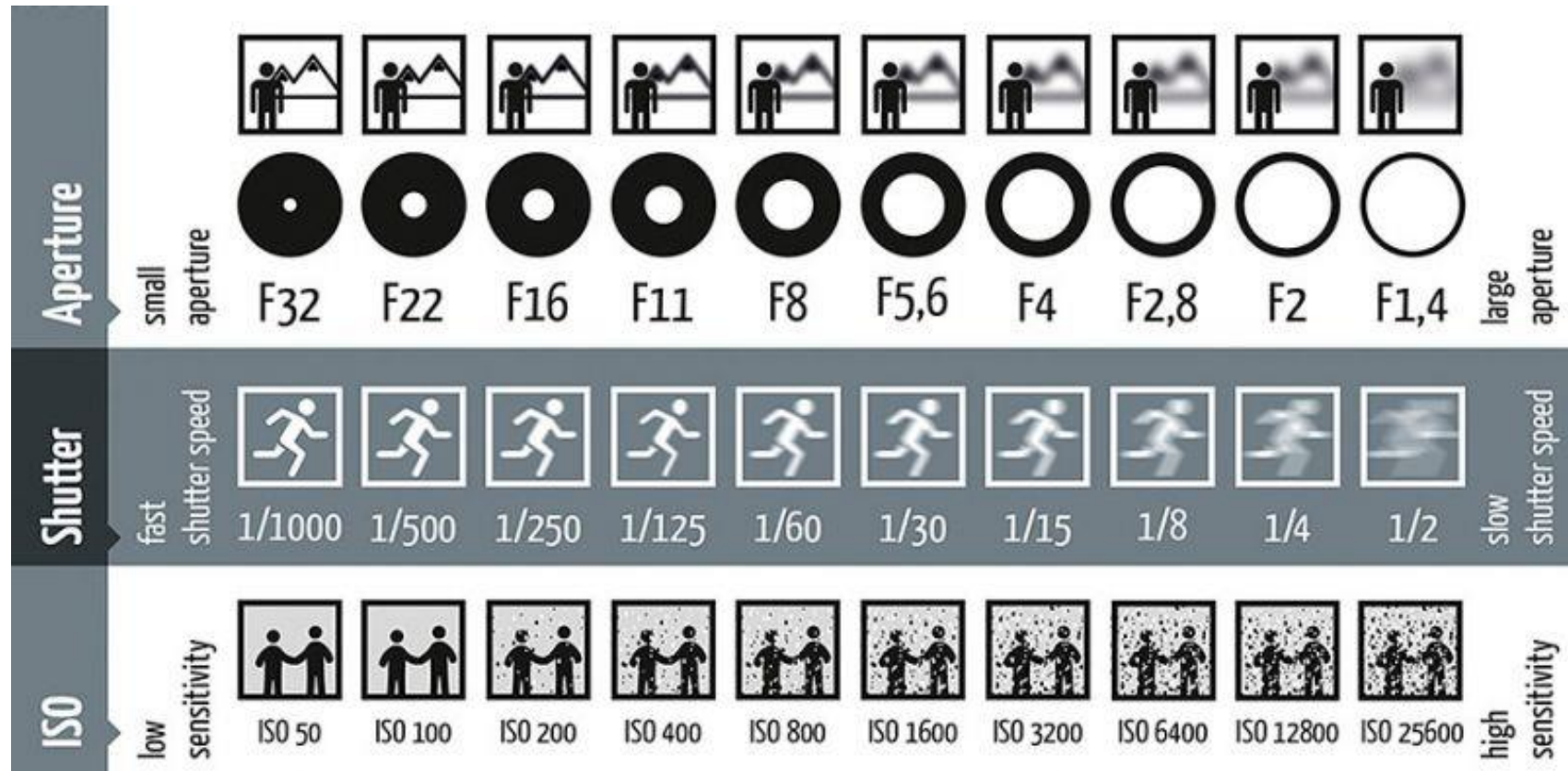
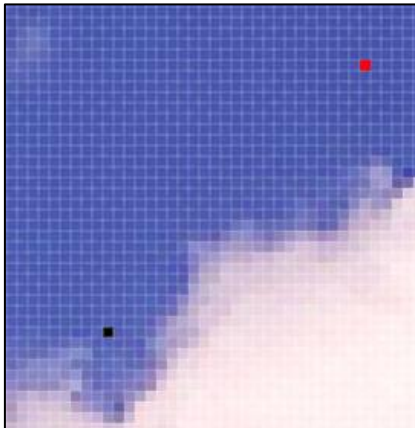
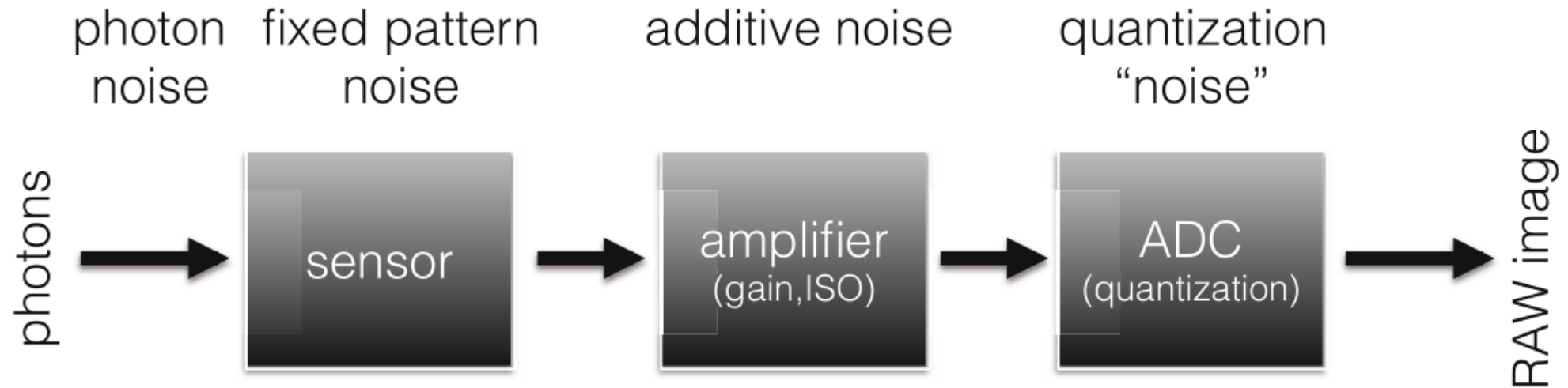


Image Acquisition

From Photons to Raw Image



Dead pixel:

permanently damaged pixel that does not receive any power

Stuck pixel:

always receive power, but not permanent like dead pixels.
they might disappear overtime.

Image Acquisition

Lens as Optical Filter

Away from focal plane: Out of Focus "Blur"

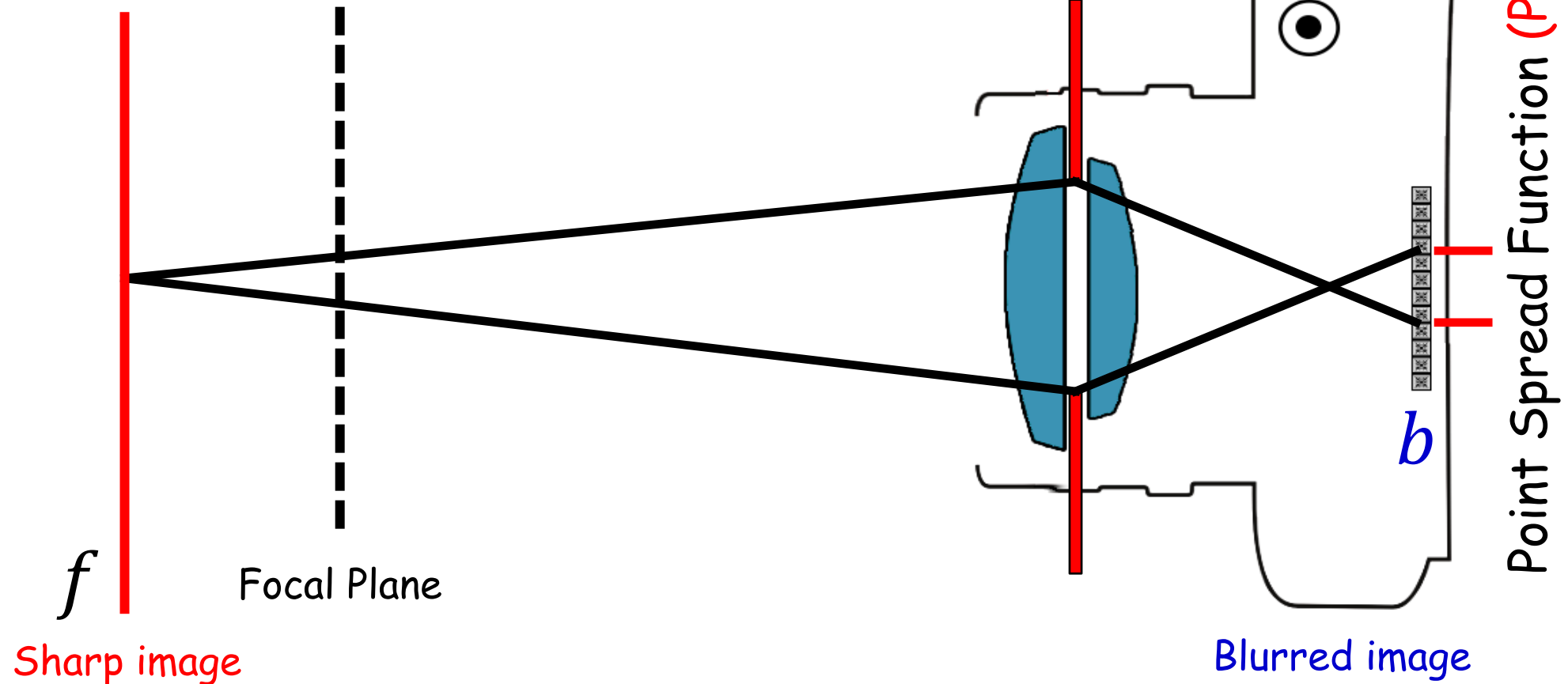


Image Acquisition

Lens as Optical Filter

Shift-invariant convolution

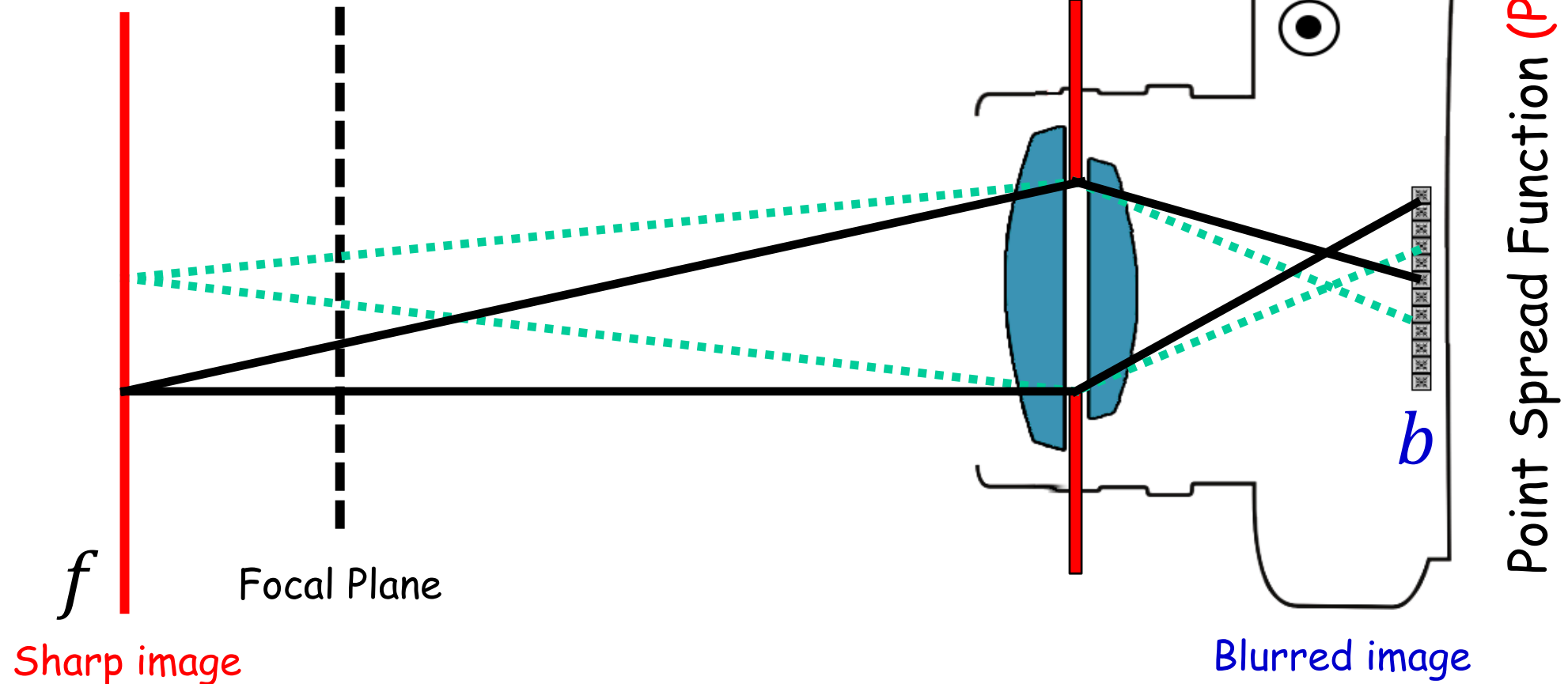
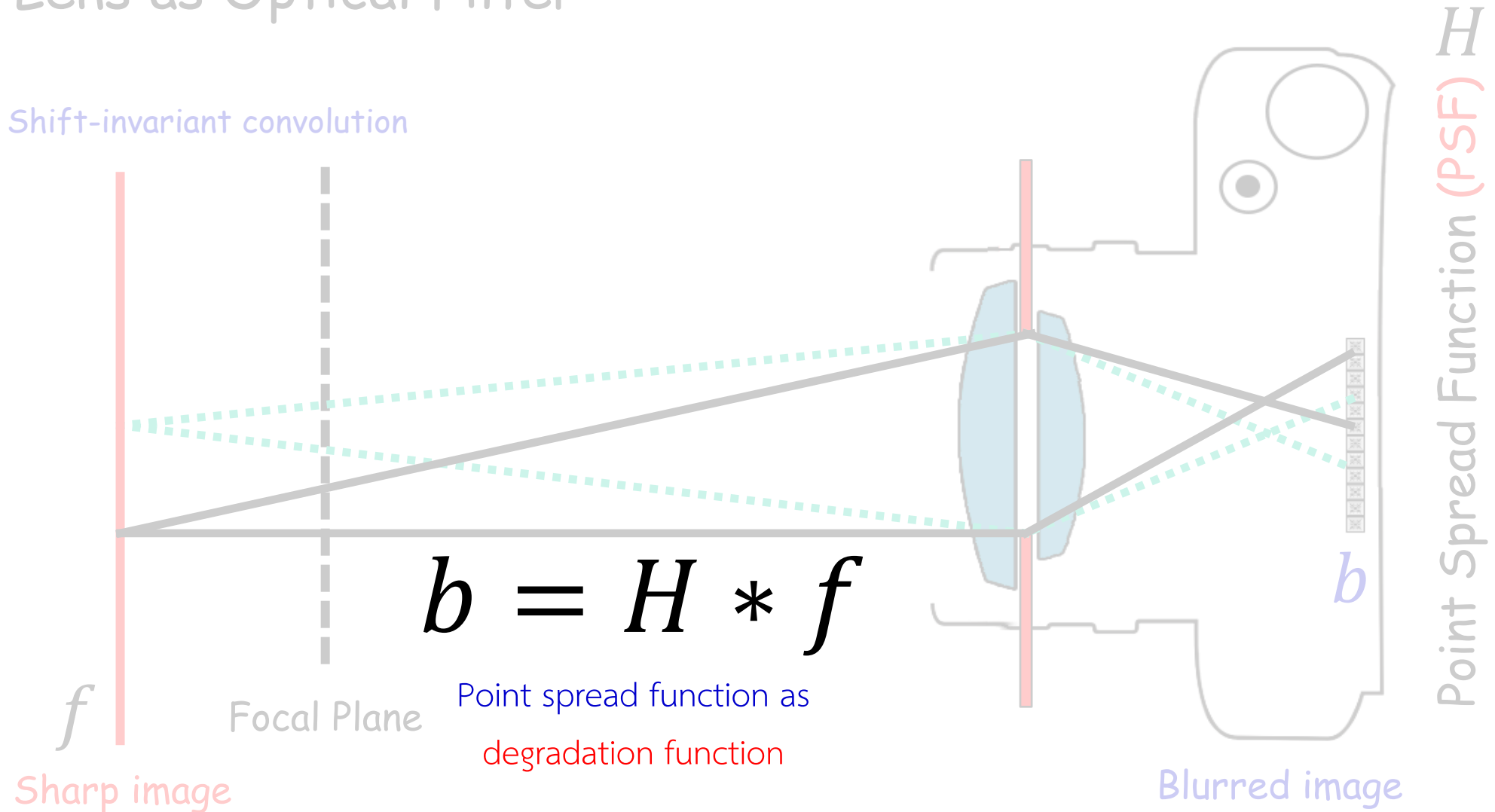


Image Acquisition

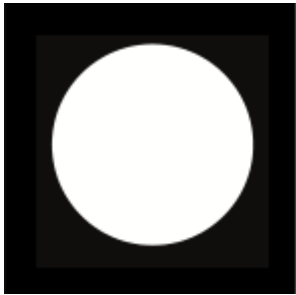
Lens as Optical Filter

Shift-invariant convolution

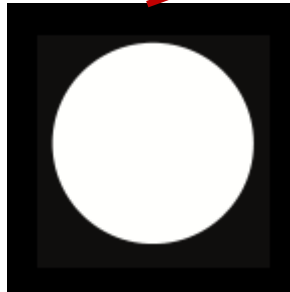


Lens as Optical Filter

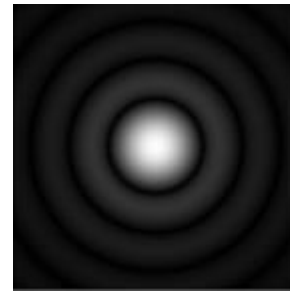
- Point Spread Function (PSF)
- Optical Transfer function (OTF)
- Modulation Transfer function (MTF)



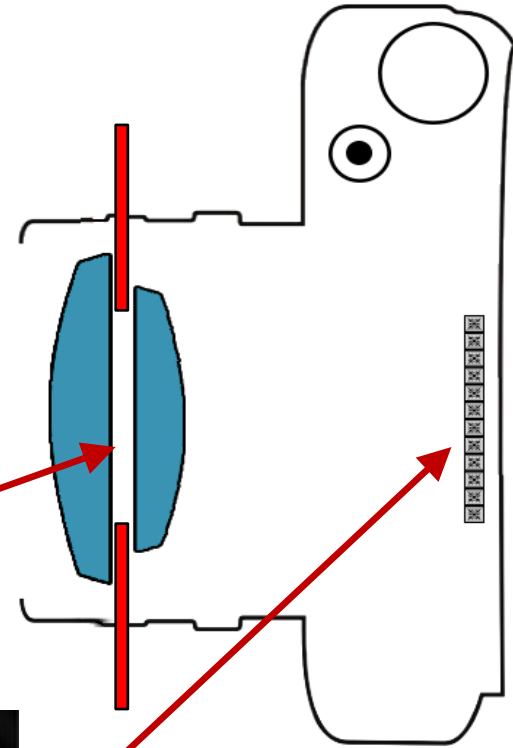
$$\text{MTF} = |\text{OTF}|$$



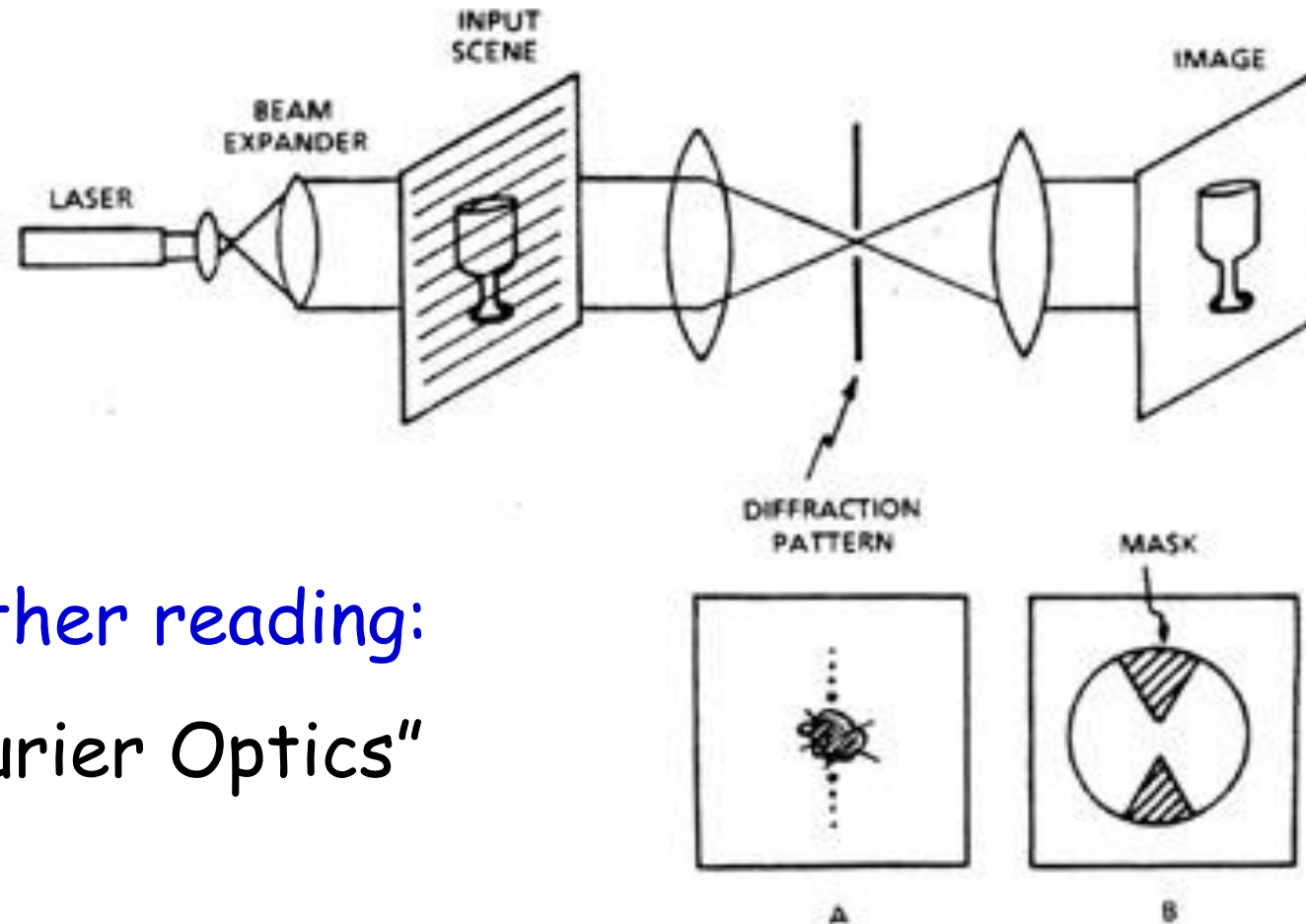
$$\text{OTF} = \mathcal{F}\{\text{PSF}\}$$



PSF



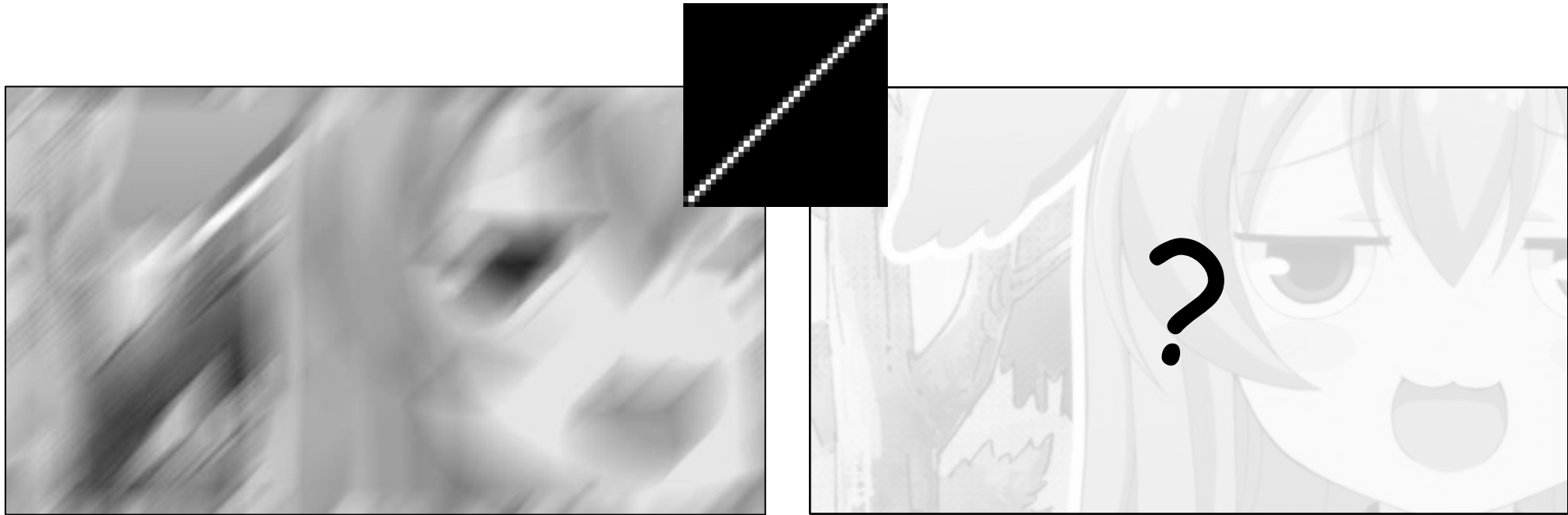
Lens as Optical Filter



Further reading:
"Fourier Optics"

Image Acquisition

So... that why our image blurry.



$$b = H * f$$

In Image Processing Class: Given measurement $[b]$ and convolution kernel $[H]$, what is $[f]$?

===== Spoil Warning =====

Deblurring: Regularization with Gradient descent



Demonstrate with the following parameters

Step size = 1.0

Number of iterations = 500



Wiener (known noise & signal)



Just as planed !!!

===== Spoil Warning =====

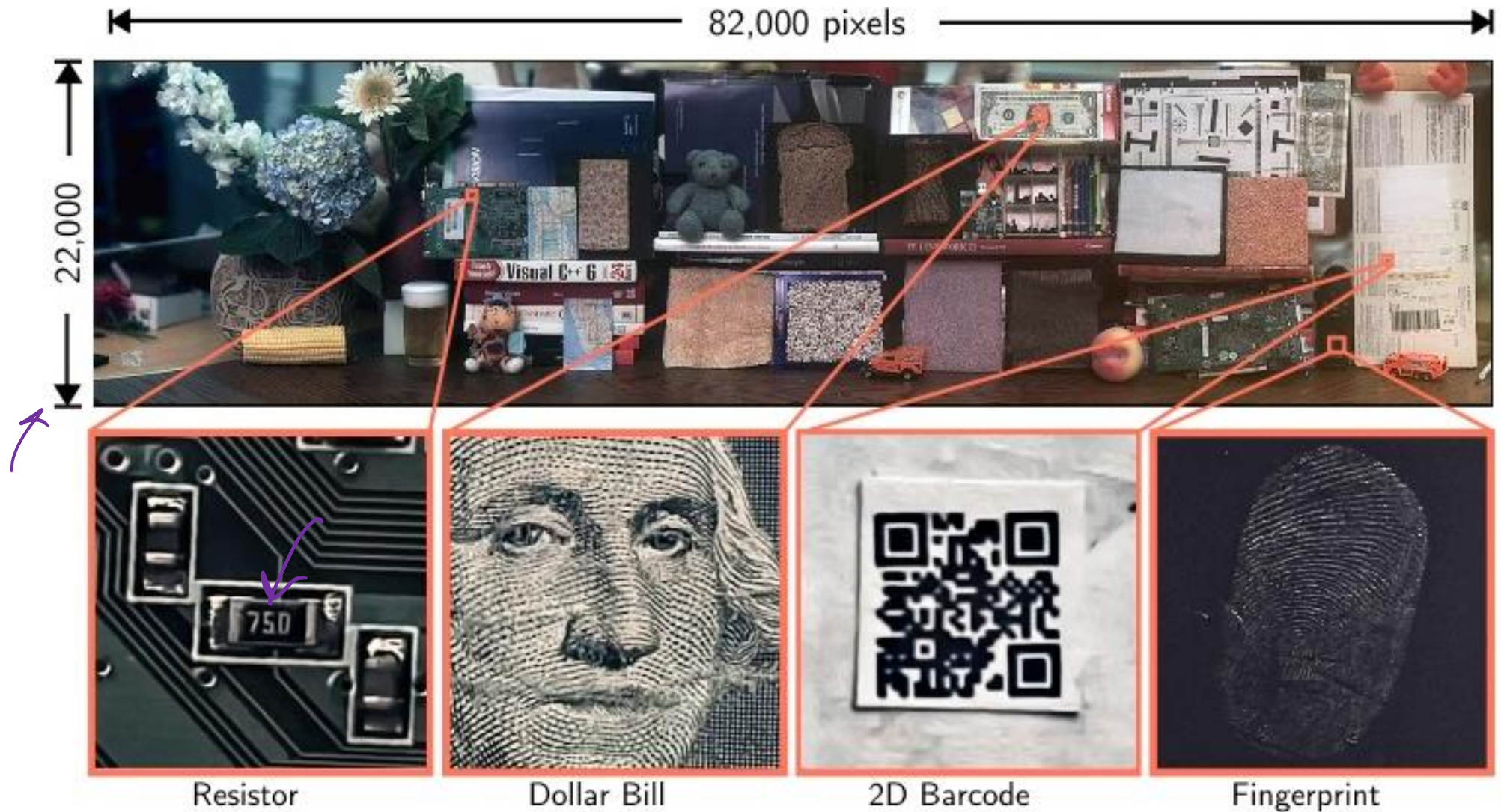
Deblurring: Wiener Filter



Not too bad, have some noise

Need to know Signal & Noise Power

heuristic in real case



Super Video

The gigapixel camera that takes **ten frames per second** will generate ten gigabytes of data every second !!!

US\$100,000 (Prototype) → US\$1,000 (Largescale)

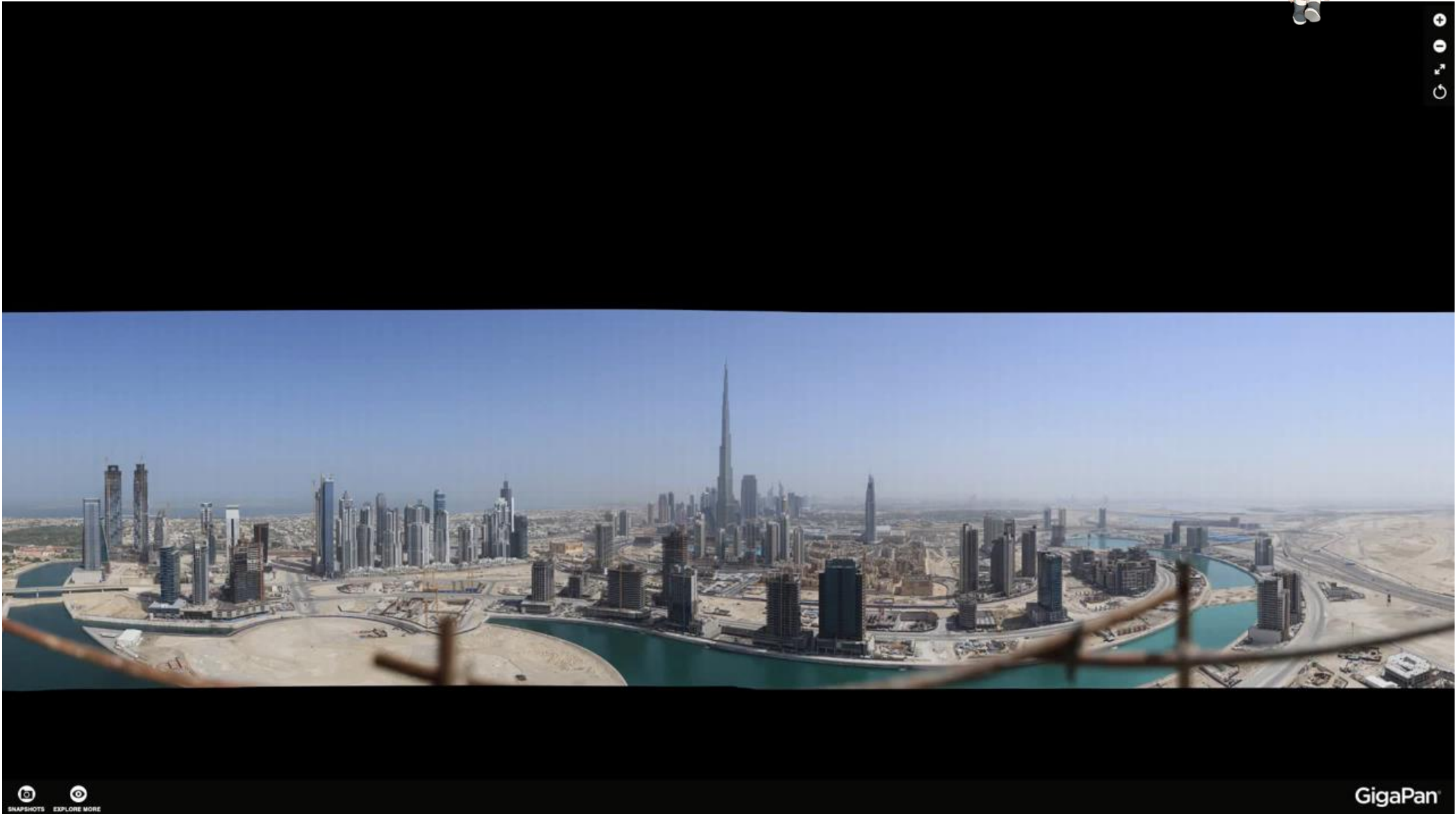
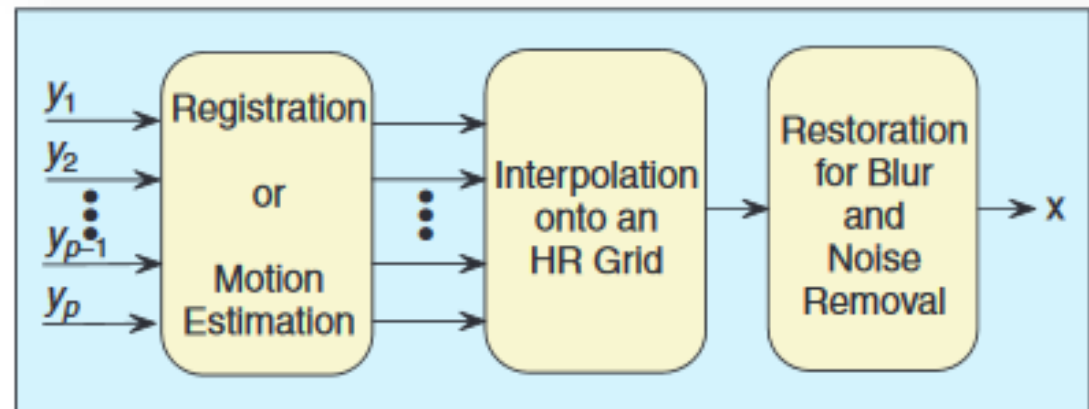
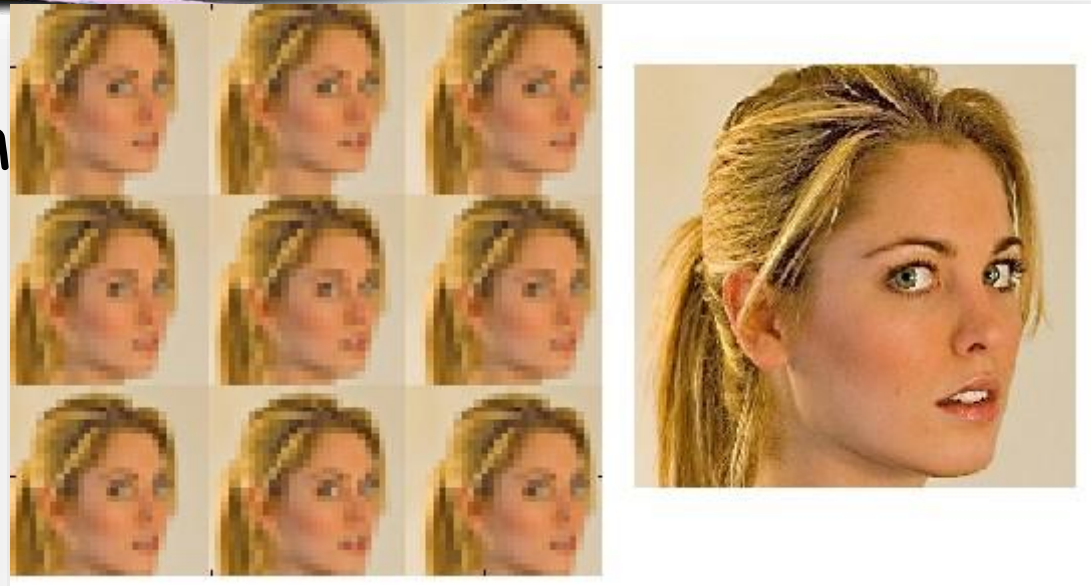


Image Restoration

Fun things to do with im



Design adaptive filter to deblur moving objects



▲ 6. Scheme for super resolution.

Super resolution image reconstruction

Image Restoration

Fun things to do with image restoration tool

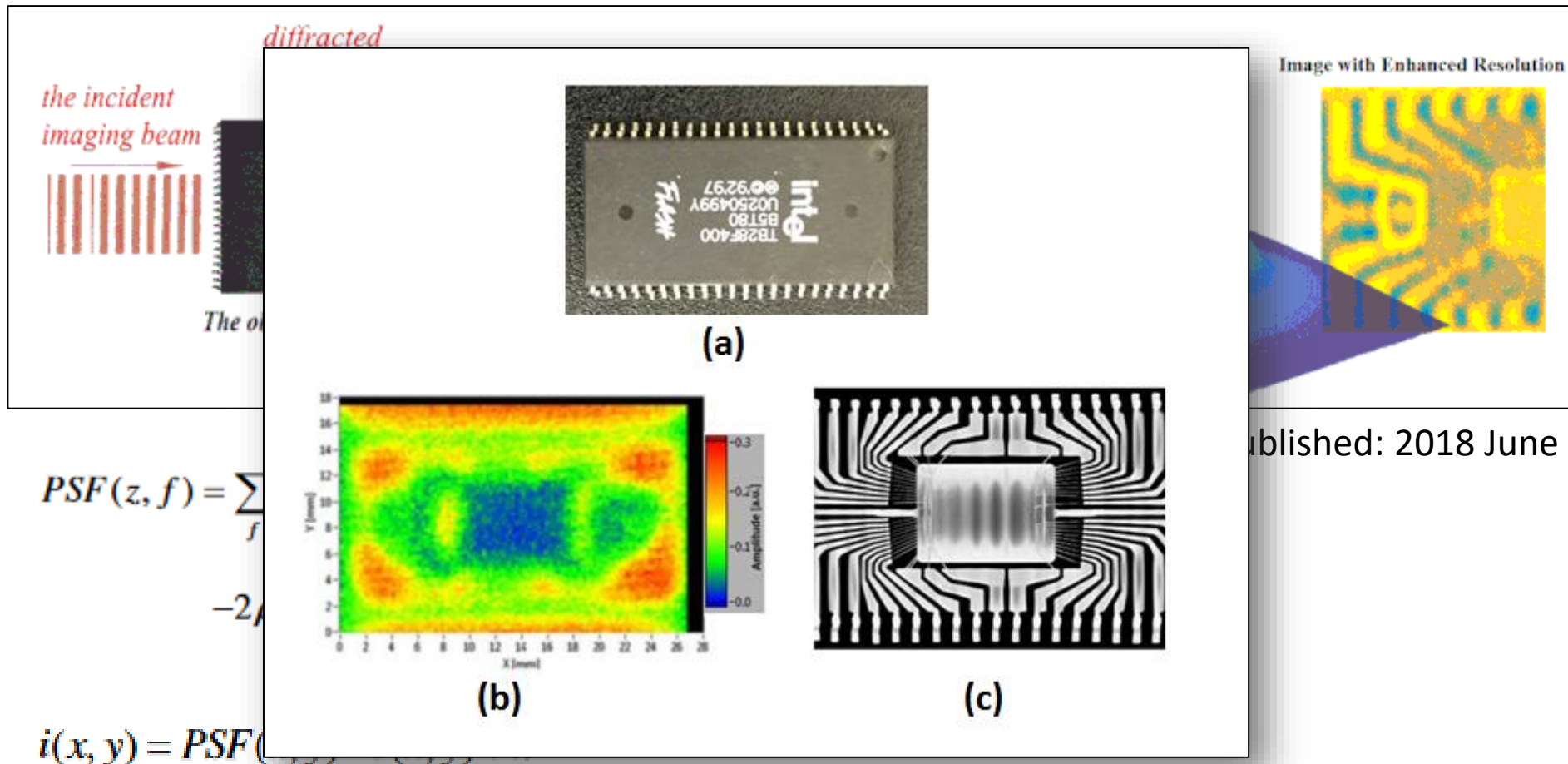


Image Restoration

~~Fun things~~ to do with image restoration tool

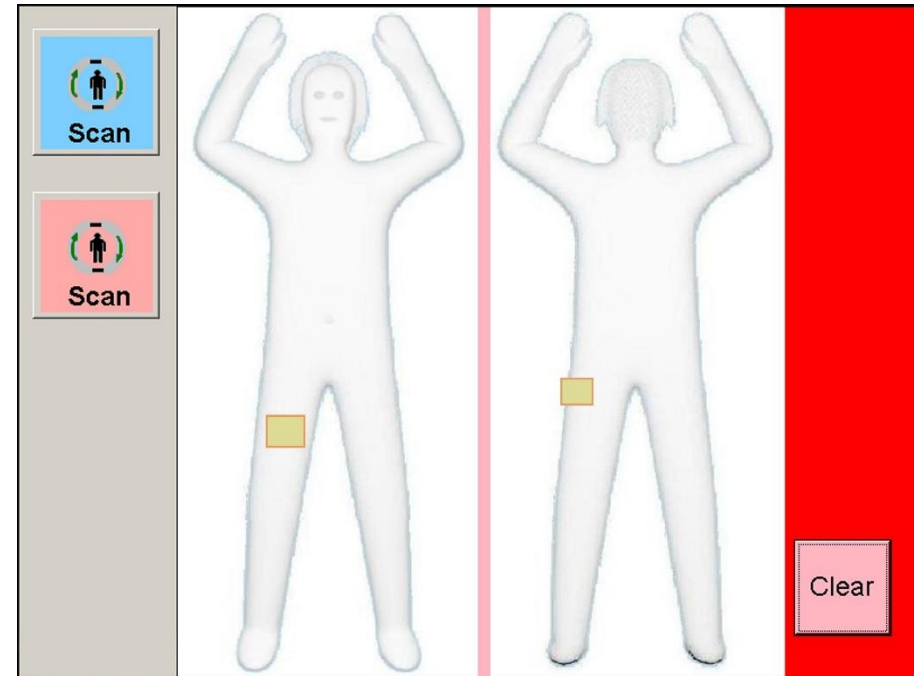


Beware

Morality and Ethics in research &
development

Image Restoration

~~Fun things~~ to do with image restoration tool



June 2013

Update software

change → Non-nude (Cookie image)

Image Restoration

~~Fun things~~ to do with image restoration tool



Ordinary photograph



Long-wavelength infrared

a trash bag is transparent, but glass (the man's spectacles) is opaque

Infrared Camera

starring Daniel

Image Restoration

Fun things to do with image restoration tool

Digital Signal Processing (DSP)

Digital Image Processing (DIP)

Echo Removal



Image Deblurring

“The larger the island of knowledge, the longer the shoreline of mystery”

PhD. Student

Image Restoration



กิจกรรมนักศึกษา

<https://canva.link/kibk4pk0q1k3vci>

แบบทดสอบหลังเรียน

